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(54) **BI- OR MULTI-SPECIFIC ANTIBODY**

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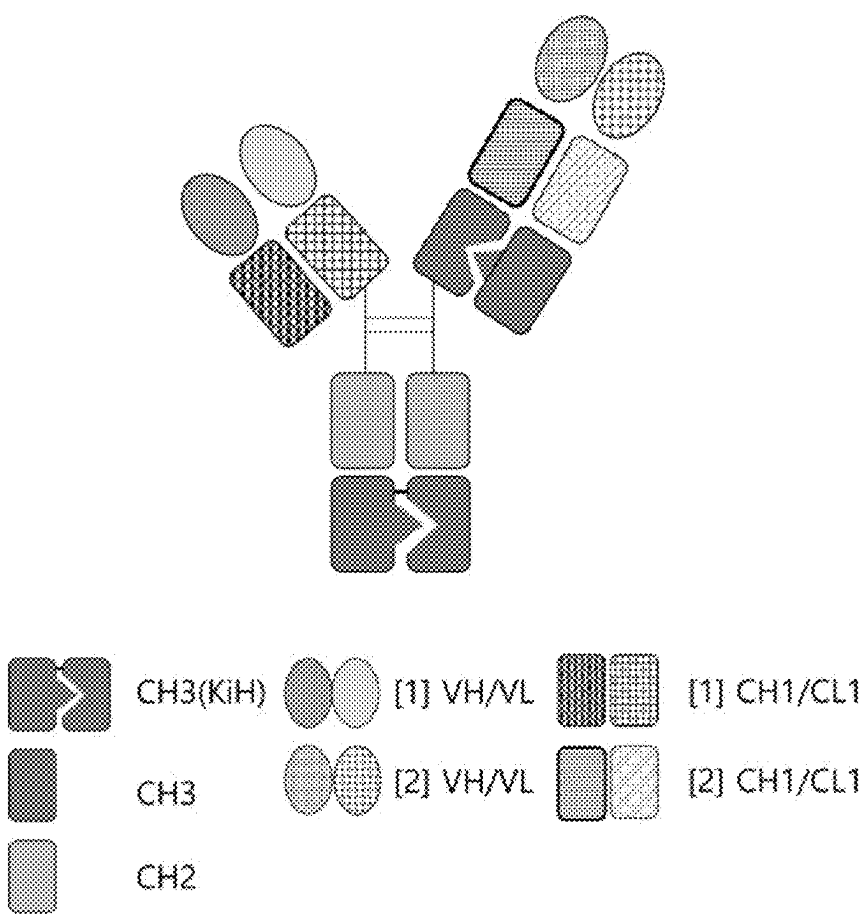
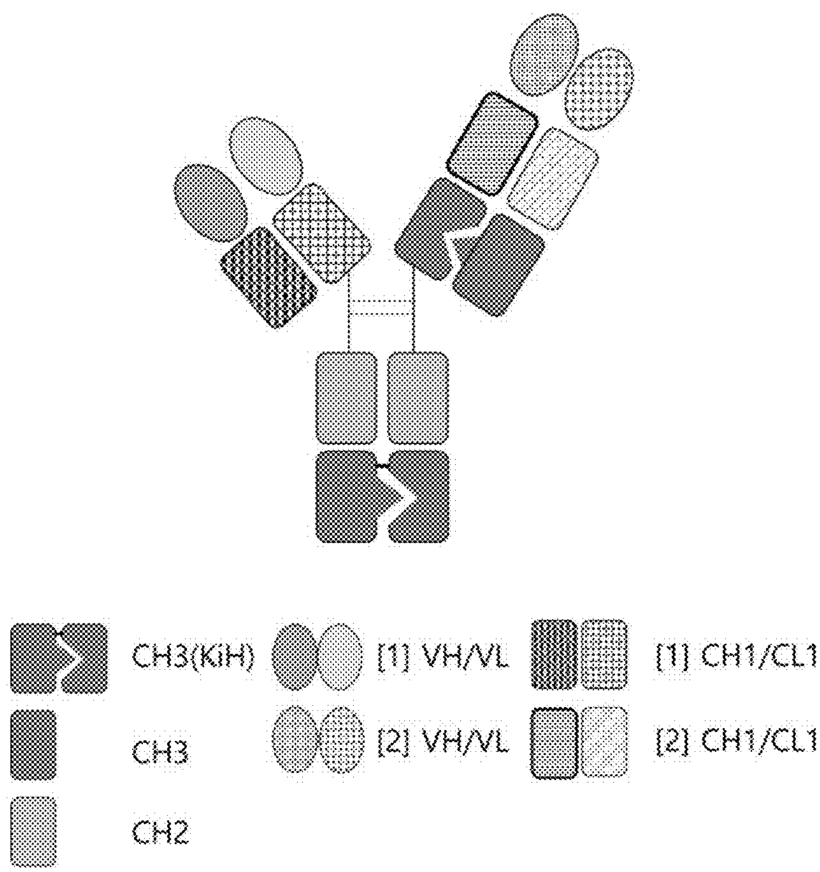
(57) **ABSTRACT**

The present invention relates to a bi- or multi-specific antibody in a novel format comprising a polypeptide in which a CH3 dimer has been introduced into a portion of a Fab region. The bi- or multi-specific antibody in the novel format forms heterodimers while exhibiting almost no non-specific binding between heavy chains and light chains, and produces almost no homodimers, and thus can be highly expressed through animal cells. In addition, the bi- or multi-specific antibody can be obtained through a purification process of an existing monoclonal antibody, and has stability equal to or higher than that of the monoclonal antibody.

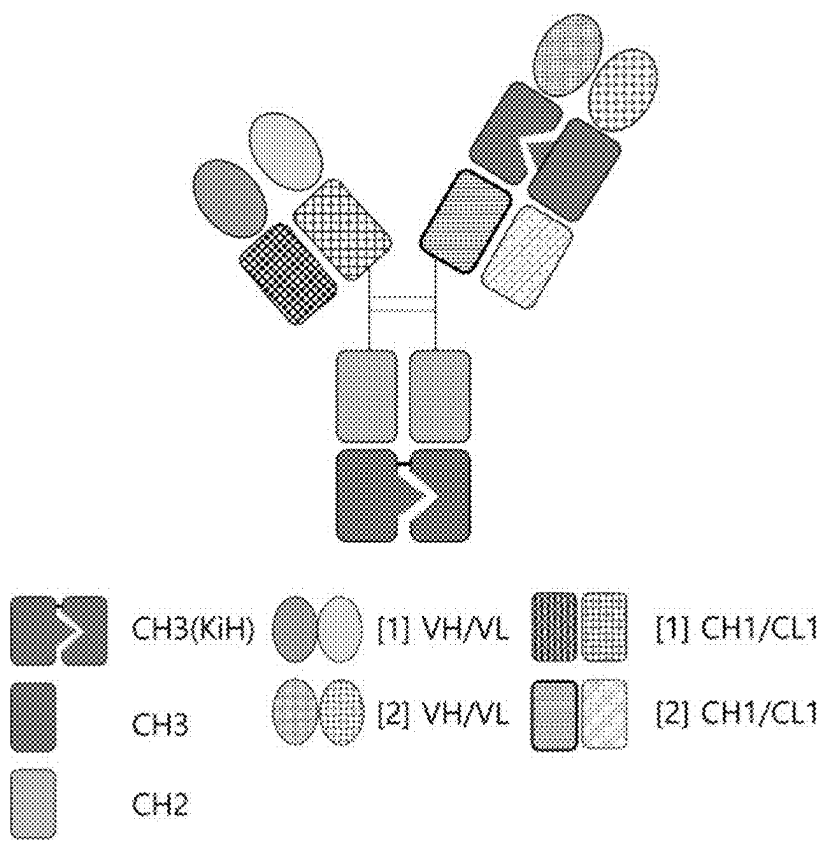
Specification includes a Sequence Listing.



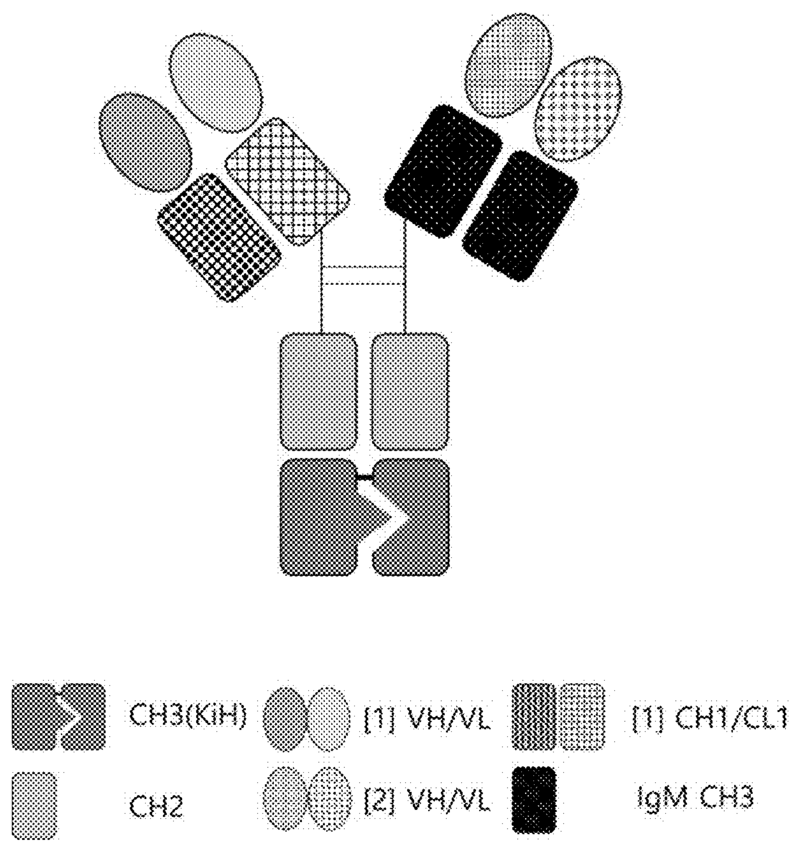
【Fig. 1】



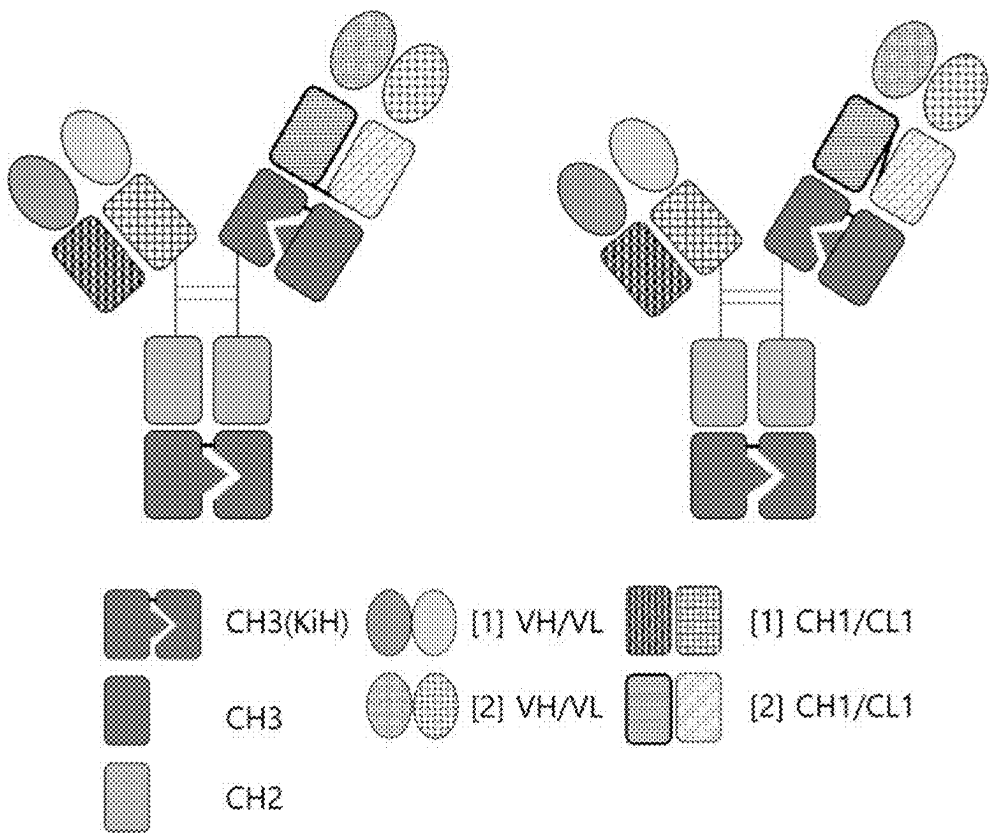
【Fig. 2】



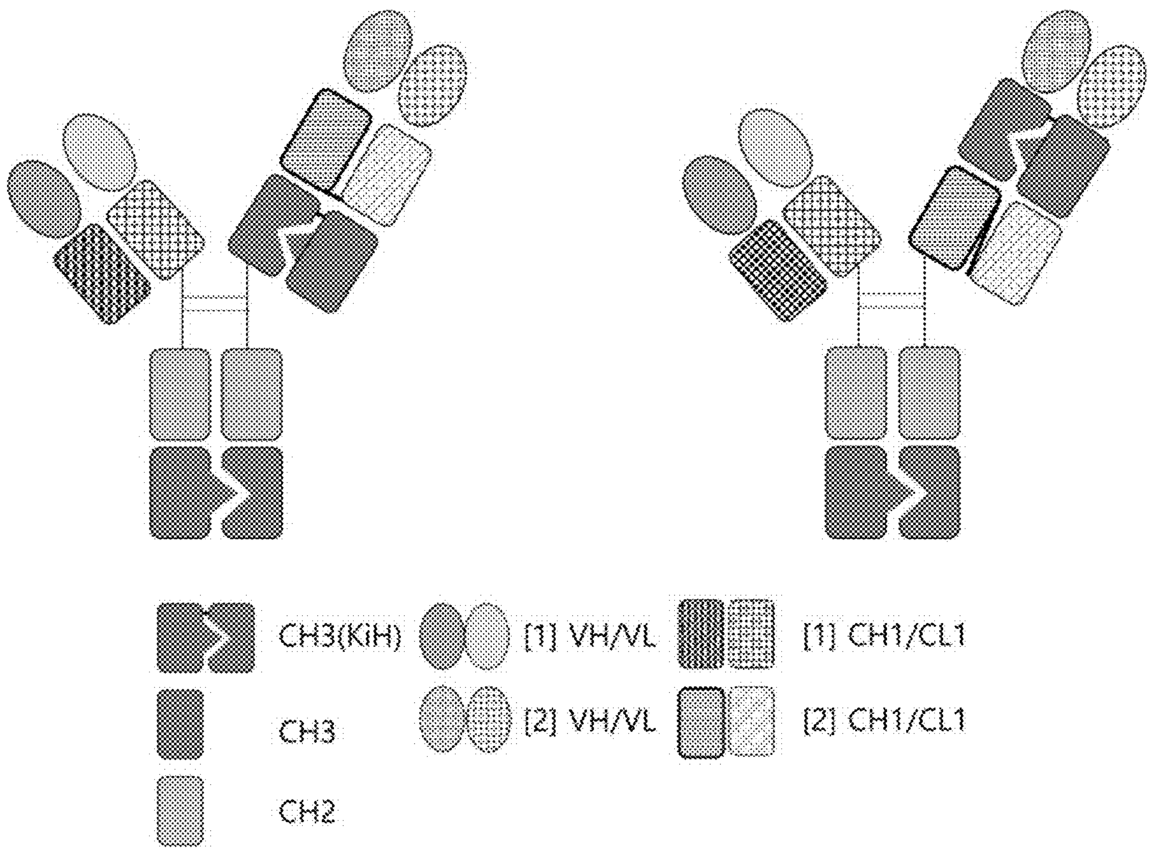
【Fig. 3】



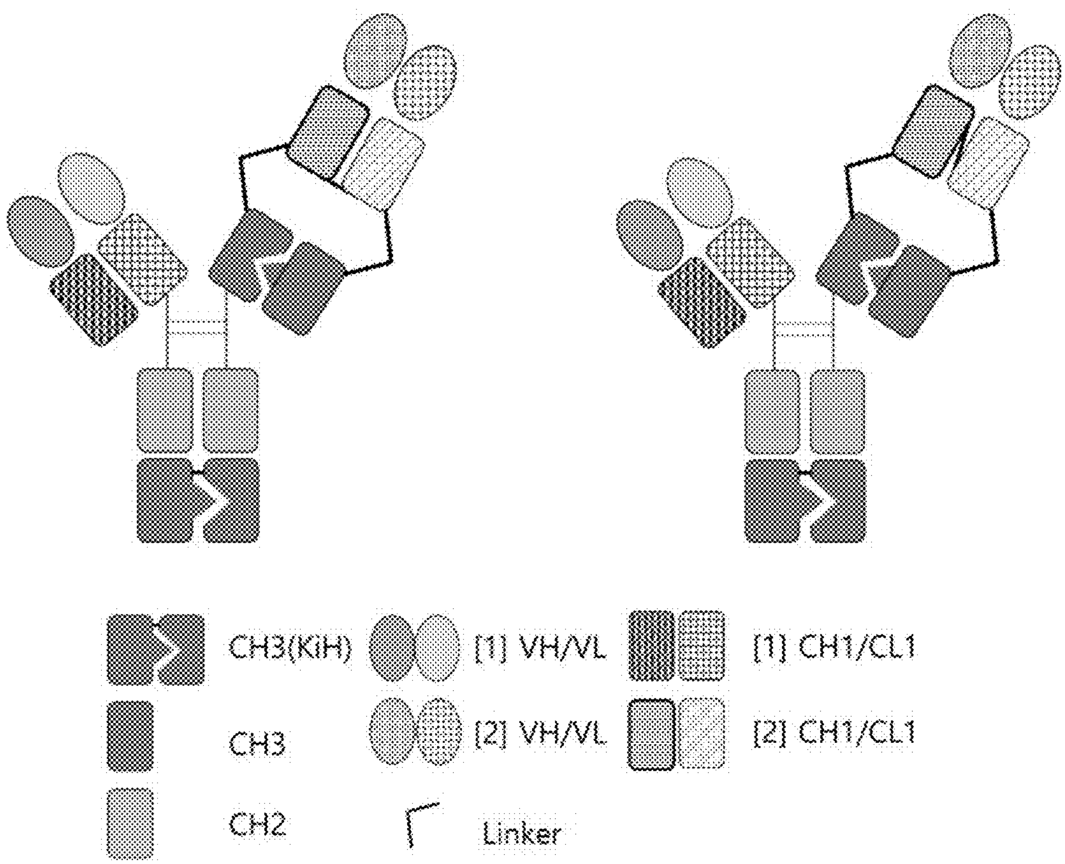
【Fig. 4】



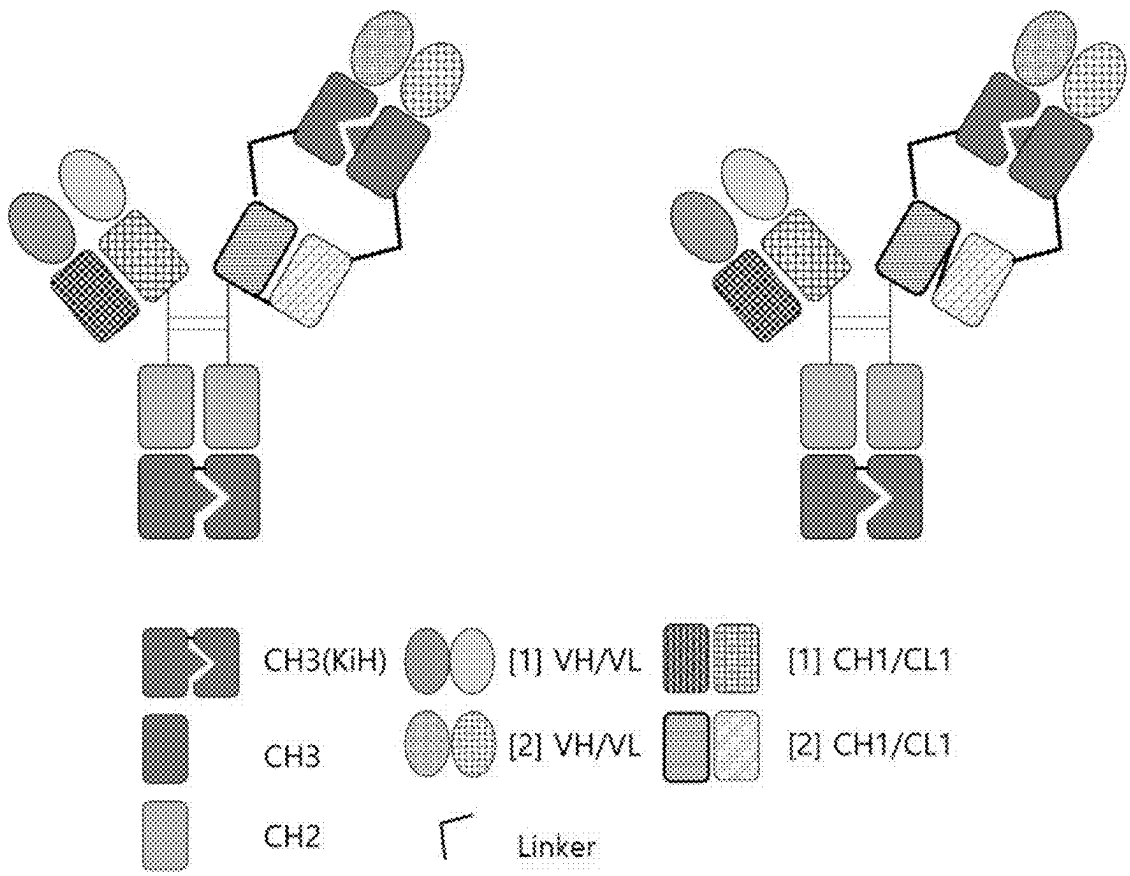
【Fig. 5】



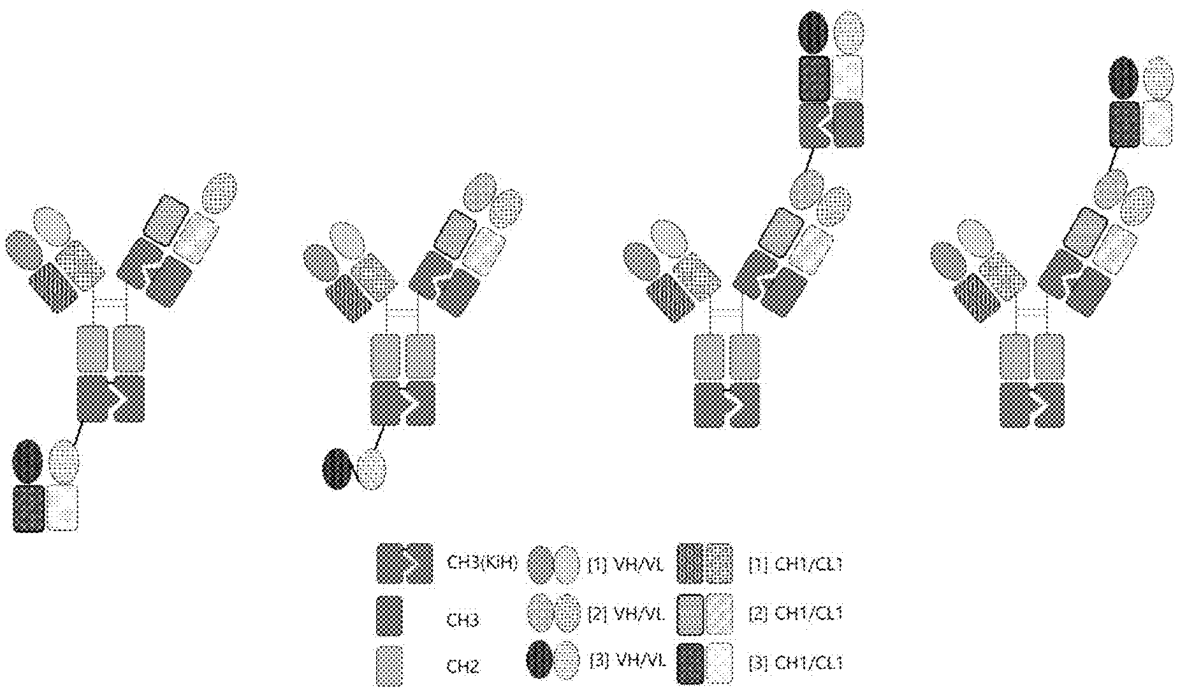
【Fig. 6】



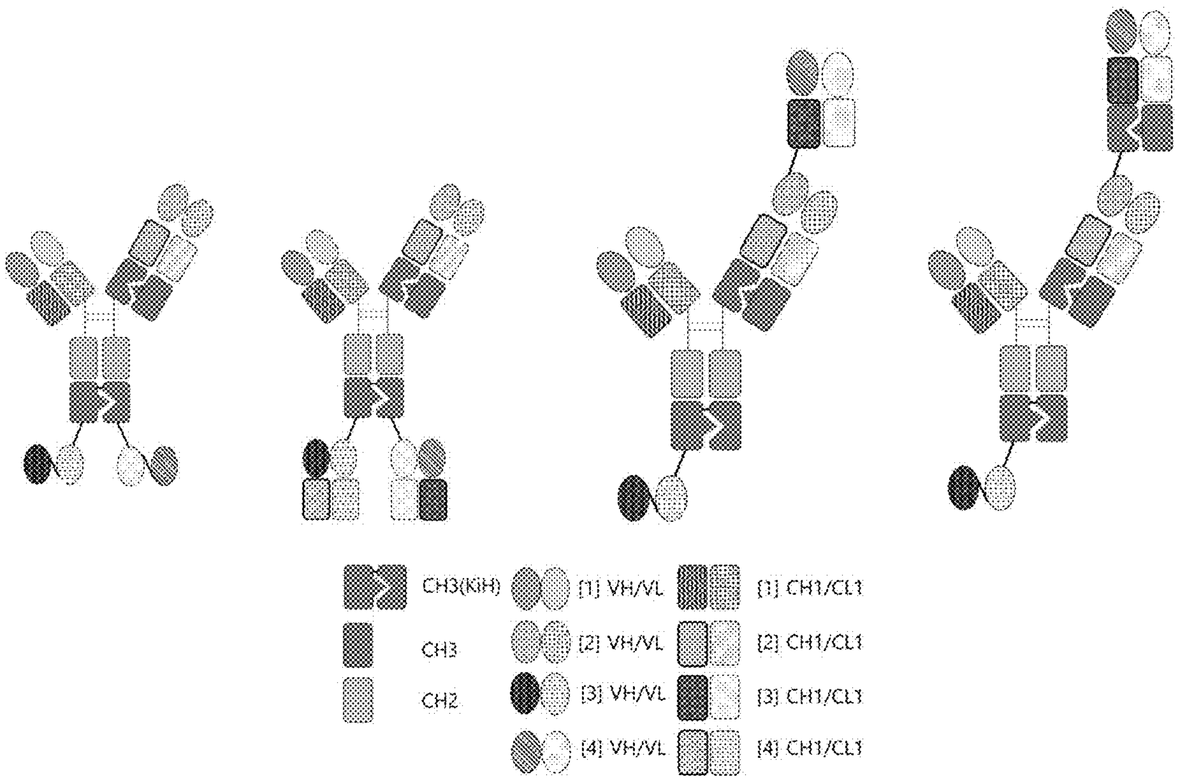
【Fig. 7】



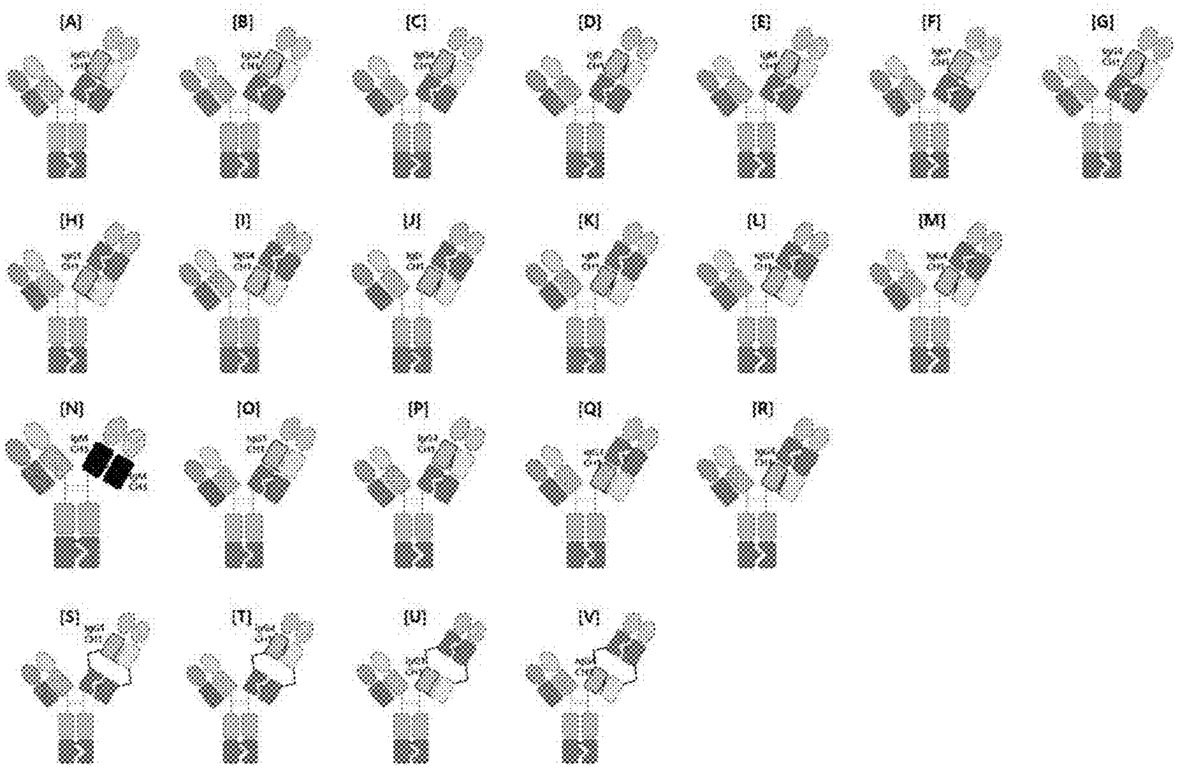
【Fig. 8】



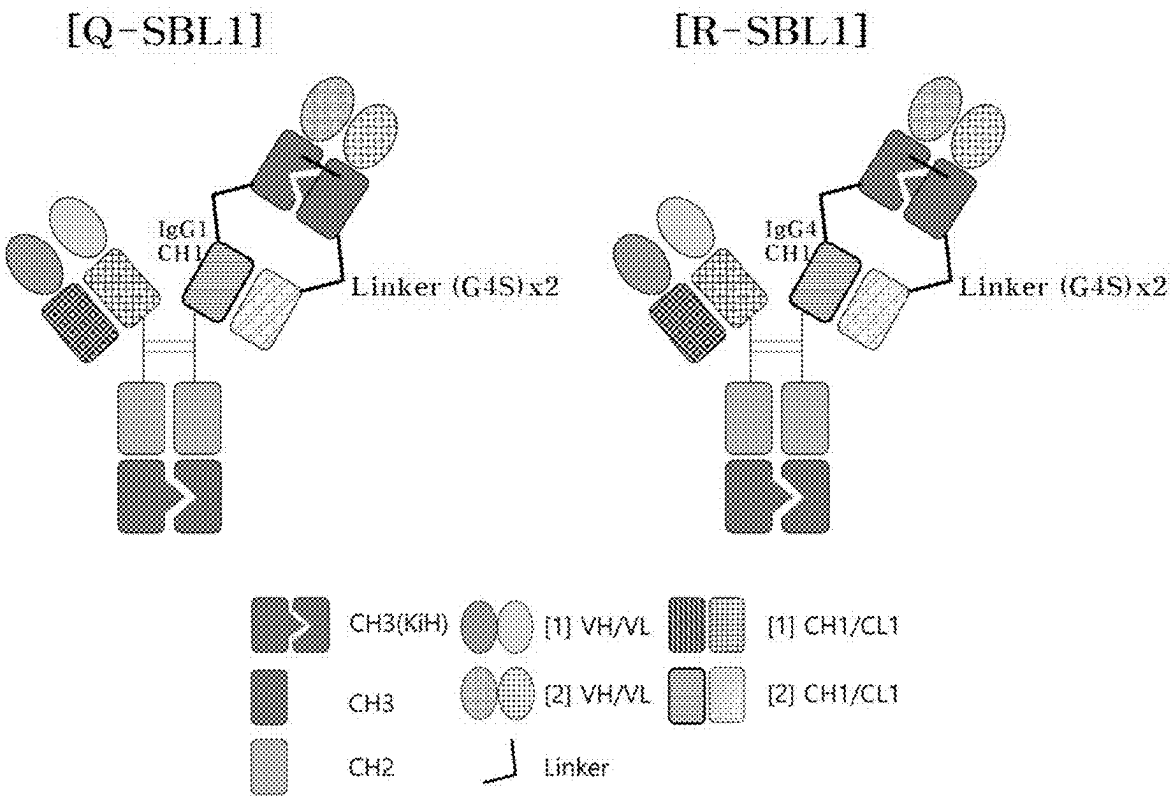
【Fig. 9】


CH3(KiH)
CH3
CH2
[1] VH/VL
[2] VH/VL
[3] VH/VL
[4] VH/VL
[1] CH1/CL1
[2] CH1/CL1
[3] CH1/CL1
[4] CH1/CL1

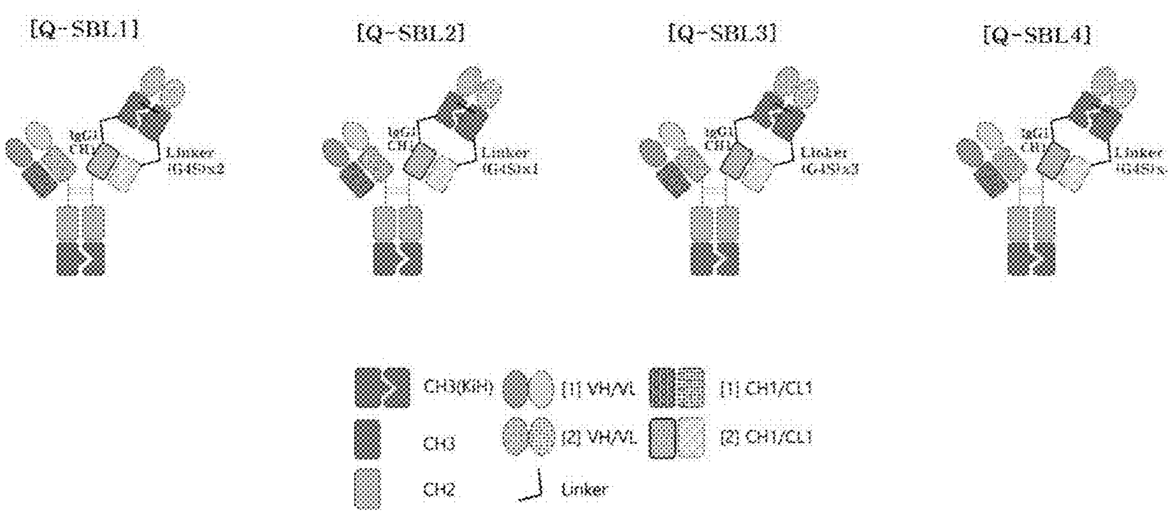
【Fig. 10】



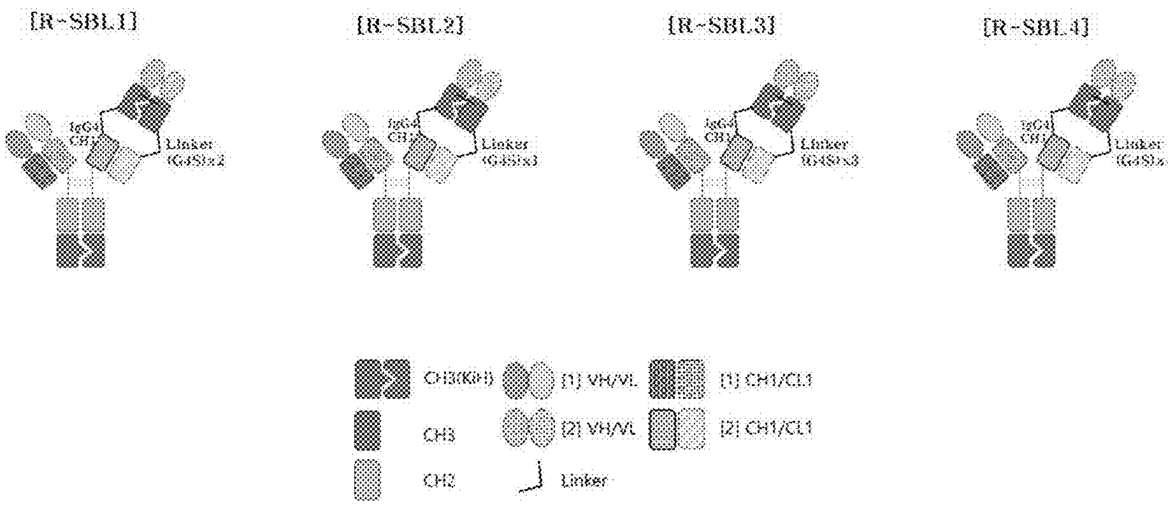
【Fig. 11】



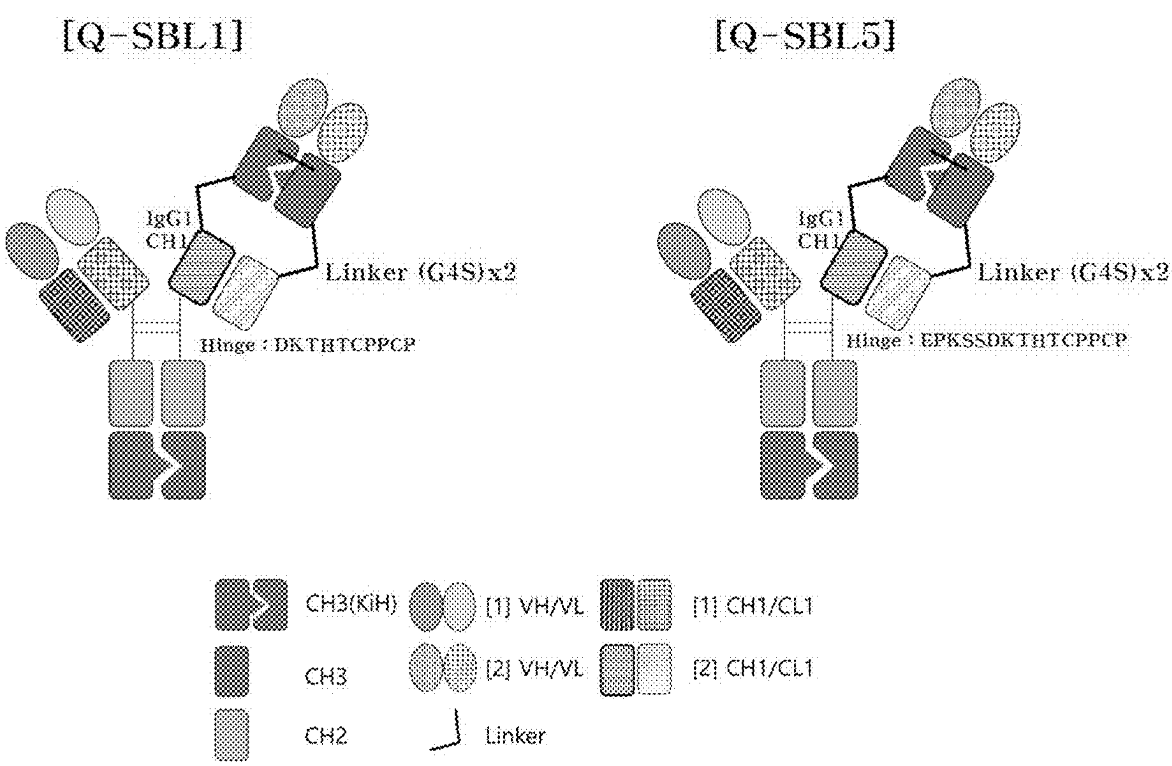
【Fig. 12a】



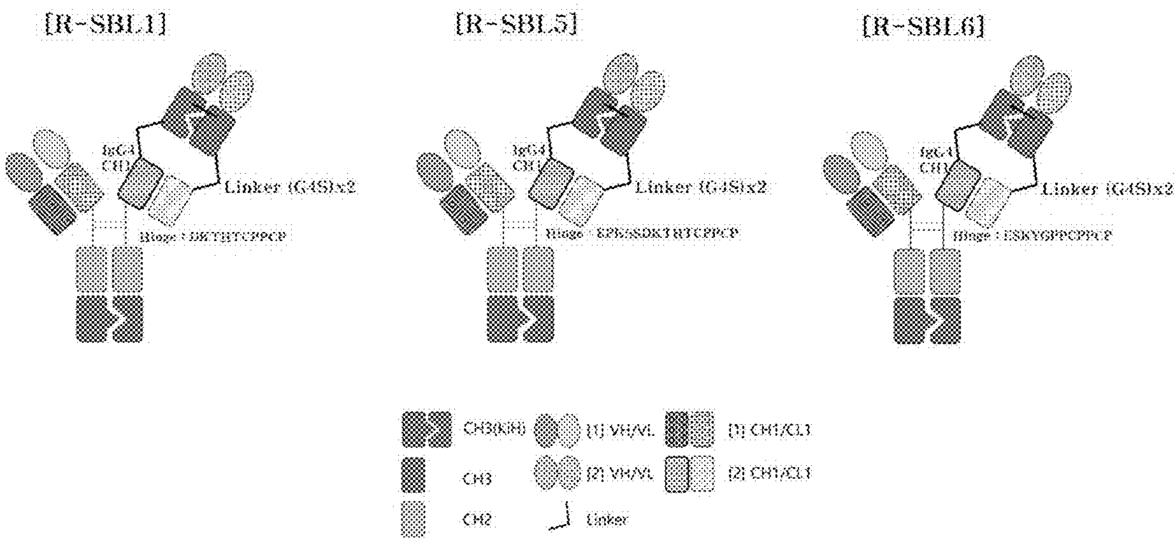
【Fig. 12b】



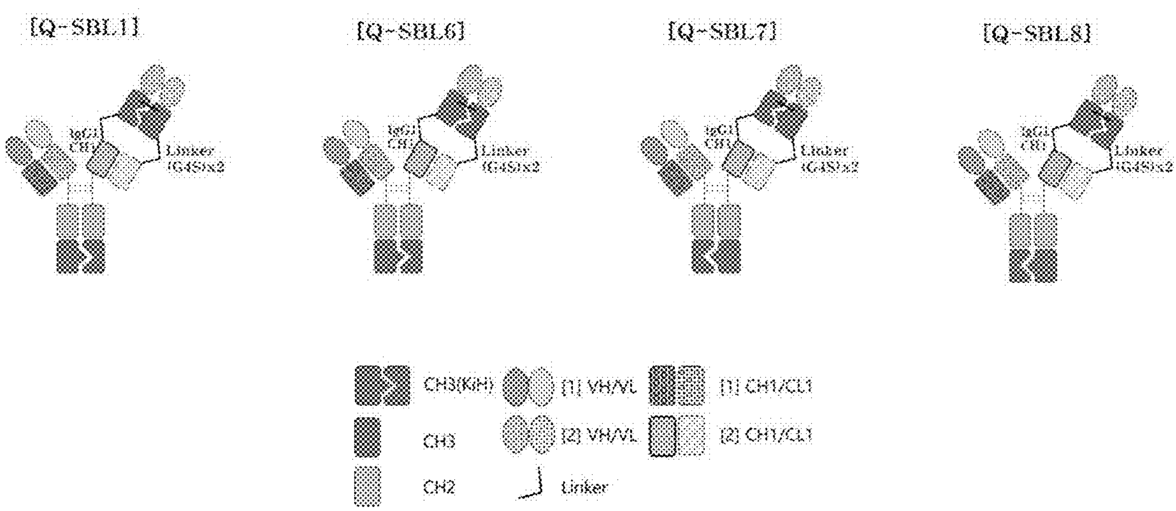
【Fig. 13a】



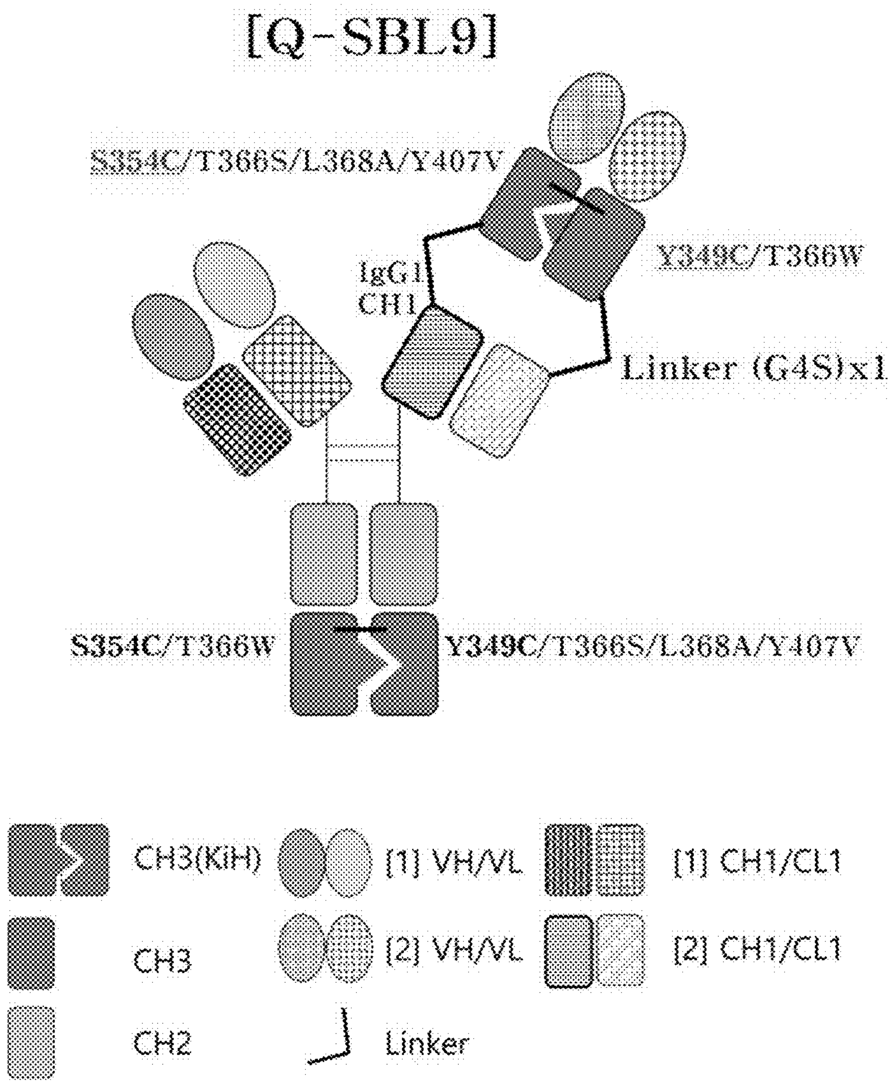
【Fig. 13b】



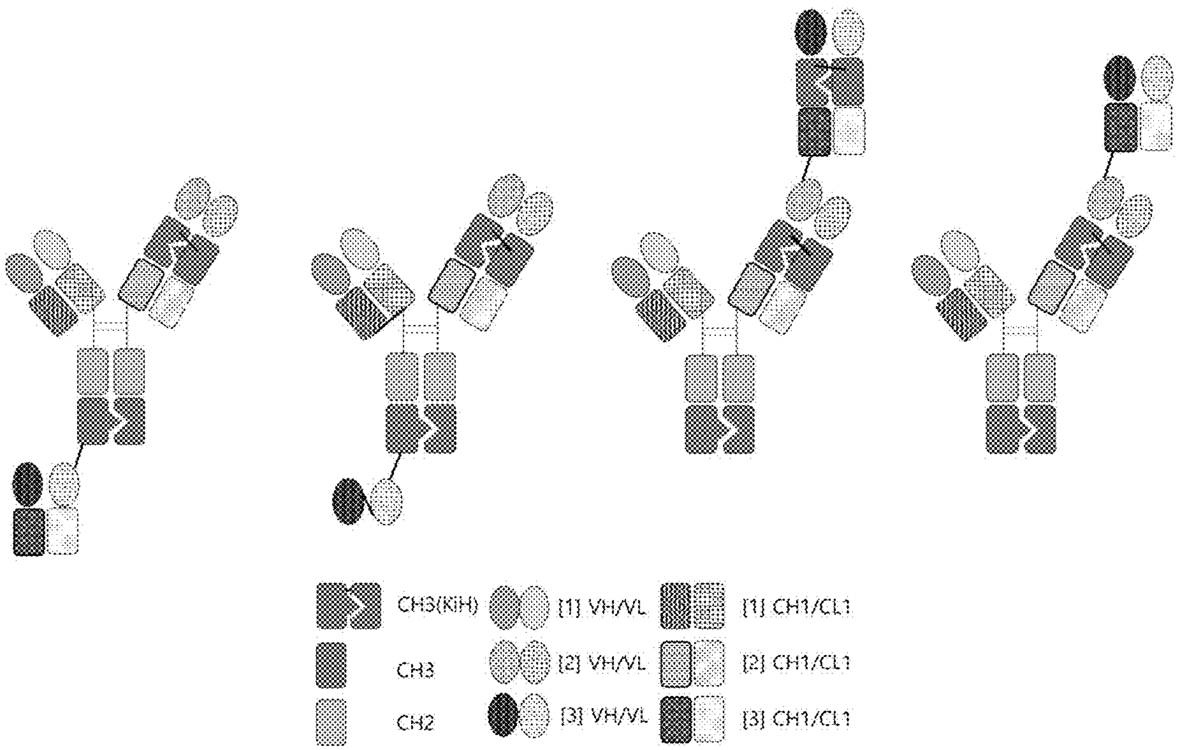
【Fig. 14】



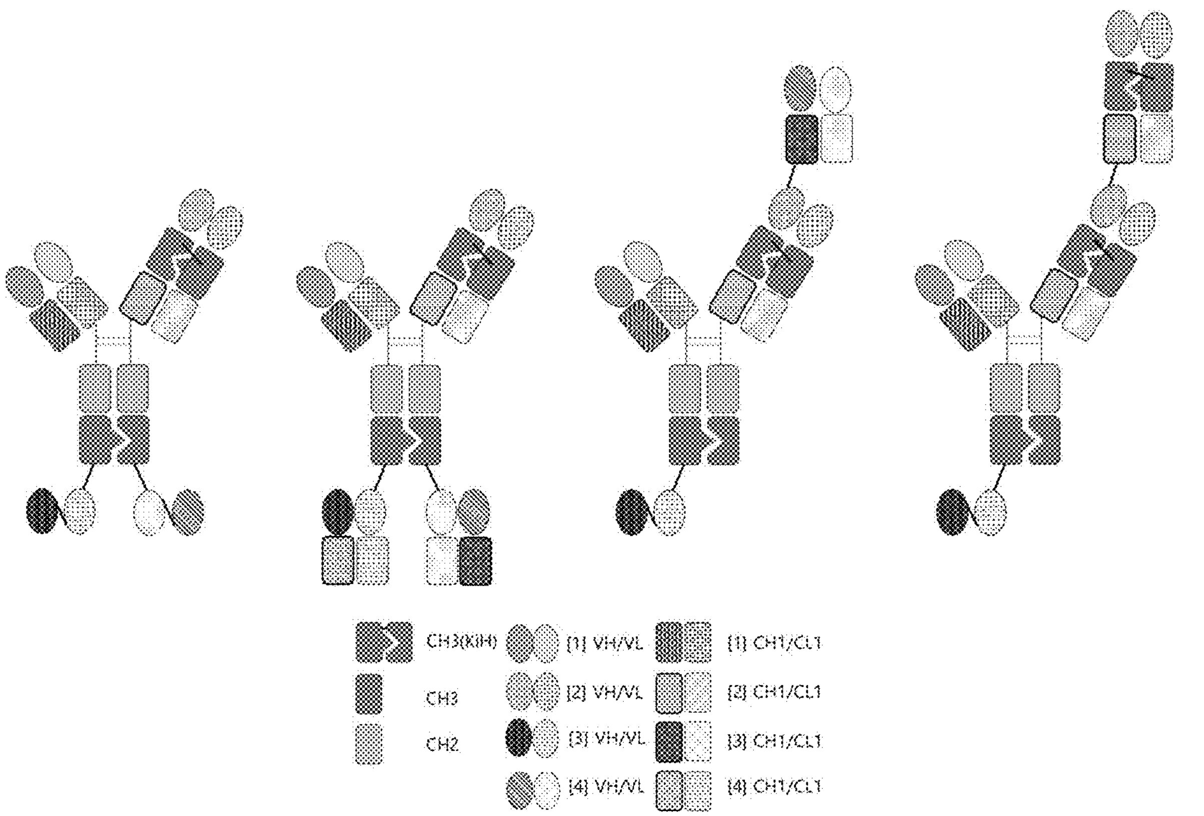
【Fig. 15】


[Q-SBL9]
S354C/T366S/L368A/Y407V
Y349C/T366W
IgG1
CH1
Linker (G4S)x1
S354C/T366W
Y349C/T366S/L368A/Y407V
CH3(KiH)
[1] VH/VL
[1] CH1/CL1
CH3
[2] VH/VL
[2] CH1/CL1
CH2
Linker

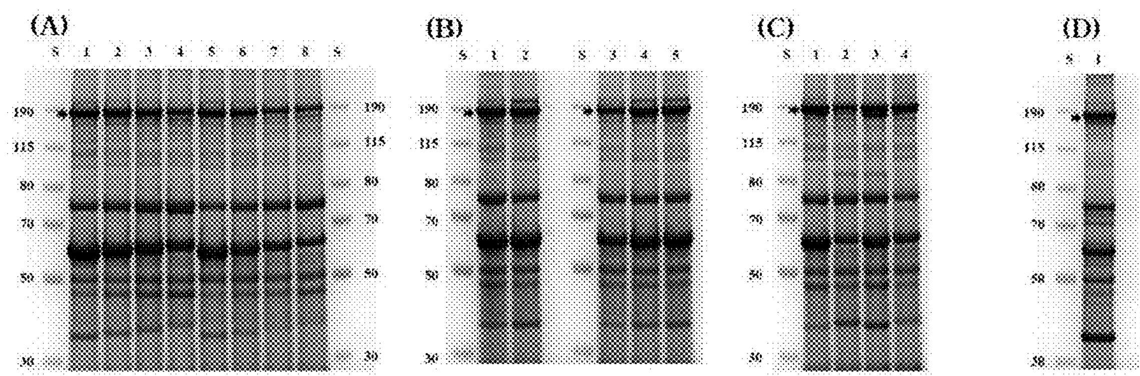
【Fig. 16】



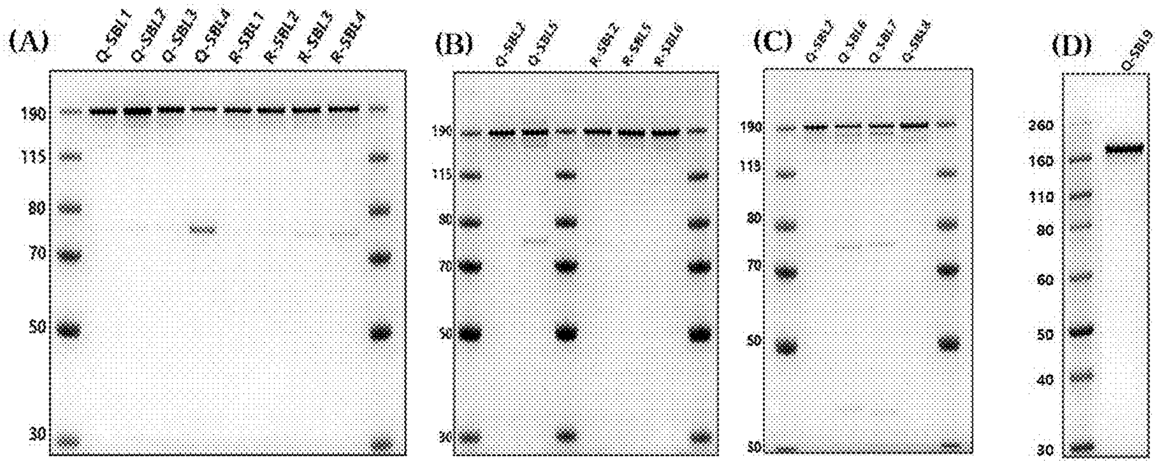
【Fig. 17】


CH3(KiH)
CH3
CH2
[1] VH/VL
[2] VH/VL
[3] VH/VL
[4] VH/VL
[1] CH1/CL1
[2] CH1/CL1
[3] CH1/CL1
[4] CH1/CL1

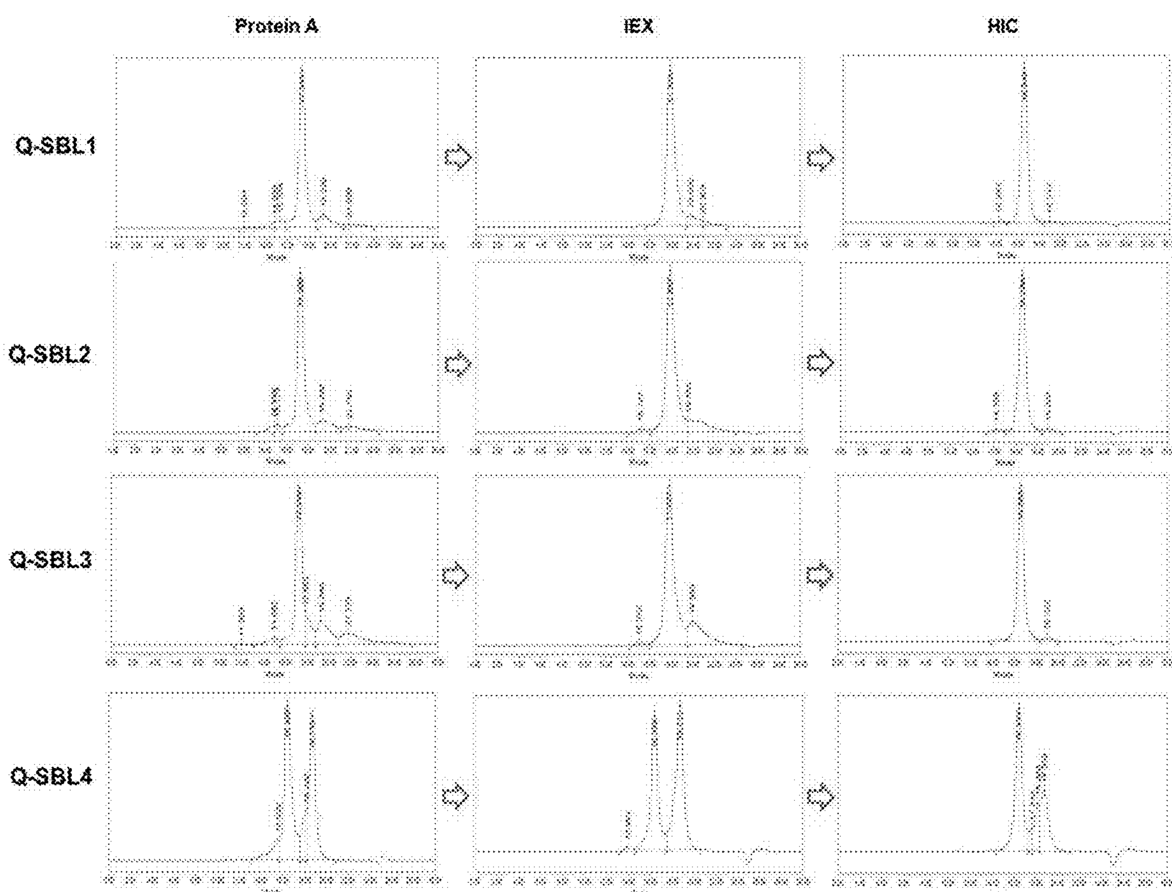
【Fig. 18】


(A)
(B)
(C)
(D)

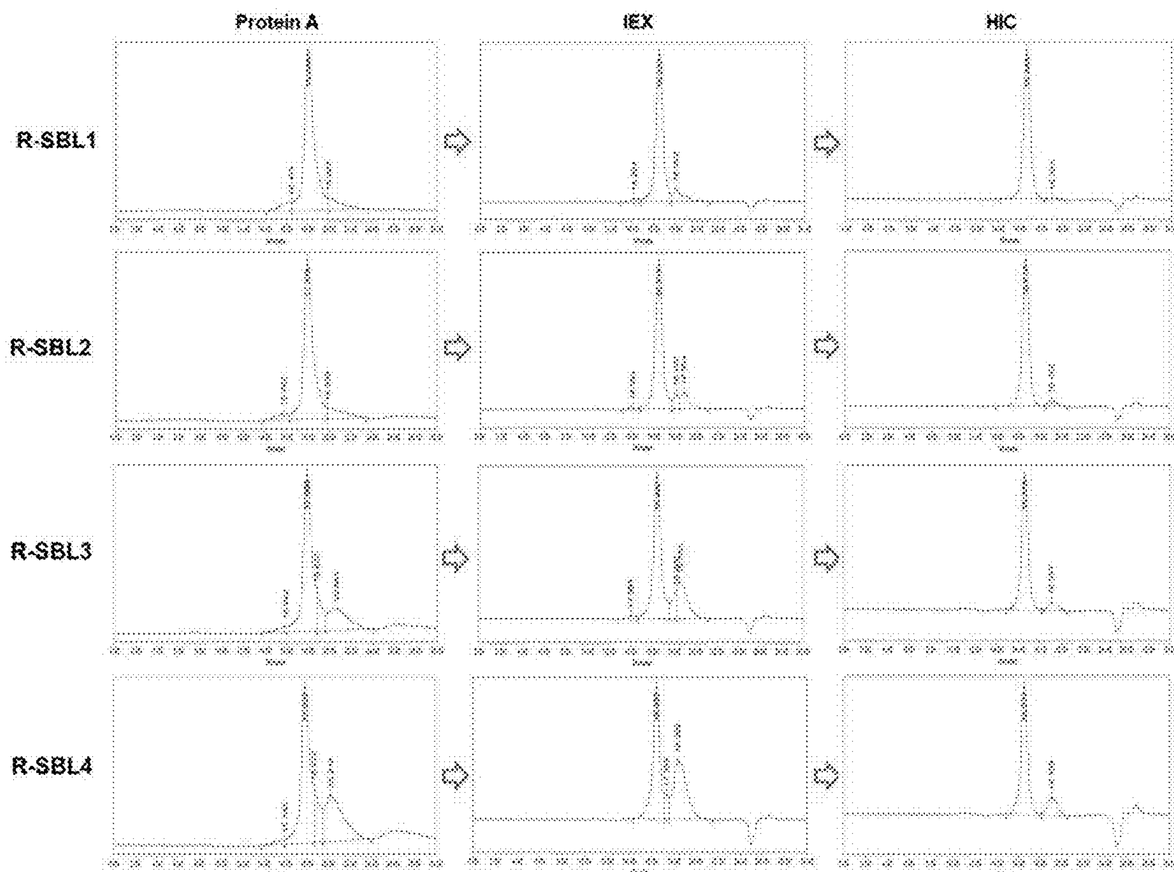
【Fig. 19】



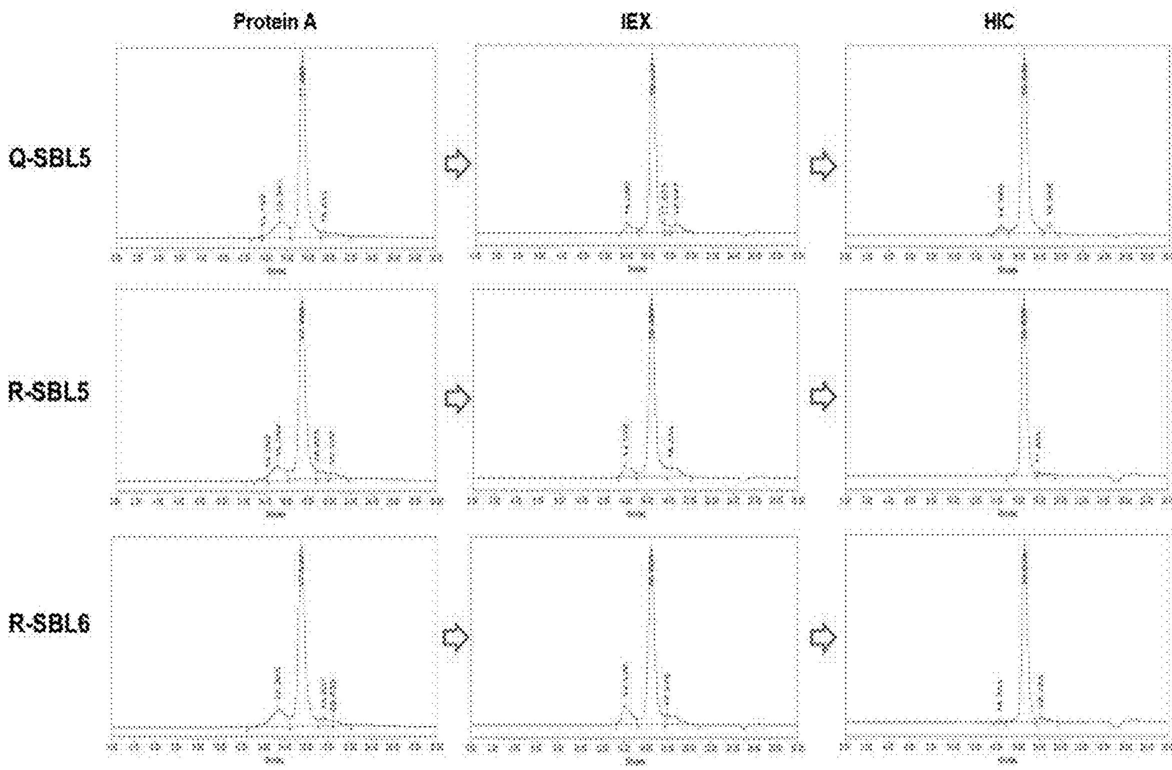
【Fig. 20a】



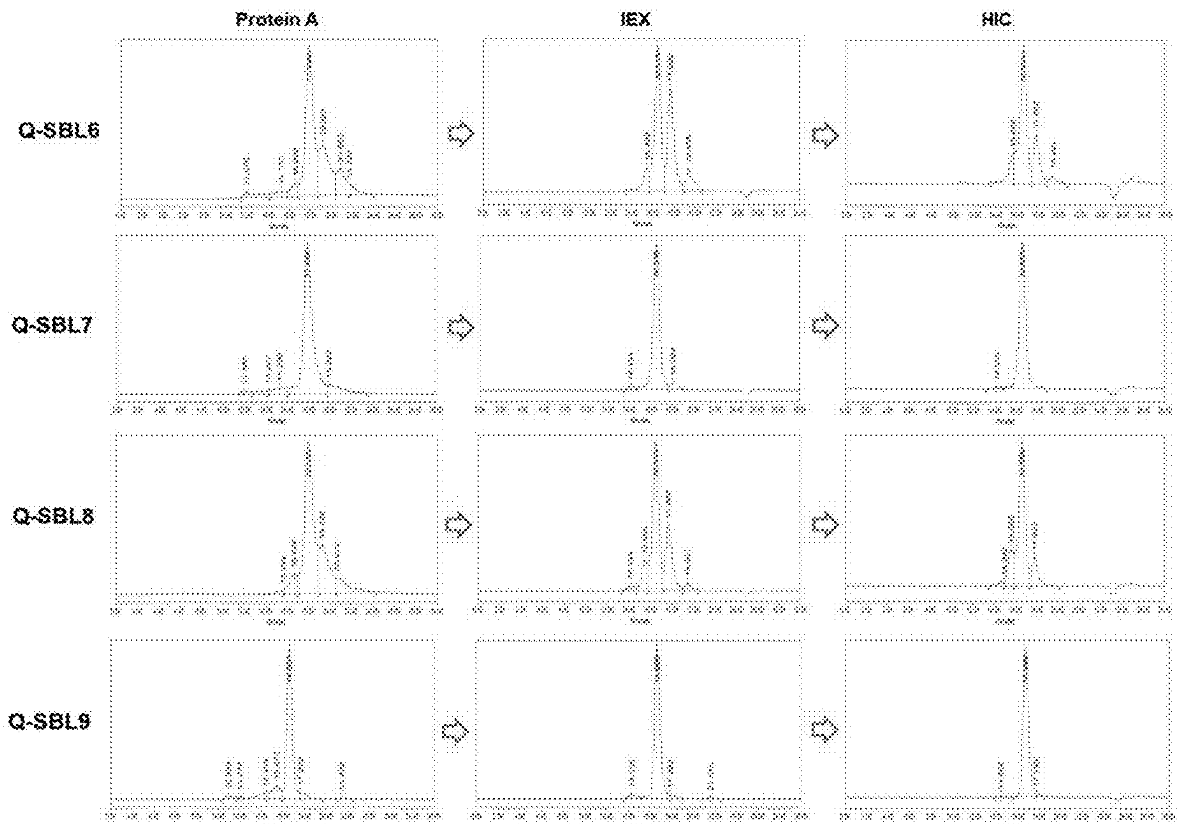
【Fig. 20b】



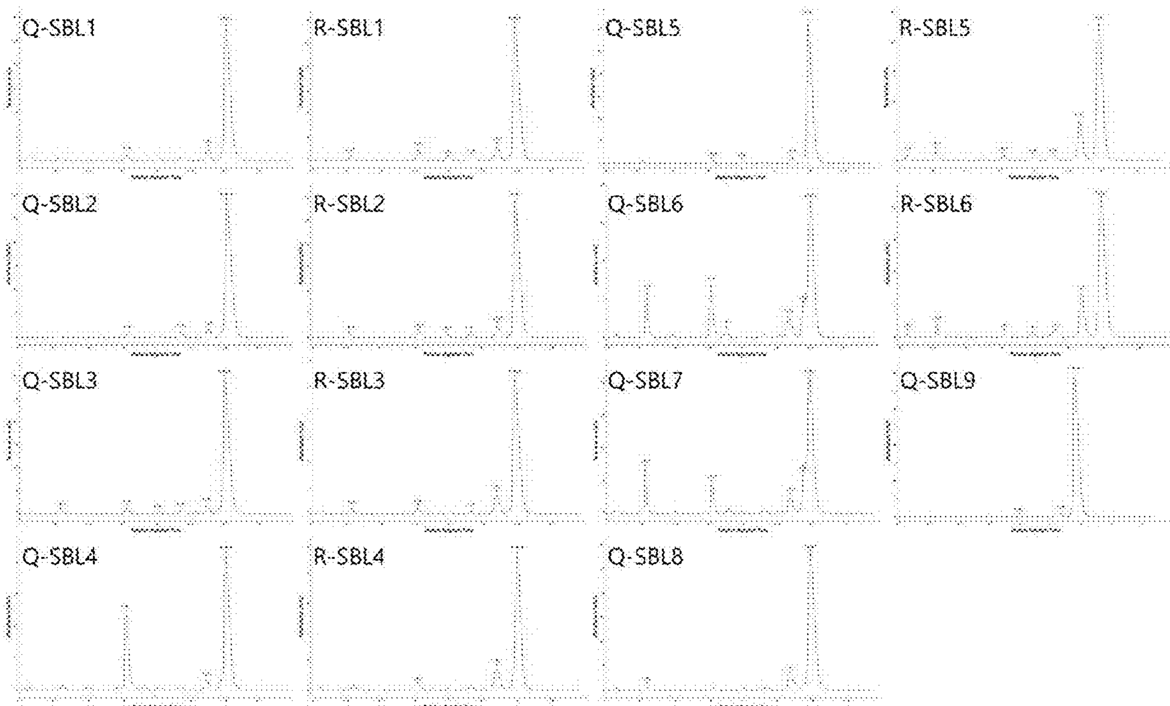
【Fig. 20c】



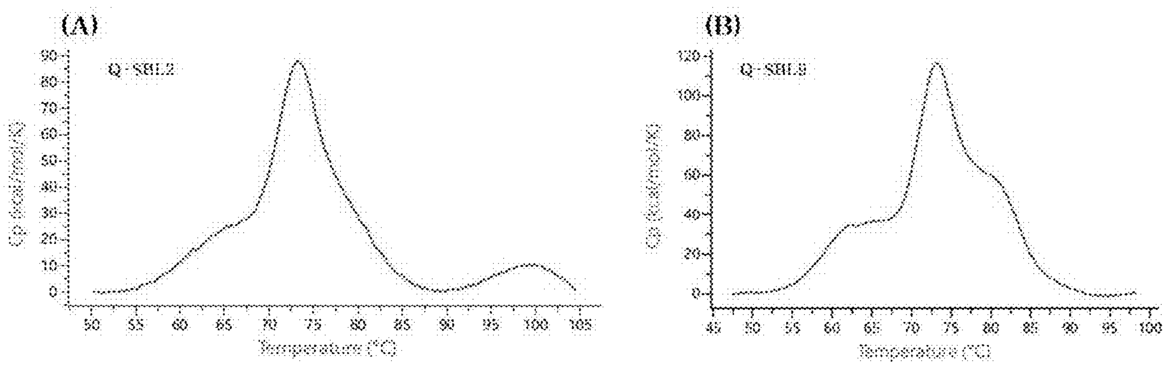
【Fig. 20d】



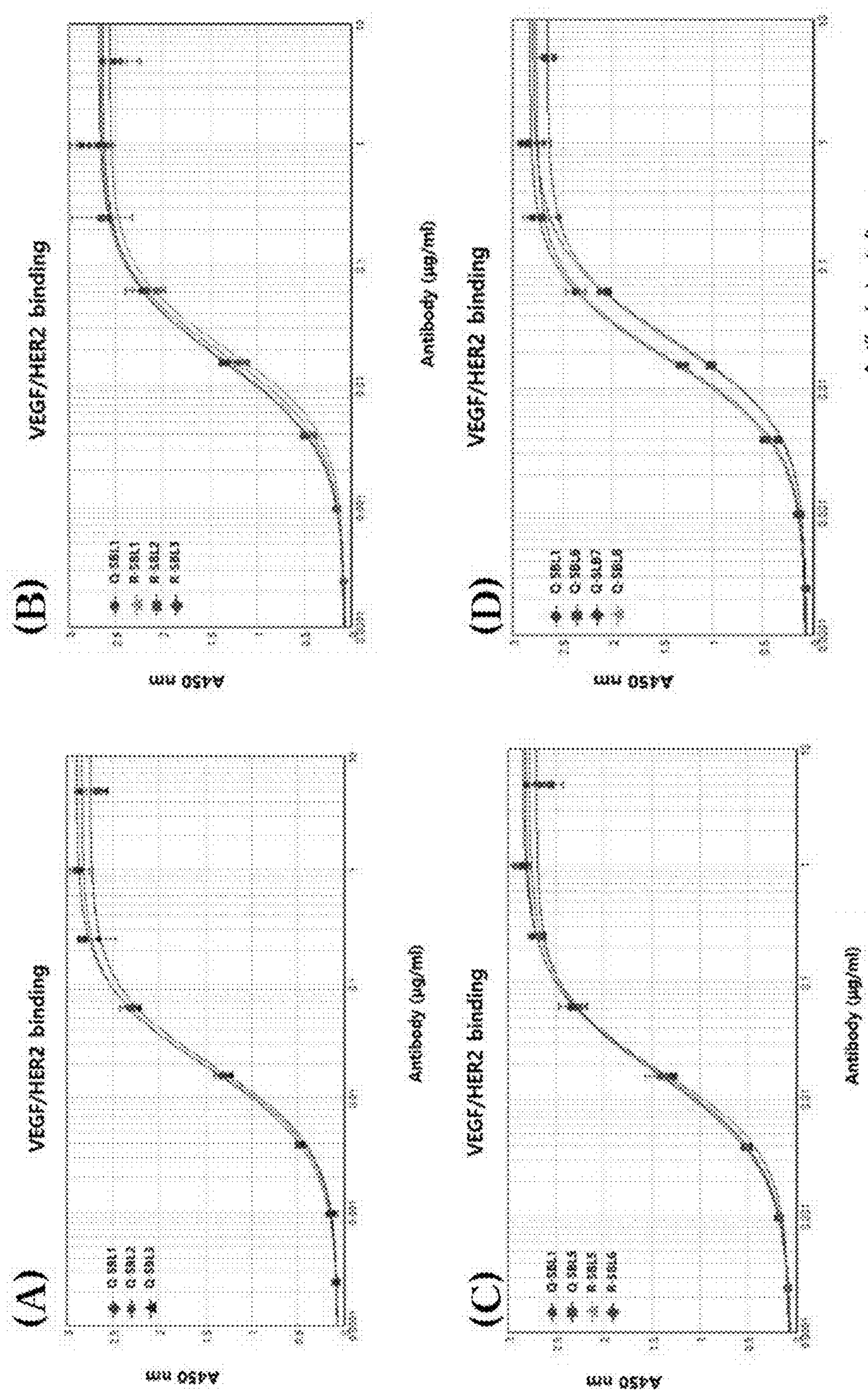
【Fig. 21】


Q-SBL1
R-SBL1
Q-SBL5
R-SBL5
Q-SBL2
R-SBL2
Q-SBL6
R-SBL6
Q-SBL3
R-SBL3
Q-SBL7
Q-SBL9
Q-SBL4
R-SBL4
Q-SBL8

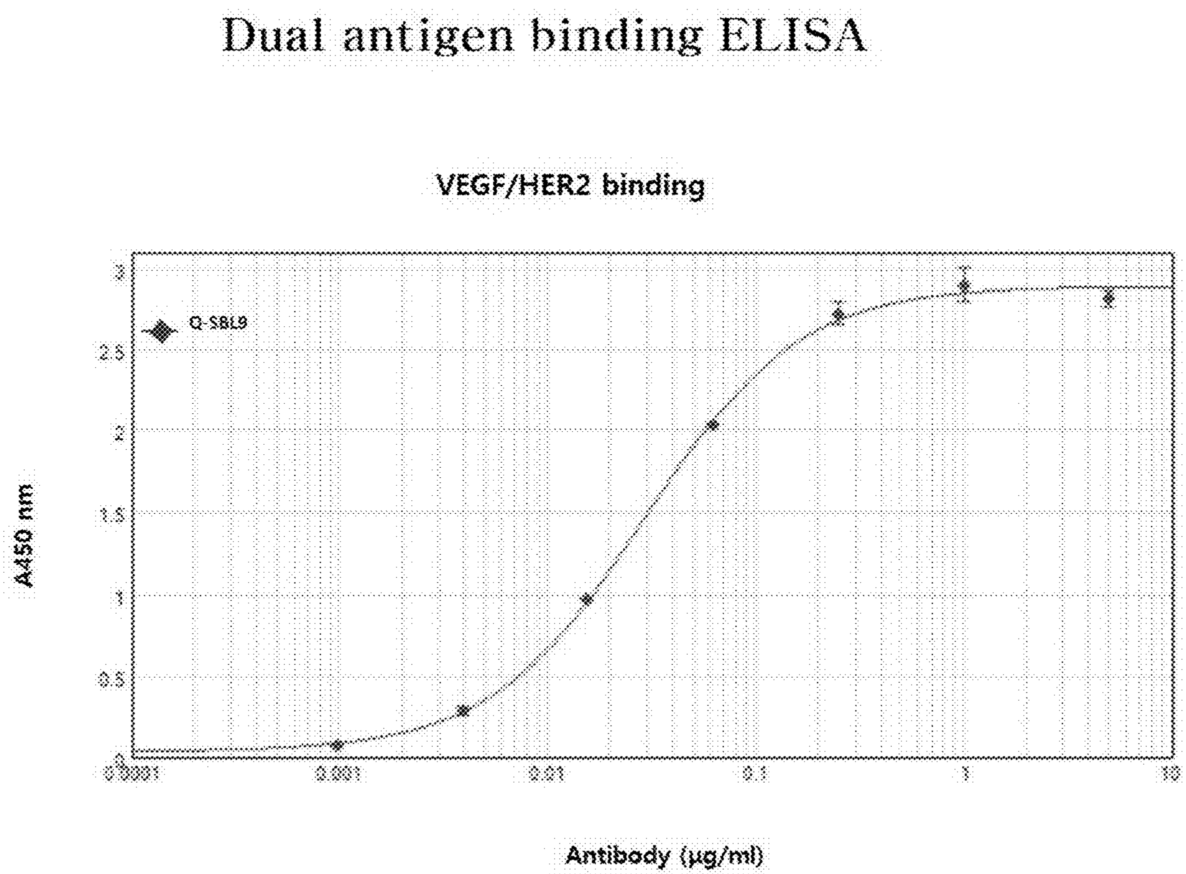
【Fig. 22】


(A)
Q-SBL2
Temperature (°C)
(B)
Q-SBL8
Temperature (°C)

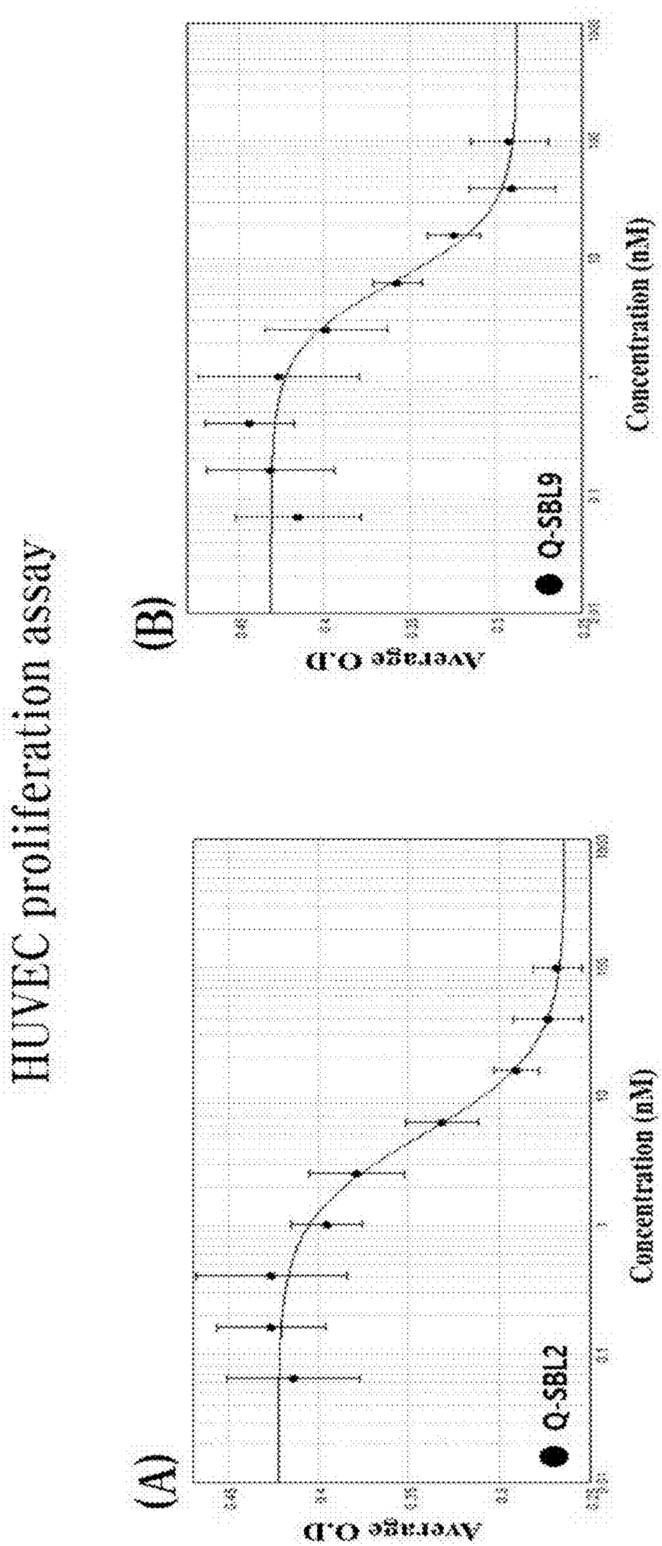
【Fig. 23a】


(A)
VEGF/HER2 binding
A450 nm
Antibody (μg/ml)
(B)
VEGF/HER2 binding
A450 nm
Antibody (μg/ml)
(C)
VEGF/HER2 binding
A450 nm
Antibody (μg/ml)
(D)
VEGF/HER2 binding
A450 nm
Antibody (μg/ml)

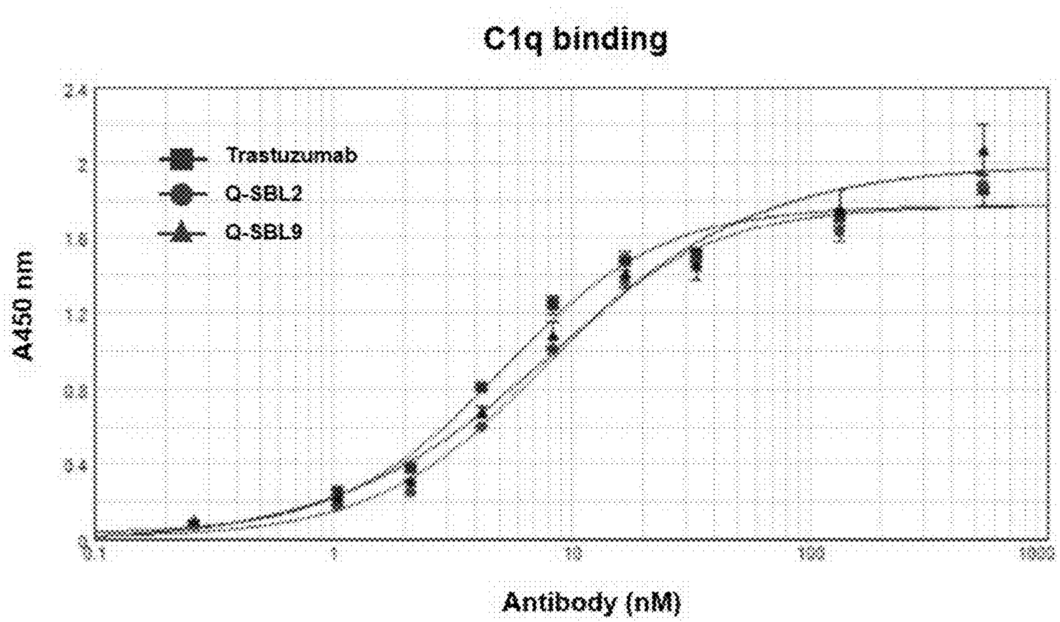
【Fig. 23b】


Dual antigen binding ELISA
VEGF/HER2 binding
Q-SBL9
A450 nm
0
0.5
1
1.5
2
2.5
3
0.0001
0.001
0.01
0.1
1
10
Antibody (μg/ml)

【Fig. 24】


HUVEC proliferation assay
(A)
Q-SBL2
Average O.D
Concentration (nM)
(B)
Q-SBL9
Average O.D
Concentration (nM)

【Fig. 25】



BI- OR MULTI-SPECIFIC ANTIBODY

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This is a United States national phase under 35 USC § 371 of international Patent Application No. PCT/KR2021/020103 filed Dec. 29, 2021, which in turn claims priority under 35 USC § 119 of Korean Patent Application No. 10-2020-0185664 filed Dec. 29, 2020. The disclosures of all such applications are hereby incorporated herein by reference in their respective entireties, for all purposes.

REFERENCE TO SEQUENCE LISTING

[0002] This application includes an electronically submitted sequence listing in .txt format. The .txt file contains a sequence listing entitled "684_SeqListing_ST25.txt" created on Jun. 24, 2023 and is 157,342 bytes in size. The sequence listing contained in this .txt file is part of the specification and is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0003] The present invention relates to a novel bispecific or multispecific antibody and relates to a bispecific or multispecific antibody including a polypeptide having a CH3 dimer introduced into a portion of a Fab region.

BACKGROUND ART

[0004] As various causes and mechanisms of one indication have recently been identified, approaches from one type of target to multiple types of targets are emerging in the development of therapeutic agents. Accordingly, in the development of therapeutic agents using antibodies, various studies have been conducted for several decades to impart the characteristics of monospecific antibodies to bispecific or multispecific mechanisms that can specifically bind to two or more antigenic proteins.

[0005] Most bioantibody drugs are composed of a single target antibody backbone, such as IgG1 or IgG4. However, such bioantibody drugs have limitations such as efficacy, convenience, and side effects as antibody therapeutic agents for a single target, since mechanisms for various targets act on diseases.

[0006] At present, various clinical stages focus on combination therapy of two monoclonal antibodies. Accordingly, there is a need to develop antibodies that allow one protein to bind to two or more targets.

[0007] Several research papers associated therewith have been published in the last 20 years or so. Currently, there are a small number of licensed bispecific antibodies, a large number of candidate materials are researched/developed in the clinical stage, and examples of licensed bispecific antibodies include Removab, Blincyto (BiTE™), and Hemlibra.

[0008] However, there is still a need for improvement of production yield, productivity, solubility, aggregation, and stability of the developed bi- or multi-target antibodies.

[0009] Under this technical background, as a result of extended efforts to develop a novel format of bispecific or multispecific antibodies, the present inventors developed a bispecific or multispecific antibody including a polypeptide having a CH3 dimer introduced into the Fab region. Based thereon, the present invention was completed.

DISCLOSURE

Technical Problem

[0010] Therefore, the present invention has been made in view of the above problems, and it is an object of the present invention to provide a novel bi- or multi-specific antibody platform.

Technical Solution

[0011] In accordance with one aspect of the present invention, the above and other objects can be accomplished by the provision of a bispecific antibody including a first arm binding to a first antigen and including VH1-CHa-Fc1 and VL1-CLb, and a second arm binding to a second antigen and including VH2-CH1-Fc2 and VL2-CL,

[0012] wherein the VH1 and the VH2 are each heavy-chain variable regions including the same or different antigen-binding regions,

[0013] the VL1 and the VL2 are each light-chain variable regions including the same or different antigen-binding regions,

[0014] the CHa includes i) an IgG heavy-chain constant region or an IgD heavy-chain constant region CH1, and an IgG heavy-chain constant region CH2 or CH3, or ii) an IgM heavy-chain constant region CH3,

[0015] the CLb includes i) a CL1 including an IgG light-chain constant region λ or κ, and at least one selected from the group consisting of IgG heavy-chain constant regions CH1, CH2, and CH3, or ii) an IgM heavy-chain constant region CH3,

[0016] the CHa and the CLb are linked to each other to form a dimer,

[0017] the CH1 is an IgG heavy-chain constant region CH1 and the CL is an IgG light-chain constant region CL, and

[0018] the Fc1 of the first arm is linked to the Fc2 of the second arm to form a heavy-chain constant region dimer.

[0019] In accordance with another aspect of the present invention, provided is a bispecific antibody including a first arm binding to a first antigen and including VH1-CHa-Fc1 and VL1-CLb, and a second arm binding to a second antigen and including VH2-CH1-Fc2 and VL2-CL,

[0020] wherein the VH1 and the VH2 are each heavy-chain variable regions including the same or different antigen-binding regions,

[0021] the VL1 and the VL2 are each light-chain variable regions including the same or different antigen-binding regions,

[0022] the CHa includes an IgG heavy-chain constant region CH3 and an IgG heavy-chain constant region CH1,

[0023] the CLb includes a CL1 including an IgG light-chain constant region λ or κ, and a CH3 including an IgG heavy-chain constant region,

[0024] the CHa and the CLb are linked to each other to form a dimer,

[0025] the CH1 is an IgG heavy-chain constant region CH1 and the CL is an IgG light-chain constant region CL, and

[0026] the Fc1 of the first arm is linked to the Fc2 of the second arm to form a heavy-chain constant region dimer.

[0027] In accordance with another aspect of the present invention, provided is a multispecific antibody including the bispecific antibody.

DESCRIPTION OF DRAWINGS

[0028] FIG. **1** illustrates the structure of a first bispecific antibody candidate.

[0029] FIG. **2** illustrates the structure of a second bispecific antibody candidate.

[0030] FIG. **3** illustrates the structure of a third bispecific antibody candidate.

[0031] FIG. **4** illustrates the structure of a fourth bispecific antibody candidate.

[0032] FIG. **5** illustrates the structure of a fifth bispecific antibody candidate.

[0033] FIG. **6** illustrates the structure of a sixth bispecific antibody candidate.

[0034] FIG. **7** illustrates the structure of a seventh bispecific antibody candidate.

[0035] FIG. **8** illustrates the structure of a multispecific antibody that binds to three targets.

[0036] FIG. **9** illustrates the structure of a multispecific antibody that binds to four targets.

[0037] FIG. **10** illustrates an exemplary bivalent bispecific antibody format.

[0038] FIG. **11** is a schematic diagram illustrating bispecific antibody formats of Q-SBL1 and R-SBL1 constructs, respectively.

[0039] FIGS. **12**A to **12**B are schematic diagrams illustrating various bispecific antibody formats of Q-SBL1 (FIG. **12**A) and R-SBL1 (FIG. **12**B) depending on the number of linkers.

[0040] FIGS. **13**A to **13**B are schematic diagrams illustrating various bispecific antibody formats of Q-SBL1 (FIG. **13**A) and R-SBL1 (FIG. **13**B) depending on hinge region.

[0041] FIG. **14** is a schematic diagram illustrating various bispecific antibody formats of Q-SBL1 upon interchange of knob-CH3/hole-CH3 domains.

[0042] FIG. **15** is a schematic diagram illustrating the bispecific antibody format of each Q-SBL9.

[0043] FIG. **16** illustrates an exemplary trivalent bispecific antibody format.

[0044] FIG. **17** illustrates an exemplary tetravalent bispecific antibody format.

[0045] FIG. **18** illustrates non-reducing SDS-PAGE of bispecific antibody candidates obtained by transient production of ExpiCHO-S culture supernatants, wherein (A) is SDS-PAGE gel images of Q-SBL-1,2,3,4 and R-SBL-1,2,3,4 (S: size marker (kDa), Lane 1: Q-SBL-2, Lane 2: Q-SBL-1, Lane 3: Q-SBL-3, Lane 4: Q-SBL-4, Lane 5: R-SBL-2, Lane 6: R-SBL-1, Lane 7: R-SBL-3, Lane 8: R-SBL-4), (B) is an SDS-PAGE gel image of Q-SBL-1, Q-SBL-5, R-SBL-1, R-SBL-5, and R-SBL-6 (S: size marker (kDa), Lane 1: Q-SBL-1, Lane 2: Q-SBL-5, Lane 3: R-SBL-1, Lane 4: R-SBL-5, Lane 5: R-SBL-6), (C) is SDS-PAGE gel images of Q-SBL-1, Q-SBL-6, Q-SBL-7, Q-SBL-8 (S: size marker (kDa), Lane 1: Q-SBL-1, Lane 2: Q-SBL-6, Lane 3: Q-SBL-7, Lane 4: Q-SBL-8), and (D) is an SDS-PAGE gel image of Q-SBL9 (S: size marker (kDa), Lane 1: Q-SBL9).

[0046] FIG. **19** illustrates the result of non-reducing SDS-PAGE analysis of the bispecific antibody product obtained by three-step purification.

[0047] FIGS. **20**A to **20**D illustrate the SEC-HPLC profile of the bispecific antibody product obtained in each purification step (FIG. **20**A: Q-SBL1, 2, 3, and 4, FIG. **20**B: R-SBL1, 2, 3, and 4, FIG. **20**C: Q-SBL5, R-SBL5 and 6, and FIG. **20**D: Q-SBL6, 7, 8 and 9).

[0048] FIG. **21** illustrates the result of CE-SDS analysis of antibodies obtained by the three-step purification under non-reduction conditions.

[0049] FIG. **22** illustrates differential scanning calorimetry (DSC) thermograms of Q-SBL2 (A) and Q-SBL9 (B) obtained by the three-step purification.

[0050] FIGS. **23**A and **23**B illustrates simultaneous dual binding of bispecific antibodies to VEGF and HER2 (FIG. **23**A, in (A) thereof): ♦: Q-SBL1, •: Q-SBL2, ▲: Q-SBL3, (FIG. **23**A, in (B) thereof) •: Q-SBL1, ■: R-SBL1, ▲: R-SBL2, ♦: R-SBL3, (FIG. **23**A, in (C) thereof) •: Q-SBL1, ■: Q-SBL5, ▲: R-SBL5, ♦: R-SBL6, (FIG. **23**A, in (D) thereof) •: Q-SBL1, ■: Q-SBL6, ♦: Q-SBL7, ▲: Q-SBL8, (FIG. **23**B) ♦: Q-SBL9.

[0051] FIG. **24** illustrates the inhibitory effect of bispecific antibodies on human umbilical vein endothelial cells (HUVEC) using Q-SBL9, wherein different concentrations of the bispecific antibody are incubated for 3 days and then treated with a CCK-8 solution and each sample was run in triplicate.

[0052] FIG. **25** illustrates the results of assay of binding of bispecific antibodies to the C1q protein; ■: Trastuzumab, •: Q-SBL2, ♦: Q-SBL9.

BEST MODE

[0053] Unless defined otherwise, all technical and scientific terms used herein have the same meanings as appreciated by those skilled in the field to which the present invention pertains. In general, the nomenclature used herein is well-known in the art and is ordinarily used.

[0054] In one aspect, the present invention is directed to a bispecific antibody including a first arm binding to a first antigen and including VH1-CHa-Fc1 and VL1-CLb, and a second arm binding to a second antigen and including VH2-CH1-Fc2 and VL2-CL,

[0055] wherein the VH1 and the VH2 are each heavy-chain variable regions including the same or different antigen-binding regions,

[0056] the VL1 and VL2 are each light-chain variable regions including the same or different antigen-binding regions,

[0057] the CHa includes i) an IgG heavy-chain constant region or an IgD heavy-chain constant region CH1, and an IgG heavy-chain constant region CH2 or CH3, or ii) an IgM heavy-chain constant region CH3,

[0058] the CLb includes i) a CL1 including an IgG light-chain constant region λ or κ, and at least one selected from the group consisting of IgG heavy-chain constant regions CH1, CH2, and CH3, or ii) an IgM heavy-chain constant region CH3,

[0059] the CH1 is an IgG heavy-chain constant region CH1 and the CL is an IgG light-chain constant region CL, and

[0060] the Fc1 of the first arm is linked to the Fc2 of the second arm to form a heavy-chain constant region dimer.

[0061] As used herein, the term "bispecific" refers to an ability of a binding protein to specifically bind to two different types of targets to control the activity of the targets.

For example, bispecificity can be obtained by conjugation of a monoclonal antibody or fragment thereof that specifically binds to each target, possesses two distinct antigen-binding arms (arm: specific for two targets) and has monovalence for each bound antigen.

[0062] VH1 and VH2 are each heavy-chain variable regions including the same or different antigen-binding regions. VL1 and VL2 are each light-chain variable regions including the same or different antigen-binding regions.

[0063] The term "polypeptide" refers to any polymeric chain of amino acids. The terms "peptide" and "protein" are used interchangeably with the term "polypeptide", which also refer to any polymeric chain of amino acids. The term "polypeptide" includes natural or synthetic proteins, protein fragments, and polypeptide analogs of protein sequences. Polypeptides may be monomeric or polymeric.

[0064] The term "specific binding" or "specifically binding" with respect to the interaction of an antibody, polypeptide, protein or peptide means that the interaction depends on the presence of a specific construct (e.g., antigenic determinant or epitope) of a chemical species. For example, antibodies generally recognize and bind to specific protein constructs rather than proteins. If an antibody is specific for epitope "A", the presence of a molecule containing epitope A (or released unlabeled A) in the reaction involving labeled "A" and the antibody reduces the amount of labeled A bound to the antibody.

[0065] An antibody refers to any immunoglobulin (Ig) molecule consisting of four polypeptide chains, two heavy (H) chains and two light (L) chains, or any functional fragment, mutant, variant or derivative having the indispensable epitope-binding property of such an Ig molecule. Specific examples of such a mutant, variant or derivative are described below, but are not limited thereto.

[0066] The term "monoclonal antibody" refers to an identical antibody, which is obtained from a population of substantially homogeneous antibodies, that is, each antibody constituting the population, excluding possible naturally occurring mutations that may be present in trivial amounts. Monoclonal antibodies are highly specific and are thus induced against a single antigenic site. Unlike conventional (polyclonal) antibody preparations that typically include different antibodies directed against different determinants (epitopes), each monoclonal antibody is directed against a single determinant on the antigen.

[0067] The term "epitope" refers to a protein determinant to which an antibody can specifically bind. Epitopes usually consist of a group of chemically active surface molecules, such as amino acid or sugar side chains, and generally have not only specific three-dimensional structural characteristics but also specific charge characteristics. Three-dimensional epitopes are distinguished from non-three-dimensional epitopes in that a bond to the former is broken in the presence of a denatured solvent, while a bond to the latter is not broken.

[0068] The non-human (e.g., murine) antibody of the "humanized" form is a chimeric antibody including minimal sequences derived from non-human immunoglobulins. In most cases, the humanized antibody is a human immunoglobulin (receptor antibody) in which a residue from the hypervariable region of a receptor is replaced with a residue from the hypervariable region of a non-human species (donor antibody) such as a mouse, rat, rabbit or non-human primate having the desired specificity, affinity and ability.

[0069] As used herein, the term "human antibody" refers to a molecule derived from human immunoglobulin, in which all of the amino acid sequences constituting the antibody including a complementarity-determining region and a structural region are composed of human immunoglobulins.

[0070] In a complete antibody, each heavy-chain consists of a heavy-chain variable region (HCVR or VH) and a heavy-chain constant region. The heavy-chain constant region is composed of three domains, CH1, CH2 and CH3. Each light-chain is composed of a light-chain variable region and a light-chain constant region. The light-chain constant region consists of one domain, CL. The VH and VL regions may be subdivided into hypervariable regions called "complementarity-determining regions" (CDRs) and further conserved regions called "framework regions" (FRs) are distributed in the CDRs. As used herein, the term "variable domain" refers to the light- and heavy-chain regions of an antibody molecule including the amino acid sequences of a complementarity-determining region (CDR; i.e., CDR1, CDR2, and CDR3) and a framework region (FR). VH refers to a variable domain of the heavy-chain. VL refers to a variable domain of the light-chain.

[0071] The term "complementarity-determining region" (CDR; i.e., CDR1, CDR2, and CDR3) refers to an amino acid residue of the antibody variable domain, which is necessary for antigen binding. Each variable domain typically has three CDR regions, identified as CDR1, CDR2, and CDR3.

[0072] The term "framework region" (FR) refers to a variable domain residue other than a CDR residue. Each variable domain typically has four FRs, identified as FR1, FR2, FR3, and FR4.

[0073] They are arranged from amino terminus to carboxy terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, and FR4. An immunoglobulin molecule may be of any type (e.g., IgG, IgE, IgM, IgD, IgA, or IgY), class (e.g., IgG1, IgG2, IgG3, IgG4, IgA1, or IgA2) or subclass.

[0074] A Fab fragment refers to a structure including a variable region of each of the heavy-chain and the light-chain, the constant region of the light-chain, and the first constant domain (CH1) of the heavy-chain, each having one antigen-binding site. Fab' is different from Fab in that it further includes a hinge region including at least one cysteine residue at the C-terminus of the CH1 domain of the heavy-chain. F(ab')2 is created by a disulfide bond between cysteine residues in the hinge region of Fab'. Fv is the minimal antibody fragment having only a heavy-chain variable region and a light-chain variable region. Two-chain Fv is a fragment in which the variable region of the heavy-chain and the variable region of the light-chain are linked by a non-covalent bond, and single-chain Fv (scFv) is a fragment in which the variable region of the heavy-chain and the variable region of the light-chain are generally linked by a covalent bond via a peptide linker therebetween, or are directly linked at the C-terminal, forming a dimer-shaped structure, like the two-chain Fv. Such antibody fragments may be obtained using proteases (e.g., Fab can be obtained by restriction-cleaving the complete antibody with papain, and the F(ab')2 fragment may be obtained by restriction-cleaving the complete antibody with pepsin), and may be produced using genetic recombination techniques.

[0075] The antibody according to the present invention includes "antigen-binding portion", which includes at least one antibody fragment having the ability to specifically bind to antigens and may specifically bind to other antigens, and thus may be bispecific or multispecific. In the present invention, the antigen-binding portion is included in the heavy-chain variable region of VH1 or VH2 and the light-chain variable region of VL1 or VL2, and each includes the same or different antigen-binding regions.

[0076] Two different variable regions can bind to two antigens. In order to minimize mispairing of the light-chain, an IgG CH3 dimer may be further introduced into either the C-terminal portion of the Fab or between the variable and constant regions of the Fab in one of the two Fab regions. By further introducing a CH3 dimer into one of the Fab regions, the two arms of the Fab region may be asymmetric to each other. Various types of CH1 domains may be introduced into the CH1/CL portion in one of Fab regions. As a result, it is possible to produce a bispecific or multispecific antibody having excellent productivity and stability.

[0077] According to the present invention, the antibody does not greatly deviate from the IgG-like structure. Structures that greatly deviate from the IgG structure are highly likely to make proteins unstable.

[0078] Moreover, a constant region of an immunoglobulin structure is further introduced into the mis-paired light-chain portion to greatly reduce the frequency of mispairing. Light-chain mispairing was prevented by introducing a known heavy-chain constant region domain into one of the two Fab arms.

[0079] The first bispecific antibody candidate has no structural modification in the left arm (second arm) in the two Fab parts that bind to the target and may have an Ig constant domain (CH3) further introduced into the right arm (first arm), may have IgG1 CH1 and various types of CH1 introduced instead of IgG1 CH1, and may have a knob-in-hole structure as a heterodimer structure of the heavy-chain.

[0080] In the second candidate bispecific antibody, the right Fab arm portion in the two Fab portions that bind to the target may be changed. The changed part may change the positions of the CH3 domain and the CH1 domain in the first bispecific antibody candidate.

[0081] In some cases, the third bispecific antibody candidate may have a configuration in which CH1/CL is substituted with CH3 of IgM.

[0082] The CHa includes i) an IgG heavy-chain constant region or an IgD heavy-chain constant region CH1, and an IgG heavy-chain constant region CH2 or CH3, or ii) an IgM heavy-chain constant region CH3.

[0083] The CHa may be derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM. Specifically, CHa may include at least one selected from the group consisting of heavy-chain constant regions CH1, CH2, and CH3 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM. For example, CHa may include heavy-chain constant region CH1 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM, and include heavy-chain constant region CH3 derived from IgG1, IgG2, IgG3, IgG4, IgD and IgM in order from N-terminus to C-terminus in the first arm. For example, CHa may include heavy-chain constant region CH3 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM, and include heavy-chain constant region CH1 derived from IgG1, IgG2, IgG3, IgG4, IgD and IgM in order from N-terminus to C-terminus in the first arm. In some cases, CHa may include an IgM-derived heavy-chain constant region CH3.

[0084] The CLb includes i) CL1 including an IgG light-chain constant region λ or κ, and at least one selected from the group consisting of IgG heavy-chain constant regions CH1, CH2, and CH3, or ii) an IgM heavy-chain constant region CH3.

[0085] The CLb may be derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM. Specifically, CLb may include CL1 including light-chain constant region λ or κ derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM, and at least one selected from the group consisting of heavy-chain constant regions CH1, CH2, and CH3 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM. For example, CLb may include CL1 including light-chain constant region λ or K derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM, and a heavy-chain constant region CH3 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM in order from N-terminus to C-terminus in the first arm. For example, CLb may include a heavy-chain constant region CH3 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM and CL1 including light-chain constant region λ or κ derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM in order from N-terminus to C-terminus in the first arm. In some cases, CLb may include a heavy-chain constant region CH3 derived from IgM.

[0086] The CH1 is an IgG heavy-chain constant region CH1 and the CL is an IgG light-chain constant region CL. Each of CH1 and CL may be derived from IgG1, IgG2, IgG3, IgG4 or IgD. The CH1 and CL may have a Fab fragment-like structure through a non-covalent interaction.

[0087] As used herein, the term "Fc region" is used to define the C-terminal region of an immunoglobulin heavy-chain that can be produced by papain digestion of an intact antibody. The Fc region may be a native sequence Fc region or a variant Fc region. The Fc region of an immunoglobulin generally includes two constant domains, a CH2 domain and a CH3 domain, and optionally includes a CH4 domain.

[0088] In one embodiment, each of CHa and CLb may include CH3. Specifically, each of CHa and CLb may include CH3 derived from IgG1, IgG2, IgG3, or IgG4. The CHa and CLb may form a dimer through covalent or non-covalent interaction.

[0089] The CHa and CLb may be linked through chains via or without a disulfide bond. Specifically, the dimer formed through CH3 included in each of CHa and CLb of the first arm, and the CH3 dimer in the Fab region including CH1 and CL1 included in CHa and CLb, respectively, are linked through a disulfide bond, and CH1 and CL1 may be linked without a disulfide bond.

[0090] In some cases, the CH3 domain and the CH1 domain may be linked through a linker. The linker may be a peptide linker and may include about 5-25 aa residues, or specifically about 5-10 aa residues. For example, the linker may include hydrophilic amino acids such as glycine and/or serine, but are not limited thereto.

[0091] Specifically, the linker, for example, includes a glycine linker (G, Gly)p (p is 1 to 10), or a GS linker $(G_nS)_m$ (n and m are each 1 to 10) in order to impart structural flexibility. Specifically, the linker may include GGGGS or (GGGGS)2, or (G, Gly)p 5-10 aa glycine wherein p is 5 to 10.

[0092] Fc_1 of the first arm and Fc_2 of the second arm each include CH2 and CH3 monomers of the heavy-chain constant region. The monomer means one domain among dimers formed through two constant domains CH2-CH3 having the same amino acid sequence of the heavy-chain constant region Fc. Fc_1 of the first arm and Fc_2 of the second arm are linked to each other to form a heavy-chain constant region dimer.

[0093] The dimer may include a homodimer formed by bonding between constant domains CH3 having the same amino acid sequence or a heterodimer formed by bonding between constant domains CH3 having different amino acid sequences.

[0094] Optionally, a disulfide bond may be present between CH3 of CHa and CH3 of CLb or a pair of CH3 of Fc, or an interchain linkage may be formed therebetween without a disulfide bond.

[0095] In one embodiment, the CH1 of CHa and CL1 of CLb may be linked via a disulfide bond or without a disulfide bond. Specifically, CH1 of CHa and CL1 of CLb may be linked without a disulfide bond.

[0096] The CHa and CLb each include CH3, and the CH3 of the CHa and the CH3 of the CLb are linked to form a dimer. The CHa and CLb each include CH3 and CH3 may form a dimer through a disulfide bond.

[0097] The CHa and CLb each include CH3, and the CH3 of the CHa and the CH3 of the CLb are linked to form a dimer. The CHa and CLb each include CH3, and CH3 may form a dimer through a disulfide bond.

[0098] In one embodiment, one of the dimers formed by bonding CH3 of CHa to CH3 of CLb includes at least one selected from the group consisting of Y349C, S354C, T366S, T366W, L368A, and Y407V, and at least one selected from the group consisting of S354C, Y349C, T366W, T366S, L368A and Y407V, and thus may have a knob-in-hole structure.

[0099] One of the dimers includes at least one selected from the group consisting of T366W, S354C, and Y349C, and the other includes at least one selected from the group consisting of S354C, Y349C, T366S, L368A, and Y407V, to form a knob-in-hole structure. Specifically, one of the dimers formed by bonding CH3 of CHa to CH3 of CLb includes S354C, T366S, L368A, and Y407V, and the other includes Y349C and T366W, to form a knob-in-hole structure.

[0100] Specifically, the CH1 of CHa and the CL1 of CLb are linked without a disulfide bond, i) any one of CH3 of CHa and CH3 of CLb includes at least one selected from the group consisting of Y349C, T366S, L368A and Y407V, and the other includes S354C and/or T366W, or ii) any one of CH3 of CHa and CH3 of CLb includes at least one selected from the group consisting of S354C, T366S, L368A and Y407V, and the other includes Y349C and/or T366W.

[0101] The CH1 of CHa and CL1 of CLb are linked without a disulfide bond, i) any one of CH3 of CHa and CH3 of CLb includes Y349C, T366S, L368A and Y407V, and the other includes S354C and T366W, or ii) one of CH3 of CHa and CH3 of CLb includes S354C, T366S, L368A, and Y407V, and the other includes Y349C and T366W.

[0102] In one embodiment, one of the CH3 dimers of the Fc includes at least one selected from the group consisting of Y349C, S354C, T366S, T366W, L368A and Y407V, and at least one selected from the group consisting of S354C, Y349C, T366W, T366S, L368A and Y407V, to form a knob-in-hole structure among dimers.

[0103] By forming a mutation in the Fc domain of IgG (immunoglobulin G), an asymmetric heterodimer is stably formed. In addition, the heterodimer structure of the heavy-chain may be a known knob-in-hole structure.

[0104] The knob-in-hole, which was first published by Genentech in a paper in 1997, is currently the most widely adopted structure by various developers. Although the knob-in-hole structure is introduced, 100% heterodimer is not formed in the knob-in-hole structure. Therefore, a high heterodimer ratio is obtained by establishing optimal transfection conditions to improve production yield.

[0105] By inducing mutations in the CH3 domains of two different Ig heavy-chains, a hole structure is created in the CH3 domain of one Ig heavy-chain and a knob structure is created in the CH3 domain of the other Ig heavy-chain, so that the two Ig heavy-chains form a heterodimer.

[0106] The knob-into-hole technology converts hydrophobic amino acid residues with large side chains into hydrophobic amino acids with small side chains for residues located in the hydrophobic core of the CH3 domain interaction site in one heavy-chain CH3 domain, to form a hole structure. In the other heavy-chain CH3 domain, hydrophobic amino acid residues with small side chains are substituted with hydrophobic amino acids with large side chains to form a knob structure. As a result, a heavy-chain constant region mutation pair introduced with two pairs of mutations is co-expressed to form a heterodimer heavy-chain constant region.

[0107] In one embodiment, to form the knob structure, CH3 of CHa and CLb or CH3 of Fc may include at least one selected from the group consisting of S354C, Y349C, T366W, T366S, L368A, and Y407V. For example, to form the hole structure, CH3 of CHa and CLb or CH3 of Fc may include at least one selected from the group consisting of Y349C, S354C, T366S, T366W, L368A, and Y407V.

[0108] In another embodiment, for example, CH3 of CHa and CLb or CH3 of Fc may include T366W to form the knob structure. In order to form the hole structure, for example, CH3 of CHa and CLb or CH3 of Fc may include at least one selected from the group consisting of T366S, L368A, and Y407V. Specifically, to form the knob structure, for example, Y349C and T366W may be included. In order to form the hole structure, for example, CH3 of CHa and CLb or CH3 of Fc may include at least one selected from the group consisting of S354C, T366S, L368A, and Y407V.

[0109] The VH/VL amino acid sequences of bevacizumab and trastuzumab may be used as two types of VH/VL amino acid sequences in various types of bispecific or multispecific antibodies. Amino acid numbering may be based on the IMGT numbering system (based on Eu index according to http://www.imgt.org/IMGTScientificChart/Numbering/Hu_IGHGn ber.html#refs).

[0110] The specific sequences used in each candidate are as follows.

TABLE 1

Configuration	Sequence	SEQ ID No.
Bevacizumab VH	EVQLVESGGGLVQPGGSLRLSCAASGYTFTNYGMNWVRQAPGKGLEWVG WINTYTGEPTYAADFKRRFTFSLDTSKSTAYLQMNSLRAEDTAVYYCAK YPHYYGSSHWYFDVWGQGTLVTVSS	1
Bevacizumab VL	DIQMTQSPSSLSASVGDRVTITCSASQDISNYLNWYQQKPGKAPKVLIY FTSSLHSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYSTVPWTF GQGTKVEIK	2
Trastuzumab VH	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVA RIYPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSR WGGDGFYAMDYWGQGTLVTVSS	3
Trastuzumab VL	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIY SASFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTF GQGTKVEIK	4

TABLE 2

Configuration	Sequence	SEQ ID No.
IgG1 CH1	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSG VHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKV	5
IgG1 CH2	APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYV DGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKAL PAPIEKTISKAK	6
IgG1 CH3	GQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPEN NYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQ KSLSLSPGK	7
IgG1 CH3-Knob	GQPREPQVYTLPPSREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQPEN NYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQ KSLSLSPGK	8
IgG1 CH3-Hole	GQPREPQVYTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPEN NYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQ KSLSLSPGK	9
IgG1 CH3-Knob (S354 C)	GQPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQPEN NYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQ KSLSLSPGK	10
IgG1 CH3-Hole (Y349C)	GQPREPQVCTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPEN NYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQ KSLSLSPGK	11
IgG1 CH3-Knob (Y349C)	GQPREPQVCTLPPSREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQPEN NYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQ KSLSLSPGK	12
IgG1 CH3-Hole (S354C)	GQPREPQVYTLPPCREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPEN NYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQ KSLSLSPGK	13
CL-Kappa	RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQS GNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNRGEC	14
CL-Kappa C214S	RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQS GNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNRGES	15

TABLE 2-continued

Configuration	Sequence	SEQ ID No.
CL-Lambda	GQPKANPTVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADGSPVKAGVETTKPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVEKTVAPTECS	16
CL-Lambda-C214S	GQPKANPTVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADGSPVKAGVETTKPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVEKTVAPTESS	17
IgG4 CH1	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTKTYTCNVDHKPSNTKVDKRV	18
IgG4 CH1-C131S	ASTKGPSVFPLAPSSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTKTYTCNVDHKPSNTKVDKRV	19
IgD CH1	APTKAPDVFPIISGCRHPKDNSPVVLACLITGYHPTSVTVTWYMGTQSQPQRTFPEIQRRDSYYMTSSQLSTPLQQWRQGEYKCVVQHTASKSKKEIF	20
IgG2 CH1	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSNFGTQTYTCNVDHKPSNTKVDKTV	66
IgM CH1	GSASAPTLFPLVSCENSPSDTSSVAVGCLAQDFLPDSITFSWKYKNNSDISSTRGFPSVLRGGKYAATSQVLLPSKDVMQGTDEHVVCKVQHPNGNKEKNVPLPV	67
IgM CH3	TAIRVFAIPPSFASIFLTKSTKLTCLVTDLTTYDSVTISWTRQNGEAVKTHTNISESHPNATFSAVGEASICEDDWNSGERFTCTVTHTDLPSPLKQTISRPKG	68

TABLE 3

Configuration	Sequence	SEQ ID No.
IgG1 Hinge-wild type	EPKSCDKTHTCPPCP	21
IgG1 Hinge-del EPKSC	DKTHTCPPCP	22
IgG1 Hinge-EPKSS	EPKSSDKTHTCPPCP	23
IgG4 Hinge	ESKYGPPCPPCP	24
Elbow sequence-1	AS	25
Elbow sequence-2	RT	26
Linker-1	GGGGS	27
Linker-2	GGGGSGGGGS	28
Linker-3	GGGGSGGGGSGGGGS	29
Linker-4	GGGGSGGGGSGGGGSGGGGS	30

[0111] As can be seen from FIG. **10**, [A] has a knob structure in the CH3 domain of the heavy-chain region that binds to one (second arm on the left) epitope, and has S354C and T366W mutations to introduce a disulfide bond. In some cases, disulfide bonds may not be introduced.

[0112] In the heavy-chain region (first arm on the right) binding to another epitope, the CH3 domain linked to the C-terminus of the CH1 domain has a hole structure and has T366S, L368A, and Y407V mutations. In some cases, a disulfide bond may be introduced. At this time, the CH3 domain has the Y349C mutation. The CH3 domain bonded to the C-terminus of the CH2 domain of the heavy-chain region has a hole structure and has Y349C, T366S, L368A, and Y407V mutations through a disulfide bond. In some cases, disulfide bonds may not be introduced. The CH3 domain linked to the C-terminus of the CL domain of the light-chain region has a knob structure and has a T366W mutation. In some cases, a disulfide bond may be introduced. At this time, the CH3 domain has the S354C mutation. Various types of CH1 domains may be introduced into the heavy-chain region CH1 domain.

[0113] E216, P217, K218, S219, and C220 may be added to the C-terminus of the IgG1 CH1 domain of the heavy-chain region (first arm on the right) that binds to another epitope. This aims at forming a disulfide bond with the CL domain. The CH3 domain sequence is based on the Uniprot site (https://www.uniprot.org/uniprot/P01857). The IgD CH1 domain sequence corresponds to the amino acid sequence from amino acids 1 to 98 in https://www.uniprot.org/uniprot/P01880. The IgM CH1 domain sequence corresponds to the amino acid sequence from amino acids 1 to 105 in https://www.uniprot.org/uniprot/P01871.

[0114] [H] has a knob structure in the CH3 domain of the heavy-chain region that binds to one (second arm on the left) epitope, and has S354C and T366W mutations to introduce a disulfide bond.

[0115] In some cases, a disulfide bond may not be introduced and S354 may be maintained.

[0116] The CH3 domain linked to the C-terminus of the VH domain in the heavy-chain region (first arm on the right) that binds to another epitope has a hole structure and has T366S, L368A, and Y407V mutations. In some cases, a disulfide bond may be introduced. The CH3 domain has the Y349C mutation. The CH3 domain linked to the C-terminus of the CH2 domain of the heavy-chain region has a hole structure and has Y349C, T366S, L368A, and Y407V mutations to introduce a disulfide bond. In some cases, a disulfide bond may not be introduced and Y349 may be maintained. The CH3 domain linked to the C-terminus of the VL domain of the light-chain region has a knob and has a T366W mutation. In some cases, a disulfide bond may be introduced. The CH3 domain has a S354C mutation. Various types of CH1 domains may be introduced into the heavy-chain region CH1 domain.

[0117] Elbow sequences A118 and S119 are added to the C-terminus of the VH domain of the heavy-chain region (first arm on the right) that binds to another epitope and various types of CH3 domains are bonded thereto next. Here, the sequence of the CH3 domain includes from P343 to K447, without the elbow sequences G341 and Q342, from the sequence shown in https://www.uniprot.org/uniprot/P01857. Elbow sequences R108 and T109 are added to the C-terminus of the VL domain of the light-chain region and various types of CH3 domains are bonded thereto next.

[0118] [N] has a knob structure in the CH3 domain of the heavy-chain region that binds to one (second arm on the left) epitope, and has S354C and T366W mutations to introduce a disulfide bond.

[0119] Elbow sequences A118 and S119 are added to the C-terminus of the VH domain of the heavy-chain region (first arm on the right) that binds to another epitope and an IgM CH3 domain is bonded thereto next. Elbow sequences R108 and T109 are added to the C-terminus of the VL domain of the light-chain region and an IgM CH3 domain is bonded thereto next. The IgM CH3 domain sequence corresponds to the amino acid sequence from amino acids 220 to 323 in https://www.uniprot.org/uniprot/P01871.

[0120] [S, U] has a knob structure in the CH3 domain of the heavy-chain region that binds to one (second arm on the left) epitope, and has S354C and T366W mutations to introduce a disulfide bond. Elbow sequences A118 and S119 are added to the C-terminus of the VH domain of the heavy-chain region (first arm on the right) that binds to another epitope and a GGGGS linker is bonded thereto next. In addition, various types of CH3 domains are bonded thereto. The GGGGS linker includes one to five GGGGS repeat sequences with various lengths.

[0121] The specific sequence for each candidate is as follows.

TABLE 4

	Heavy Chain-1	Light Chain-1	Heavy Chain-2	Light Chain-2
[A]	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?		?	?
	?		?	?
	?		?	?
	?		?	?
	?		?	?
	?		?	
	?		?	
	?		?	

TABLE 4-continued

	Heavy Chain-1	Light Chain-1	Heavy Chain-2	Light Chain-2
	?		?	
	?			
	?			

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TABLE 5

[B]	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
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	?		?	?
	?		?	?
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TABLE 6

[C]	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
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	?		?	?
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TABLE 7

[D]	?	?	?	?
	?	?	?	?
	?	?	?	?
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	?	?	?	?
	?	?	?	?

TABLE 7-continued

	?	?	?	?
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	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
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TABLE 8

[E]	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
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	?		?	?
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	?		?	
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TABLE 9

[F]	?	?	?	?
	?	?	?	?

TABLE 9-continued

	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
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	?	?	?	?
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TABLE 10

[G]	?	?	?	?
	?	?	?	?
	?	?	?	?
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TABLE 11

[H]	?	?	?	?
	?	?	?	?
	?	?	?	?
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	?		?	?
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	?		?	

TABLE 11-continued

	(?)		(?)	
	(?)		(?)	
	(?)			
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TABLE 12

[I]	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
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TABLE 13

[J]	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
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TABLE 14

[K]	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
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	(?)	(?)	(?)	(?)

TABLE 14-continued

	?	?	?	?
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TABLE 15

[L]	?	?	?	?
	?	?	?	?
	?	?	?	?
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TABLE 16

[M]	?	?	?	?
	?	?	?	?
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TABLE 17

[N]	?	?	?	?
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TABLE 18

[O]	?	?	?	?
	?	?	?	?
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TABLE 19

[P]	?	?	?	?
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TABLE 19-continued

	(?)	(?)
	(?)	(?)
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TABLE 20

[Q]	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
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TABLE 21

[R]	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
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TABLE 22

[S]	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)
	(?)	(?)	(?)	(?)

TABLE 22-continued

	?	?	?	?
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	?	?	?	?
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	?			
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TABLE 23

[T]	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
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	?			

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TABLE 24

[U]	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
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TABLE 24

[V]	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
	?	?	?	?
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[0122] In another aspect, the present invention is directed to a bispecific antibody including a first arm binding to a first antigen and including VH1-CHa-Fc1 and VL1-CLb, and a second arm binding to a second antigen and including VH2-CH1-Fc2 and VL2-CL,

- **[0123]** wherein the VH1 and the VH2 are each heavy-chain variable regions including the same or different antigen-binding regions,
- **[0124]** the VL1 and VL2 are each light-chain variable regions including the same or different antigen-binding regions,
- **[0125]** the CHa includes an IgG heavy-chain constant region CH3 and an IgG heavy-chain constant region CH1,
- **[0126]** the CLb includes a CL1 including an IgG light-chain constant region λ or κ, and an IgG heavy-chain constant region CH3,
- **[0127]** the CH1 is an IgG heavy-chain constant region CH1 and the CL is an IgG light-chain constant region CL, and
- **[0128]** the Fc1 of the first arm is linked to the Fc2 of the second arm to form a heavy-chain constant region dimer.

[0129] The IgG heavy-chain constant region CH3 may include heavy-chain constant region CH3 derived from IgG1, IgG2, IgG3 or Ig4.

[0130] The IgG heavy-chain constant region CH1 may include heavy-chain constant region CH1 derived from IgG1, IgG2, IgG3 or IgG4. Specifically, the IgG heavy-chain constant region CH1 may include heavy-chain constant region CH1 derived from IgG1 or IgG4.

[0131] The CHa may include an IgG heavy-chain constant region CH1 and an IgG heavy-chain constant region CH3 in order from the N-terminus to the C-terminus of the first arm. The CHa may include an IgG heavy-chain constant region CH3 and an IgG heavy-chain constant region CH1 in the order from the N-terminus to the C-terminus of the first arm.

[0132] The CLb may include CL1 including an IgG light-chain constant region λ or κ, and heavy-chain constant region CH3 derived from IgG in order from the N-terminus to the C-terminus of the first arm. For example, CLb may include heavy-chain constant region CH3 derived from IgG and CL1 including light-chain constant region λ or κ derived from IgG in the order from N-terminus to C-terminus in the first arm.

[0133] In one embodiment, the CH1 of CHa and CL1 of CLb may be linked via a disulfide bond or without a disulfide bond. Specifically, CH1 of CHa and CL1 of CLb may be linked without a disulfide bond.

[0134] The CHa and CLb each include CH3 and the CH3 of the CHa and the CH3 of the CLb are linked to form a dimer. The CHa and CLb each include CH3 and CH3 may form a dimer through a disulfide bond.

[0135] Specifically, the dimer formed through CH3 included in each of CHa and CLb of the first arm, and the CH3 dimer in the Fab region including CH1 and CL1 included in each of CHa and CLb are linked through a disulfide bond, and CH1 and CL1 are linked without a disulfide bond.

[0136] One of the dimers includes at least one selected from the group consisting of T366W, S354C, and Y349C, and the other includes at least one selected from the group consisting of S354C, Y349C, T366S, L368A, and Y407V, to form a knob-in-hole structure. Specifically, one of the dimers formed by bonding CH3 of CHa to CH3 of CLb includes S354C, T366S, L368A, and Y407V, and the other includes Y349C and T366W, to form a knob-in-hole structure.

[0137] Specifically, the CH1 of CHa and the CL1 of CLb are linked without a disulfide bond, i) any one of CH3 of CHa and CH3 of CLb includes at least one selected from the group consisting of Y349C, T366S, L368A and Y407V, and the other includes S354C and/or T366W, or ii) any one of CH3 of CHa and CH3 of CLb includes at least one selected from the group consisting of S354C, T366S, L368A and Y407V, and the other includes Y349C and/or T366W.

[0138] The CH1 of CHa and CL1 of CLb are linked without a disulfide bond, i) any one of CH3 of CHa and CH3 of CLb includes Y349C, T366S, L368A and Y407V, and the other includes S354C and T366W, or ii) any one of CH3 of CHa and CH3 of CLb includes S354C, T366S, L368A, and Y407V, and the other includes Y349C and T366W.

[0139] CH3 of CHa or CH3 of CLb may include the sequence of SEQ ID NOs: 8 to 13. CH3 of CHa or CH3 of CLb may include a mutation of knob (T366W) (SEQ ID NO: 8) or hole (T366S/L368A/Y407V) (SEQ ID NO: 9) of the IgG1 CH3 domain. The IgG1 CH3 domain may include a mutation, namely, a knob (S354C/T366W) (SEQ ID NO: 10), a hole (Y349C/T366S/L368A/Y407V) (SEQ ID NO: 11), knob (Y349C/T366W) (SEQ ID NO: 12), or hole (S354C/T366S/L368A/Y407V) (SEQ ID NO: 13) to generate a cysteine for a disulfide bond.

[0140] In some cases, CH3 and CH1 of CHa, and CH3 and CL1 of CLb may be linked through a linker. The linker may be a peptide linker and may include about 5-25 aa residues, or specifically about 5-10 aa residues. For example, the linker may include hydrophilic amino acids such as glycine and/or serine, but are not limited thereto.

[0141] Specifically, the linker, for example, includes a glycine linker (G, Gly)p (p is 1 to 10), or a GS linker $(G_nS)_m$ (n and m are each 1 to 10) in order to impart structural flexibility. Specifically, the linker may include GGGGS or (GGGGS)2, or (G, Gly)p 5-10 aa glycine wherein p is 5 to 10.

[0142] Among dimers formed by Fc_1 of the first arm and Fc_2 of the second arm, the CH3 dimer may include monomers that may be linked with or without a disulfide bond. In one embodiment, one of the CH3 dimers of the Fc includes at least one selected from the group consisting of Y349C, S354C, T366S, T366W, L368A and Y407V, and includes at least one selected from the group consisting of S354C, Y349C, T366W, T366S, L368A and Y407V, to form a knob-in-hole structure among dimers.

[0143] The first arm and the second arm may be linked through a hinge. The first arm and the second arm may be linked through a hinge including one or more sequences selected from the group consisting of:

[0144] DKTHTCPPCP;

[0145] EPKSSDKTHTCPPCP; and

[0146] ESKYGPPCPPCP.

[0147] The configuration of the bispecific antibody format that simultaneously binds to the first antigen and the second antigen is as follows. The bispecific antibody format is formed by combination of a total of four polypeptides, two heavy (H) chains and two light chains. The configuration of the heavy chain and the light chain of the first arm that binds to the first antigen is as follows. The heavy chain includes VH-CH3a-CH1a-Hinge-CH2-CH3b or VH-CH1a-CH3a-Hinge-CH2-CH3b. Here, CH3a may include a sequence of an IgG1 CH3 domain, and CH1a may include a sequence of an IgG1 CH1 domain (SEQ ID NO: 5) or an IgG4 CH1 domain (SEQ ID NOS: 18 and 19) or an IgD CH1 domain (SEQ ID NO: 20). CH3a or CH3b may include a mutation of a knob (T366W) (SEQ ID NO: 8) or a hole (T366S/L368A/Y407V) (SEQ ID NO: 9) of the IgG1 CH3 domain. The IgG1 CH3 domain may include a mutation, namely a knob (S354C/T366W) (SEQ ID NO: 10), hole (Y349C/T366S/L368A/Y407V) (SEQ ID NO: 11), knob (Y349C/T366W) (SEQ ID NO: 12), or hole (S354C/T366S/L368A/Y407V) (SEQ ID NO: 13) to generate a cysteine for a disulfide bond. The light chain of the first arm that binds to the first antigen is composed of VL-CH3c-CLb or VL-CLb-CH3c. Here, CH3c represents the IgG1 CH3 domain and includes a mutation of a knob (T366W) (SEQ ID NO: 8) or a hole (T366S/L368A/Y407V) (SEQ ID NO: 9). The IgG1 CH3 domain may include a mutation, namely a knob (S354C/T366W) (SEQ ID NO: 10) or a hole (Y349C/T366S/L368A/Y407V) (SEQ ID NO: 11) or a knob (Y349C/T366W) (SEQ ID NO: 12) or a hole (S354C/T366S/L368A/Y407V) (SEQ ID NO: 13) to generate a cysteine for a disulfide bond. CLb may be a kappa type (SEQ ID NOs: 14 and 15) or lambda type (SEQ ID NOs: 16 and 17).

[0148] The configuration of the heavy chain and the light chain of the second arm that binds to the second antigen is as follows. The configuration includes VH-CH1-Hinge-CH2-CH3d or VH-CH1-Hinge-CH2-CH3d. The IgG1 CH3d domain may include a mutation of a knob (T366W) (SEQ ID NO: 8) or a hole (T366S/L368A/Y407V) (SEQ ID NO: 9). The IgG1 CH3d domain may also include a mutation to generate cysteines for disulfide bonds, namely a knob (S354C/T366W) (SEQ ID NO: 10) or a hole (Y349C/T366S/L368A/Y407V) (SEQ ID NO: 11). The configuration of the light chain may include VL-CLb. CLb may be a kappa type (SEQ ID NO: 14) or a lambda type (SEQ ID NO: 16).

[0149] In the Q-SBL1 (SEQ ID NO: 31, 32, 62, 63) bispecific antibody format, the configuration of heavy- and light-chains of the first arm binding to the first antigen is as follows. The configuration of the heavy chain of the first arm includes VH-CH3a-Linker-CH1a-Hinge-CH2-CH3b. Here, the amino acid sequence of the linker was determined as GGGGSGGGGS (SEQ ID NO: 28). The CH1a domain may be an IgG1 CH1 domain (SEQ ID NO: 5). In order to remove the disulfide bond between the CH1a domain and CLb, the hinge region may have an amino acid sequence of DKTHTCPPCP (SEQ ID NO: 22). The CH3a domain includes a hole mutation and a mutation for forming a disulfide bond with CH3c of the light chain, and the mutation of the CH3a domain may be a hole (Y349C/T366S/L368A/Y407V) (SEQ ID NO: 11). The CH3b domain may include a hole mutation (T366S/L368A/Y407V) (SEQ ID NO: 9). An elbow sequence AS may be added between the VH region and the CH3a region (SEQ ID NO: 25).

[0150] The configuration of the first arm light chain may include VL-CH3c-Linker-CLb. The CH3c domain includes a knob mutation and a mutation for a disulfide bond with CH3a of the heavy chain, and the mutation of the CH3c domain may be a knob (S354C/T366W) (SEQ ID NO: 10). The amino acid sequence of the linker may be GGGGSGGGGS (SEQ ID NO: 28). The CLb domain may be a Kappa type and may include a C216S (Eu numbering) mutation to remove a disulfide bond with the CH1 domain of the first arm heavy chain (SEQ ID NO: 17). When the CLb domain is a lambda type, it may include a C214S (Eu numbering) mutation in order to remove the disulfide bond with the CH1 domain of the first arm heavy chain. This means that the disulfide bond to form the CH1a/CLb dimer is completely removed. An elbow sequence RT was added between the VL region and the CH3a region (SEQ ID NO: 26).

[0151] The configuration of the heavy chain and the light chain of the second arm that binds to the second antigen is as follows. The configuration of the heavy chain of the second arm may include VH-CH1-CH2-CH3d and the CH3d domain may include a knob (T366W) (SEQ ID NO: 8) mutation. The configuration of the light chain of the second arm is VL-CLb. CLb may be a kappa type (SEQ ID NO: 14) or a lambda type (SEQ ID NO: 16)

		VH	Elbow sequence	CH3a	linker	CH1	Hinge	CH2	CH3b
Q-SBL1	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/ T366S/ L368A/ Y407V)	GGGGS GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 T366S/ L368A/ Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366W)
		VL	Elbow sequence	CH3c	linker	CLb			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/ T366W)	GGGGS GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	—	—	—	kappa CL			

[0152] The R-SBL1 (SEQ ID NOS: 31, 32, 62, and 63) bispecific antibody format is different from Q-SBL1 in that it has a configuration in which only the CH1a domain portion in the heavy-chain region of the first arm that binds to the first antigen and may include a C131S mutation to remove the disulfide bond with CLb in the light-chain region of the first arm (SEQ ID NO: 19).

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
R-SBL1	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/ T366S/ L368A/ Y407V)	GGGGS GGGGS	IgG4 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/ L368A/ Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPP CP	IgG1 CH2	IgG1 CH3 (T366W)
		VL	Elbow sequence	CH3	linker	CL			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/ T366W)	GGGGS GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	—	—	—	kappa CL			

[0153] In another aspect, the present invention is directed to a multispecific antibody including the bispecific antibody described above.

[0154] As used herein, the term "multispecific" refers to an ability of a binding protein to specifically bind three different types of targets to control the activity of the targets. For example, multispecificity can be obtained by conjugation of a monoclonal antibody or fragment thereof that specifically binds to each target, possesses three or more distinct antigen-binding arms and has monovalence for each bound antigen.

[0155] For example, one or more antibody fragments that bind to additional antigens may be further included at the N-terminus of the first arm or the second arm or at the Fc_1 terminus of the first arm or the Fc_2 terminus of the second arm. An antigen-binding fragment that binds to an additional antigen may be further included at the N-terminus of the first arm or the second arm, or an antigen-binding fragment that binds to an additional antigen may be further included at the Fc_1 or Fc_2 terminus of the first arm. As a result, it is possible to construct multispecific antibodies targeting three or more antigens.

[0156] The antibody fragment includes a portion of an intact antibody, or an antigen-binding or variable region of an intact antibody. For example, the antibody fragment may include a Fab, Fab', F(ab')2, Fv, scFv, or diabody.

[0157] In one embodiment, an antibody fragment including VH3-CHa and VL3-CLb binding to a third antigen may be included at the N-terminus of the first arm. In some cases, an antibody fragment in the form of scFv that binds to a third antigen may be included at the Fc terminus (FIG. **7**).

[0158] In another embodiment, an antibody fragment including VH3-CHa and VL3-CLb that bind to a third antigen is included at the N-terminus of the first arm and an scFv form antibody fragment that binds to a fourth antigen may be included at the Fc terminus. In some cases, an antibody fragment in the form of an scFv that binds to a third antigen and a fourth antigen may be included at the Fc terminus (FIG. **8**).

[0159] The bispecific target form may be expanded to a tri-valent multispecific antibody and a tetra-valent multispecific antibody. In the form of a representative bispecific target antibody, the tri-valent multispecific antibody may be constructed by linking the scFv or Fab form targeting a third antigen to the N-terminus or C-terminus of the heavy-chain region of the first arm or the heavy-chain region of the second arm through a linker (FIG. **16**). In addition, the tetra-valent multispecific antibody may be constructed by further linking a scFv or Fab form targeting a fourth antigen thereto through a linker (FIG. **17**).

[0160] As used herein, the term "mutation (variation)" may mean a mutation (variation), for example, substitution, addition and/or deletion, of an amino acid sequence constituting a heavy-chain variable region and/or light-chain variable region and may include any mutation that does not impair antigen binding and efficacy, without limitation. Introduction of mutations into the binding protein according to the present invention may be applied, for example, to an external variable region or an internal variable region, or both an external variable region and an internal variable region.

[0161] When taking into consideration mutations having biologically equivalent activity, the polypeptides, binding proteins or nucleic acid molecule encoding the same according to the present invention is interpreted to include a sequence having substantial identity with the sequence set forth in the sequence number. The term "substantial identity" means that a sequence has a homology of at least 61%, preferably a homology of at least 70%, more preferably at least 80%, and most preferably at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, and 99% when aligning the sequence of the present invention and any other sequence so as to correspond to each other as much as possible and analyzing the aligned sequence using algorithms commonly used in the art. Alignment methods for sequence comparison are well-known in the art. The NCBI Basic Local Alignment Search Tool (BLAST) is accessible through NCBI or the like, and can be used in conjunction with sequence analysis programs such as BLASTP, BLAS™, BLASTX, TBLASTN and TBLASTX over the Internet. BLAST is available at www.ncbi.nlm.nih.gov/BLAST/. A method of comparing sequence homology using this program can be found at www.ncbi.nlm.nih.gov/BLAST/blast_help.html.

[0162] Based on this, the sequence according to the present invention may have a homology of 60%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more compared to the sequence disclosed herein or the entirety thereof. Homology can be determined through sequence comparison and/or alignment by methods known in the art. For example, the percentage sequence homology of the nucleic acid or protein according to the present invention can be determined using a sequence comparison algorithm (i.e., BLAST or BLAST 2.0), manual alignment, or visual inspection.

[0163] In another aspect, the present invention is directed to a nucleic acid encoding the bispecific antibody.

[0164] The first arm and/or the second arm of the bispecific antibody may be produced in a recombinant manner. The nucleic acid is isolated and inserted into a replicable vector, followed by further cloning (amplification of DNA) or further expression. Based on this, in another aspect, the present invention is directed to a vector including the nucleic acid.

[0165] The term "nucleic acid" is intended to encompass both DNA (gDNA and cDNA) and RNA molecules, and a nucleotide, which is the basic constituent unit of nucleic acids, includes naturally derived nucleotides as well as analogues thereof, in which sugar or base moieties are modified. The sequence of the nucleic acid encoding heavy- and light-chain variable regions of the present invention can vary. Such variation includes addition, deletion, or non-conservative or conservative substitution of nucleotides.

[0166] DNA encoding the first arm and/or second arm of the binding protein may be easily separated or synthesized using conventional procedures (for example, using an oligonucleotide probe capable of specifically binding to DNA) A variety of vectors are obtainable. Vector components generally include, but are not limited to, one or more of the following components: signal sequences, replication origins, one or more marker genes, enhancer elements, promoters, and transcription termination sequences.

[0167] As used herein, the term "vector" refers to a means for expressing target genes in host cells, and includes plasmid vectors, cosmid vectors, and viral vectors such as bacteriophage vectors, adenovirus vectors, retroviral vectors, and adeno-associated viral vectors. The nucleic acid encoding the antibody in the vector is operably linked to a promoter.

[0168] The term "operably linked" means functional linkage between a nucleic acid expression regulation sequence (e.g., an array of the binding site of the promoter, signal sequence, or transcription regulator) and another nucleic acid sequence, and enables the regulation sequence to regulate transcription and/or translation of the other nucleic acid sequence.

[0169] When a prokaryotic cell is used as a host, it generally includes a potent promoter capable of conducting transcription (such as a tac promoter, lac promoter, lacUV5 promoter, lpp promoter, pLλ promoter, pRλ promoter, rac5 promoter, amp promoter, recA promoter, SP6 promoter, trp promoter, or T7 promoter), a ribosome-binding site for initiation of translation, and a transcription/translation termination sequence. In addition, for example, when a eukaryotic cell is used as a host, it includes a promoter derived from the genome of a mammalian cell (e.g., a metallothionein promoter, a β-actin promoter, a human hemoglobin promoter or a human muscle creatine promoter), or a promoter derived from a mammalian virus (e.g., an adenovirus late promoter, vaccinia virus 7.5K promoter, SV40 promoter, cytomegalovirus (CMV) promoter, HSV tk promoter, mouse mammary tumor virus (MMTV) promoter, HIV LTR promoter, Moloney virus promoter, Epstein-Barr virus (EBV) promoter, or Rous sarcoma virus (RSV) promoter), and generally has a polyadenylation sequence as a transcription termination sequence.

[0170] Optionally, the vector may be fused with another sequence in order to facilitate purification of the antibody expressed thereby. The sequence to be fused therewith may include, for example, glutathione S-transferase (Pharmacia, USA), maltose-binding protein (NEB, USA), FLAG (IBI, USA), 6× His (hexahistidine; Qiagen, USA) and the like.

[0171] The vector includes antibiotic resistance genes commonly used in the art as selectable markers, and examples thereof include genes conferring resistance to ampicillin, gentamycin, carbenicillin, chloramphenicol, streptomycin, kanamycin, geneticin, neomycin, and tetracycline.

[0172] In another aspect, the present invention is directed to a cell transformed with the above-mentioned vector. The cell used to produce the bispecific antibody of the present invention may be a prokaryote, yeast, or higher eukaryotic cell, but is not limited thereto.

[0173] Prokaryotic host cells such as *Escherichia coli*, strains of the genus *Bacillus*, such as *Bacillus subtilis* and *Bacillus thuringiensis, Streptomyces* spp., *Pseudomonas* spp. (for example, *Pseudomonas putida*), *Proteus mirabilis* and *Staphylococcus* spp. (for example, *Staphylococcus carnosus*) may be used.

[0174] Interest in animal cells is the greatest, and examples of useful host cell lines include, but are not limited to, COS-7, BHK, CHO, CHOK1, DXB-11, DG-44, CHO/-DHFR, CV1, COS-7, HEK293, BHK, TM4, VERO, HELA, MDCK, BRL 3A, W138, Hep G2, SK-Hep, MMT, TRI, MRC 5, FS4, 3T3, RIN, A549, PC12, K562, PER.C6, SP2/0, NS-0, U20S, and HT1080.

[0175] In another aspect, the present invention is directed to a method of producing a bispecific antibody including (a) culturing the cells, and (b) recovering a bispecific antibody from the cultured cells.

[0176] The cells may be cultured in various media. Any commercially available medium can be used as a culture medium without limitation. All other essential supplements well-known to those skilled in the art may be included in appropriate concentrations. Culture conditions such as temperature and pH are those that are conventionally used with the host cells selected for expression, which will be apparent to those skilled in the art.

[0177] The recovery of the bispecific antibody may be carried out, for example, by centrifugation or ultrafiltration to remove impurities and further purification of the resulting product using, for example, affinity chromatography. Other additional purification techniques such as anion or cation exchange chromatography, hydrophobic interaction chromatography and hydroxyapatite (HA) chromatography may be used.

[0178] Hereinafter, the present invention will be described in more detail with reference to examples. However, it will be obvious to those skilled in the art that these examples are provided only for illustration of the present invention and should not be construed as limiting the scope of the present invention. Accordingly, the substantial scope of the present invention will be defined by the appended claims and equivalents thereto.

Example 1. Anti-VEGF× Anti-HER2 Bispecific Antibody Design

[0179] The first antigen was determined as a HER2 protein, and the heavy-chain variable region amino acid sequence (SEQ ID NO: 3) and the light-chain variable region amino acid sequence (SEQ ID NO: 4) of trastuzumab that bind to HER2 were used for the VH or VL in a bispecific antibody format. The second antigen was determined as a VEGF-A target and the heavy-chain variable region amino acid sequence (SEQ ID NO: 1) and the light-chain variable region amino acid sequence (SEQ ID NO: 2) of bevacizumab were used for the VH or VL in the bispecific antibody format. Amino acid sequence information was obtained through https://go.drugbank.com/.

[0180] Various candidates were established to select the optimal bispecific antibody formats. A total of three engineering attempts were made using Q-SBL1 and R-SBL1 as basic format antibodies and diversity was given to the length of the linker between the CH3 dimer and CH1/CL in the Fab region of the first arm (FIGS. **12**A, and **12**B). i) GGGGS (SEQ ID NO: 27), ii) GGGGSGGGGS (SEQ ID NO: 28), iii) GGGGSGGGGSGGGGS (SEQ ID NO: 29), and iv) GGGGSGGGGSGGGGSGGGGS (SEQ ID NO: 30) were applied to the linker of Q-SBL1 (SEQ ID NOS: 31, 32, 62, 63) and R-SBL1 (SEQ ID NOS: 49, 50, 62, and 63). The candidates, to which the linker of GGGGS was applied, were Q-SBL2 (SEQ ID NOS: 33, 34, 62, and 63) and R-SBL2 (SEQ ID NOS: 51, 52, 62, and 63), the candidates to which the linker of GGGGSGGGGS was applied were Q-SBL1 (SEQ ID NOS: 33, 34, 62, and 63), and R-SBL1 (SEQ ID NOS: 49, 50, 62, and 63), and the candidates, to which the linker of GGGGSGGGGSGGGGS was applied, were Q-SBL3 (SEQ ID NOS: 35, 36, 62, and 63), and R-SBL3 (SEQ ID NOS: 53, 54, 62, and 63), and the candidates, to which the linker of GGGGSGGGGSGGGGSGGGGS was applied, were Q-SBL4 (SEQ ID NOS: 37, 38, 62, and 63) and R-SBL4 (SEQ ID NOS: 55, 56, 62, and 63).

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
Q-SBL1	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/ T366S/ L368A/ Y407V)	GGGGS GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/ L368A/ Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPP CP	IgG1 CH2	IgG1 CH3 (T366W)

-continued

		VL	Elbow sequence	CH3	linker	CL			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/ T366W)	GGGGS GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	—	—	—	kappa CL			
		VH	**Elbow sequence**	**CH3**	**linker**	**CH1**	**Hinge**	**CH2**	**CH3**
Q-SBL2	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/ T366S/ L368A/ Y407V)	GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/ L368A/ Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPP CP	IgG1 CH2	IgG1 CH3 (T366W)
		VL	**Elbow sequence**	**CH3**	**linker**	**CL**			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/ T366W)	GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	—	—	—	kappa CL			
		VH	**Elbow sequence**	**CH3**	**linker**	**CH1**	**Hinge**	**CH2**	**CH3**
Q-SBL3	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/ T366S/ L368A/ Y407V)	GGGGS GGGGS GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/ L368A/ Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPP CP	IgG1 CH2	IgG1 CH3 (T366W)
		VL	**Elbow sequence**	**CH3**	**linker**	**CL**			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/ T366W)	GGGGS GGGGS GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	—	—	—	kappa CL			

-continued

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
Q-SBL4	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/ T366S/ L368A/ Y407V)	GGGGS GGGGS GGGGS GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/ L368A/ Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPP CP	IgG1 CH2	IgG1 CH3 (T366W)

		VL	Elbow sequence	CH3	linker	CL
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/ T366W)	GGGGS GGGGS GGGGS GGGGS	kappa CL
	2nd arm light-chain sequence	VL	—	—	—	kappa CL

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
R-SBL1	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/ T366S/ L368A/ Y407V)	GGGGS GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/ L368A/ Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPP CP	IgG1 CH2	IgG1 CH3 (T366W)

		VL	Elbow sequence	CH3	linker	CL
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/ T366W)	GGGGS GGGGS	kappa CL
	2nd arm light-chain sequence	VL	—	—	—	kappa CL

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
R-SBL2	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/ T366S/ L368A/ Y407V)	GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/ L368A/ Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPP CP	IgG1 CH2	IgG1 CH3 (T366W)

-continued

		VL	Elbow sequence	CH3	linker	CL
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/T366W)	GGGGS	kappa CL
	2nd arm light-chain sequence	VL	—	—	—	kappa CL

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
R-SBL3	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/T366S/L368A/Y407V)	GGGGS GGGGS GGGGS	IgG4 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/L368A/Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366W)

		VL	Elbow sequence	CH3	linker	CL
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/T366W)	GGGGS GGGGS GGGGS	kappa CL
	2nd arm light-chain sequence	VL	—	—	—	kappa CL

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
R-SBL4	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/T366S/L368A/Y407V)	GGGGS GGGGS GGGGS GGGGS	IgG4 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/L368A/Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366W)

		VL	Elbow sequence	CH3	linker	CL
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/T366W)	GGGGS GGGGS GGGGS GGGGS	kappa CL
	2nd arm light-chain sequence	VL	—	—	—	kappa CL

[0181] Second, all candidate groups were derived under the same conditions by introducing an IgG1 hinge (EPKSSDKTHTCPPCP) (SEQ ID NO: 23) or an IgG4 hinge (ESKYGPPCPPCP) (SEQ ID NO: 24) into the hinge region amino acid sequence of the first arm in Q-SBL1 and R-SBL1, and the amino acid sequence of the hinge region of the second arm was EPKSCDKTHTCPPCP (SEQ ID NO: 21) (FIGS. **13**A and **13**B). When an IgG1 hinge (EPKSSDKTHTCPPCP) (SEQ ID NO: 23) was used as the hinge region of the heavy-chain region of the first arm in Q-SBL1, Q-SBL5 (SEQ ID NOS: 39, 40, 62, and 63) was obtained. When the IgG1 hinge (EPKSSDKTHTCPPCP)

(SEQ ID NO: 23) was used as the hinge region of the heavy-chain region of the first arm in R-SBL1, R-SBL5 (SEQ ID NOS: 57, 58, 62, 63) was obtained, and when an IgG4 hinge (ESKYGPPCPPCP) (SEQ ID NO: 24) was used as the hinge region of the heavy-chain region of one arm, R-SBL6 (SEQ ID NOS: 59, 60, 62, and 63) was obtained. Here, when the IgG1 hinge of the first arm was used, the C220S (Eu numbering) mutation was present (SEQ ID NO: 23). The mutation was formed in order to remove the disulfide bond with CLb.

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
Q-SBL1	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/ T366S/ L368A/ Y407V)	GGGGS GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/ L368A/ Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPP CP	IgG1 CH2	IgG1 CH3 (T366W)
		VL	Elbow sequence	CH3	linker	CL			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/ T366W)	GGGGS GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	—	—	—	kappa CL			

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
Q-SBL5	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/ T366S/ L368A/ Y407V)	GGGGS GGGGS	IgG1 CH1	EPKSSDKTHTCPPC P	IgG1 CH2	IgG1 CH3 (T366S/ L368A/ Y407V)
	2nd arm heavy-chain sequence	VH	—	—	—	IgG1 CH1	EPKSCDKTHTCPP CP	IgG1 CH2	IgG1 CH3 (T366W)
		VL	Elbow sequence	CH3	linker	CL			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/ T366W)	GGGGS GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	—	—	—	kappa CL			

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
R-SBL1	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/ T366S/ L368A/ Y407V)	GGGGS GGGGS	IgG4 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/ L368A/ Y407V)

-continued

	2nd arm heavy-chain sequence	VH	null	null	null	IgG1 CH1	EPKSCDKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366W)
		VL	Elbow sequence	CH3	linker	CL			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/T366W)	GGGGS GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	null	null	null	kappa CL			
		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
R-SBL5	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/T366S/L368A/Y407V)	GGGGS GGGGS	IgG4 CH1	EPKSSDKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/L368A/Y407V)
	2nd arm heavy-chain sequence	VH	null	null	null	IgG1 CH1	EPKSCDKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366W)
		VL	Elbow sequence	CH3	linker	CL			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/T366W)	GGGGS GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	null	null	null	kappa CL			
		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
R-SBL6	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/T366S/L368A/Y407V)	GGGGS GGGGS	IgG4 CH1	ESKYGPPCPPCP	IgG1 CH2	IgG1 CH3 (T366S/L368A/Y407V)
	2nd arm heavy-chain sequence	VH	null	null	null	IgG1 CH1	EPKSCDKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366W)
		VL	Elbow sequence	CH3	linker	CL			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/T366W)	GGGGS GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	null	null	null	kappa CL			

[0182] Third, a candidate group was derived by exchanging the positions of the CH3 domain mutated to the knob or hole included in the CH3 domain (FIG. **14**). More specifically, in Q-SBL1, the CH3 domain of the Fab region in the heavy-chain region of the first arm includes a hole (Y349C/T366S/L368A/Y407V) (SEQ ID NO: 11), and the CH3 domain of the Fc region includes a hole (T366S/L368A/Y407V) (SEQ ID NO: 9). The CH3 domain of the Fab region in the light-chain region of the first arm includes a knob (S354C/T366W) (SEQ ID NO: 10). In addition, the CH3 domain of the Fc region in the heavy-chain region of the second arm includes a knob (T366W) (SEQ ID NO: 8).

[0183] In Q-SBL6 (SEQ ID NOS: 41, 42, 62, and 63), the CH3 domain of the Fab region in the heavy-chain region of the first arm includes a knob (S354C/T366W) (SEQ ID NO: 10), and the CH3 domain of the Fc region includes a hole (T366S/L368A/Y407V) (SEQ ID NO: 9). The CH3 domain of the Fab region in the light-chain region of the first arm includes a hole (Y349C/T366S/L368A/Y407V) (SEQ ID NO: 11). In addition, the CH3 domain of the Fc region in the heavy-chain region of the second arm includes a knob (T366W) (SEQ ID NO: 8).

[0184] In Q-SBL7 (SEQ ID NOS: 43, 44, 62, and 64), the CH3 domain of the Fab region in the heavy-chain region of the first arm includes a hole (Y349C/T366S/L368A/Y407V) (SEQ ID NO: 11) and the CH3 domain of the Fc region includes a knob (T366W) (SEQ ID NO: 8). The CH3 domain of the Fab region in the light-chain region of the first arm includes a knob (S354C/T366W) (SEQ ID NO: 10). In addition, the CH3 domain of the Fc region in the heavy-chain region of the second arm includes a hole (Y349C/T366S/L368A/Y407V) (SEQ ID NO: 11). In Q-SBL8 (SEQ ID NOS: 45, 46, 62, 64), the CH3 domain of the Fab region in the heavy-chain region of the first arm includes a knob (S354C/T366W) (SEQ ID NO: 10), and the CH3 domain of the Fc region includes a knob (T366W) (SEQ ID NO: 8). The CH3 domain of the Fab region in the light-chain region of the first arm includes a hole (Y349C/T366S/L368A/Y407V) (SEQ ID NO: 11). In addition, the CH3 domain of the Fc region in the heavy-chain region of the second arm includes a hole (T366S/L368A/Y407V) (SEQ ID NO: 9).

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
Q-SBL1	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/T366S/L368A/Y407V)	GGGGS GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/L368A/Y407V)
	2nd arm heavy-chain sequence	VH	null	null	null	IgG1 CH1	EPKSCDKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366W)

		VL	Elbow sequence	CH3	linker	CL
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/T366W)	GGGGS GGGGS	kappa CL
	2nd arm light-chain sequence	VL	null	null	null	kappa CL

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
Q-SBL6	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (S354C/T366W)	GGGGS GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366S/L368A/Y407V)
	2nd arm heavy-chain sequence	VH	null	null	null	IgG1 CH1	EPKSCDKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366W)

		VL	Elbow sequence	CH3	linker	CL
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (Y349C/T366S/L368A/Y407V)	GGGGS GGGGS	kappa CL

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	2nd arm light-chain sequence	VL	null	null	null	kappa CL			
		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
Q-SBL7	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (Y349C/T366S/L368A/Y407V)	GGGGS GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366W)
	2nd arm heavy-chain sequence	VH	null	null	null	IgG1 CH1	EPKSCDKTHTCPP CP	IgG1 CH2	IgG1 CH3 (T366W/L368A/Y407V))
		VL	Elbow sequence	CH3	linker	CL			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (S354C/T366W)	GGGGS GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	null	null	null	kappa CL			
		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
Q-SBL8	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (S354C/T366W)	GGGGS GGGGS	IgG1 CH1	DKTHTCPPCP	IgG1 CH2	IgG1 CH3 (T366W)
	2nd arm heavy-chain sequence	VH	null	null	null	IgG1 CH1	EPKSCDKTHTCPP CP	IgG1 CH2	IgG1 CH3 (T366W)
		VL	Elbow sequence	CH3	linker	CL			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (Y349C/T366S/L368A/Y407V)	GGGGS GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	null	null	null	kappa CL			

[0185] Fourth, Q-SBL9 (SEQ ID NOS: 47, 48, 61, and 62) has disulfide bonds formed through point mutations in other amino acids of the knob and hole parts in the CH3 domain of the heavy-chain region of the first arm, and this is most prominent feature (FIG. **15**). More specifically, the Q-SBL9 bispecific antibody format has the following structure of the heavy chain and the light chain of the first arm that binds to the first antigen. The heavy chain of the first arm has the structure of VH-CH3a-Linker-CH1-Hinge-CH2-CH3b. Here, the amino acid sequence of the linker was determined as GGGGS (SEQ ID NO: 27). The CH1 region used herein was the IgG1 CH1 domain. The hinge region amino acid sequence to remove the disulfide bond with the CH1 domain was EPKSSDKTHTCPPCP (SEQ ID NO: 23). The CH3a domain includes a hole mutation and a mutation for a disulfide bond with CH3c of the light chain, and the mutation portion of the CH3a domain is the hole (S354C/T366S/L368A/Y407V) (SEQ ID NO: 13). The CH3b domain includes a mutation in the hole (Y349C/T366S/L368A/Y407V) (SEQ ID NO: 11). The CH3b domain of the first arm forms a disulfide bond with the CH3d domain of the second arm. An elbow sequence was added between the VH region and the CH3a region (SEQ ID NO: 25).

[0186] The light chain of the first arm has the configuration of VL-CH3c-Linker-CLb. The CH3c domain of the light chain includes a knob mutation and a mutation for a disulfide bond with CH3a of the heavy chain, and the mutated portion of the CH3c domain is the knob (Y349C/T366W) (SEQ ID NO: 12). The amino acid sequence of the linker is GGGGS (SEQ ID NO: 27). The CLb domain is a kappa type (SEQ ID NO: 15), and the CLb domain includes a C216S (Eu numbering) mutation to remove a disulfide bond with the CH1 domain of the first arm heavy chain. An elbow sequence was added between the VL region and the CH3a region (SEQ ID NO: 26).

[0187] The heavy chain and the light chain of the second arm that binds to the second antigen have the following configuration. The heavy chain of the second arm has VH-CH1-CH2-CH3d, and the CH3d domain includes a knob (S354C/T366W) (SEQ ID NO: 10) mutation. This part forms a disulfide bond with the CH3b domain of the heavy-chain region of the first arm. The light chain of the second arm has the structure of VL-CLb. CLb is a kappa type (SEQ ID NO: 14) or lambda type (SEQ ID NO: 16).

		VH	Elbow sequence	CH3	linker	CH1	Hinge	CH2	CH3
SBL9	1st arm heavy-chain sequence	VH	AS	IgG1 CH3 (S354C/T366S/L368A/Y407V)	GGGGS	IgG1 CH1	EPKSSDKTHTCPPCP	IgG1 CH2	IgG1 CH3 (Y349C/T366S/L368A/Y407V)
	2nd arm heavy-chain sequence	VH	null	null	null	IgG1 CH1	EPKSCDKTHTCPPCP	IgG1 CH2	IgG1 CH3 (S354C/T366W)
		VL	Elbow sequence	CH3	linker	CL			
	1st arm light-chain sequence	VL	RT	IgG1 CH3 (Y349C/T366W)	GGGGS	kappa CL			
	2nd arm light-chain sequence	VL	null	null	null	kappa CL			

[0188] The specific configuration of each candidate is as follows.

Configuration	Sequence	No.
Q-SBL1 HC (First arm heavy-chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARIYPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGDGFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK	31
Q-SBL1 LC (First arm light - chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSASFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGTKVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGES	32
Q-SBL2 HC (First arm heavy-chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARIYPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGDGFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGKGGGGSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVDKTHTCPPCPAPELLGGPSVFLFPPKP	33

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Configuration	Sequence	No.
	KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNS TYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKTTPPVLDS DGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK	
Q-SBL2 LC (First arm light - chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPRE AKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYAC EVTHQGLSSPVTKSFNRGES	34
Q-SBL3 HC (First arm heavy- chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSGGGGSASTKGPSVFPLAP SSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYS LSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVDKTHTCPPCPAPELLGG PSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAK TKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENN YKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSL SLSPGK	35
Q-SBL3 LC (First arm light - chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSGTASVV CLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKA DYEKHKVYACEVTHQGLSSPVTKSFNRGES	36
Q-SBL4 HC (First arm heavy- chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSGGGGSGGGGSASTKGPSV FPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQS SGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVDKTHTCPPCPAP ELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVE VHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNG QPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHY TQKSLSLSPGK	37
Q-SBL4 LC (First arm light - chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSGGGGSGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSG TASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTL TLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGES	38
Q-SBL5 HC (First arm heavy- chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSASTKGPSVFPLAPSSKST SGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVV TVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSSDKTHTCPPCPAPELLGG PSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAK TKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENN YKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSL SLSPGK	39
Q-SBL5 LC (First arm	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT	40

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Configuration	Sequence	No.
light - chain region)	QKSLSLSPGKGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSGTASVVCLLNN FYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKH KVYACEVTHQGLSSPVTKSFNRGES	
Q-SBL6 HC (First arm heavy-chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVYTLPPCREEMTKNQVSLWCLVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSASTKGPSVFPLAPSSKST SGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVV TVPSSSLGTQTYICNVNHKPSNTKVDKKVDKTHTCPPCPAPELLGGPSVFL FPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR EPQVYTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKTTP PVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPG K	41
Q-SBL6 LC (First arm light-chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVCTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSGTASVVCLLNN FYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKH KVYACEVTHQGLSSPVTKSFNRGES	42
Q-SBL7 HC (First arm heavy-chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSASTKGPSVFPLAPSSKST SGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVV TVPSSSLGTQTYICNVNHKPSNTKVDKKVDKTHTCPPCPAPELLGGPSVFL FPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR EPQVYTLPPSREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNYKTTP PVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPG K	43
Q-SBL 7 LC (First arm light - chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSGTASVVCLLNN FYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKH KVYACEVTHQGLSSPVTKSFNRGES	44
Q-SBL8 HC (First arm heavy-chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVYTLPPCREEMTKNQVSLWCLVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSASTKGPSVFPLAPSSKST SGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVV TVPSSSLGTQTYICNVNHKPSNTKVDKKVDKTHTCPPCPAPELLGGPSVFL FPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR EPQVYTLPPSREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNYKTTP PVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPG K	45
Q-SBL8 LC (First arm light - chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVCTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSGTASVVCLLNN FYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKH KVYACEVTHQGLSSPVTKSFNRGES	46
Q-SBL9 HC (First arm heavy-chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVYTLPPCREEMTKNQVSLSCAVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSASTKGPSVFPLAPSSKSTSGGTA ALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSS SLGTQTYICNVNHKPSNTKVDKKVEPKSSDKTHTCPPCPAPELLGGPSVFL	47

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Configuration	Sequence	No.
	FPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR EPQVCTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKTTP PVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPG K	
Q-SBL9 LC (First arm light - chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVCTLPPSREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPRE AKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYAC EVTHQGLSSPVTKSFNRGES	48
R-SBL1 HC (First arm heavy- chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSASTKGPSVFPLAPSSRST SESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVV TVPSSSLGTKTYTCNVDHKPSNTKVDKRVDKTHTCPPCPAPELLGGPSVFL FPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR EPQVYTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKTTP PVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPG K	49
R-SBL1 LC (First arm light - chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSGTASVVCLLNN FYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKH KVYACEVTHQGLSSPVTKSFNRGES	50
R-SBL2 HC (First arm heavy- chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVES CSVMHEALHNHYTQKSLSLSPGKGGGGSASTKGPSVFPLAPSSRSTSESTA ALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSS SLGTKTYTCNVDHKPSNTKVDKRVDKTHTCPPCPAPELLGGPSVFLFPPKP KDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNS TYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY TLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKTTPPVLDS DGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK*	51
R-SBL2 LC (First arm light- chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPRE AKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYAC EVTHQGLSSPVTKSFNRGES	52
R-SBL3 HC (First arm heavy- chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSGGGGSASTKGPSVFPLAP SSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYS LSSVVTVPSSSLGTKTYTCNVDHKPSNTKVDKRVDKTHTCPPCPAPELLGG PSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAK TKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENN YKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSL SLSPGK	53
R-SBL3 LC (First arm	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT	54

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Configuration	Sequence	No.
light - chain region)	QKSLSLSPGKGGGGSGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSGTASVV CLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKA DYEKHKVYACEVTHQGLSSPVTKSFNRGES	
R-SBL4 HC (First arm heavy- chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSGGGGSGGGGSASTKGPSV FPLAPSSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQS SGLYSLSSVVTVPSSSLGTKTYTCNVDHKPSNTKVDKRVDKTHTCPPCPAP ELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVE VHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNG QPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHY TQKSLSLSPGK	55
R-SBL4 LC (First arm light - chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSGGGGSGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSG TASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTL TLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGES	56
R-SBL5 HC (First arm heavy- chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSASTKGPSVFPLAPSSRST SESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVV TVPSSSLGTKTYTCNVDHKPSNTKVDKRVEPKSSDKTHTCPPCPAPELLGG PSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAK TKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENN YKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSL SLSPGK	57
R-SBL5 LC (First arm light - chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSGTASVVCLLNN FYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKH KVYACEVTHQGLSSPVTKSFNRGES	58
R-SBL6 HC (First arm heavy- chain region)	EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARI YPTNGYTRYADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGD GFYAMDYWGQGTLVTVSSASPREPQVCTLPPSREEMTKNQVSLSCAVKGFY PSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLVSKLTVDKSRWQQGNVFS CSVMHEALHNHYTQKSLSLSPGKGGGGSGGGGSASTKGPSVFPLAPSSRST SESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVV TVPSSSLGTKTYTCNVDHKPSNTKVDKRVESKYGPPCPPCPAPELLGGPSV FLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKP REEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQ PREPQVYTLPPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKT TPPVLDSDGSFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLS PGK	59
R-SBL6 LC (First arm light- chain region)	DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSA SFLYSGVPSRFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGT KVEIKRTPREPQVYTLPPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQ PENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYT QKSLSLSPGKGGGGSGGGGSRTVAAPSVFIFPPSDEQLKSGTASVVCLLNN FYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKH KVYACEVTHQGLSSPVTKSFNRGES	60
LC-1 HC (second arm heavy- chain region)	EVQLVESGGGLVQPGGSLRLSCAASGYTFTNYGMNWVRQAPGKGLEWVGWI NTYTGEPTYAADFKRRFTFSLDTSKSTAYLQMNSLRAEDTAVYYCAKYPHY YGSSHWYFDVWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTY ICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKD TLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL	61

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Configuration	Sequence	No.
	PPCREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDG SFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK	
LC-1 LC (second arm light - chain region)	DIQMTQSPSSLSASVGDRVTITCSASQDISNYLNWYQQKPGKAPKVLIYFT SSLHSGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQYSTVPWTFGQGT KVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA LQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSP VTKSFNRGEC	62
LC-3 HC (second arm heavy-chain region)	EVQLVESGGGLVQPGGSLRLSCAASGYTFTNYGMNWVRQAPGKGLEWVGWI NTYTGEPTYAADFKRRFTFSLDTSKSTAYLQMNSLRAEDTAVYYCAKYPHY YGSSHWYFDVWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTY ICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKD TLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL PPSREEMTKNQVSLWCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDG SFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK	63
LC-3-H-K HC (second arm heavy-chain region)	EVQLVESGGGLVQPGGSLRLSCAASGYTFTNYGMNWVRQAPGKGLEWVGWI NTYTGEPTYAADFKRRFTFSLDTSKSTAYLQMNSLRAEDTAVYYCAKYPHY YGSSHWYFDVWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVK DYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTY ICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKD TLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL PPSREEMTKNQVSLSCAVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDG SFFLVSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK	64

Example 2. Bispecific Antibody Production

[0189] Vector plasmids containing coding genes for the heavy and light-chain regions of the first arm and vector plasmids containing coding genes for the heavy and light-chain regions of the second arm were produced. Genes encoding two heavy and light chains were inserted into one vector plasmid. CMV was used as the promoter and WPRE (woodchuck hepatitis virus post-transcriptional regulatory element) was inserted after the coding gene to increase the expression level during transient expression.

[0190] ExpiCHO-S animal cells (Thermo Fisher A29127) were co-transfected with vector plasmids containing the coding genes of the heavy- and light-chain regions of the first arm and vector plasmids containing the coding genes of the heavy- and light-chain regions of the second arm. The co-transfection was performed using ExpiFectamine CHO Transfection Kit (Thermo Fisher A29130) according to the corresponding manual. Two weeks after co-transfection, the supernatant was obtained by centrifugation (10000×g, 15 mins) to obtain the bispecific antibody, and the supernatant was filtered (0.22 μm) to remove residual cell debris. FIG. **18** illustrates the expression pattern of the supernatant through SDS-PAGE under non-reducing conditions.

Example 3. Bispecific Antibody Expression

[0191] Octet quantitative assay based on a Protein A biosensor (Fortebio 18-5010) was used to assay bispecific antibodies. A calibration curve was obtained using a known IgG1 type sample as a standard material and the expression level of the sample was calculated through the calibration curve.

TABLE 26

Concentration of antibody in culture supernatant quantified by octet assay

Sample ID	Concentration (ug/ml)
Q-SBL1	471.5
Q-SBL2	415.2
Q-SBL3	354.9
Q-SBL4	343.8
R-SBL1	337.8
R-SBL2	338.0
R-SBL3	211.8
R-SBL4	223.4
Q-SBL5	493.1
R-SBL5	407.1
R-SBL6	351.4
Q-SBL6	272.7
Q-SBL7	438.8
Q-SBL8	284.6
Q-SBL9	322.0

Example 4. Bispecific Antibody Purification

[0192] A purified bispecific antibody having a purity of greater than 95% was isolated using AKTA avant 25/150 (Cytiva) as a protein purification system by protein A affinity chromatography, ion exchange chromatography (IEX), and hydrophobic interaction chromatography (HIC) or the like to perform various research and analysis of the bispecific antibody of the present invention. The bispecific antibody material obtained in each purification step was identified by SDS-PAGE using an 8% bis-Tris gel and MES buffer. The purity of the bispecific antibody was also measured using size exclusion high-performance liquid chromatography (SEC-HPLC) using a TSKgel G3000SWxl (Tosoh Bioscience) column.

[0193] The culture solution obtained by expressing the bispecific antibody candidates of the present invention was filtered using a 0.22 μm filter paper and then primarily purified using a MabSelect SuRe (Cytiva) column, which is a type of protein A affinity chromatography.

[0194] The primary purification was specifically performed as follows. The MabSelect Sure column was equilibrated with a 50 mM Tris (pH 7.0) buffer solution and then the filtered culture solution was loaded onto the column. Proteins not bound to the column were washed with the equilibration buffer solution for 5 cv. Then, impurities non-specifically bound to the MabSelect Sure column were removed with a 50 mM Tris (pH 7.0) buffer solution and a 20 mM Bis-Tris (pH 5.5) buffer solution containing 0.5 M sodium chloride. Then, the bispecific antibody specifically binding to the MabSelect Sure column was eluted using a 0.2 M glycine (pH 3.2) buffer solution for 4 cv. The eluted bispecific antibody sample was neutralized to pH 5.0 with a 1.0 M Tris buffer solution and then filtered through a 0.22 μm filter paper.

[0195] The next purification was performed using a Capto SP (Cytiva) column based on cation exchange chromatography in ion exchange chromatography, and the details are as follows. The Capto SP column was stabilized with a 50 mM sodium acetate (pH 5.0) buffer solution, a bispecific antibody sample neutralized to pH 5.0 was applied thereto, and impurities not bound to the column were removed with the same buffer solution. The bispecific antibody bound to the column was eluted with 0.1 M to 1.0 M sodium chloride.

[0196] In the final purification process, hydrophobic interaction chromatography was performed to remove high molecular weight (HMW) and low molecular weight (LMW) impurities from the secondarily purified bispecific antibody sample. In this process, a butyl-based Sepharose column was used and samples were prepared by substituting the bispecific antibody purified product obtained after ion exchange chromatography with a high-concentration salt buffer solution such that the salt concentration in the bispecific antibody was 1.0 M to 1.5 M. The column was equilibrated with a 50 mM sodium acetate (pH 5.0) buffer solution having the same salt concentration as the applied sample and then the prepared sample was loaded thereon. The bound bispecific antibody was eluted for 20 cv with a gradient method of 50 mM sodium acetate (pH 5.0) having no salt. The final purified bispecific antibody eluted with a purity of 95% or higher was concentrated to a concentration of 1 to 2 mg/mL using a 10 kDa molecular-weight cut-off ultrafiltration tube and then used in an appropriate buffer solution depending on analysis conditions.

[0197] FIG. **19** is a schematic diagram of an SDS-PAGE gel showing the protein content obtained by tertiary purification of the bispecific antibody candidates of the present invention using hydrophobic interaction chromatography.

[0198] FIGS. **20**A, **20**B, **20**C, and **20**D are chromatograms of the bispecific antibody candidates subjected to the final purification of the present invention analyzed using size exclusion high-performance liquid chromatography (SEC-HPLC).

TABLE 27

Analysis of purity of bispecific antibody after rPA purification and three-step purification by size exclusion high performance liquid chromatography (SEC-HPLC)

Sample ID	SEC analysis (After 1-step purification)	SEC analysis (After 3-step purification)
Q-SBL1	85	96
Q-SBL2	80	97
Q-SBL3	78	97
Q-SBL4	45	58
R-SBL1	86	95
R-SBL2	87	98
R-SBL3	67	92
R-SBL4	54	84
Q-SBL5	70	98
R-SBL5	84	98
R-SBL6	79	96
Q-SBL6	53	68
Q-SBL7	62	99
Q-SBL8	83	94
Q-SBL9	78	99

Example 5. Bispecific Antibody Characterization

5-1. CE-SDS Analysis

[0199] Maurice (Protein Simple) and Maurice CE-SDS PLUS Application Kit (Protein simple) were used for sodium dodecyl sulfate capillary electrophoresis (CE-SDS) to analyze the purity of bispecific antibody candidates under non-reducing or reducing conditions. In one analysis, 25 μL of protein was used at a maximum concentration of 2 mg/mL in the sample analysis after purification using protein A resin, and 25 μL of protein was used at a maximum protein concentration of 1 mg/mL in sample analysis after protein A resin/cation exchange chromatography (CEX)/hydrophobic binding chromatography (HIC) purification. For analysis, each sample was mixed with 2.5 μL of 250 mM iodoacetamide or 14.2 M 2-mercaptoethanol, 2 μL of an internal standard material and 25 μL of a sodium dodecyl sulfate sample buffer, followed by heating at 70° C. for 10 minutes. After completion of the analysis, the data was analyzed using Compass for iCE software version 2.2.0 provided by the manufacturer.

[0200] FIG. **21** shows data analyzed on CE-SDS under non-reducing conditions after 3-step (rPA+CEX+HIC) purification. After 3-step (rPA+CEX+HIC) purification of the bispecific antibody candidates, the purity was determined by sodium dodecyl capillary electrophoresis (CE-SDS). As a result, in the candidate group in which the linker length between the CH3 dimer and CH1/CL in the Fab region of the first arm was varied, Q-SBL4 or R-SBL4, a candidate, to which the linker of GGGGSGGGGSGGGGSGGGGS was applied, had the least purity. Second, the purity of all the candidates derived by introducing the IgG1 hinge (EPKSSDKTHTCPPCP) or the IgG4 hinge (ESKYGPPCPPCP) into the hinge region amino acid sequence of the first arm was not improved even after 3-step (rPA+CEX+HIC) purification. Finally, in the candidate group derived by exchanging the positions of the domains mutated to the knob or hole included in the CH3 domain, Q-SBL8 in which knob and hole domain positions were changed had the highest purity. Overall, Q-SBL2 and Q-SBL9 had high purity.

TABLE 28

Purity of bispecific antibodies after 1-step and 3-step purification under non-reducing conditions

Sample ID	CE-SDS (1-step purification, Non-reduced, %)	CE-SDS (3-step purification, Non-reduced, %)
Q-SBL1	57.4	86.7
Q-SBL2	65.0	88.9
Q-SBL3	53.0	88.3
Q-SBL4	45.3	61.2
R-SBL1	48.6	83.1
R-SBL2	68.4	85.6
R-SBL3	38.8	82.2
R-SBL4	38.1	79.1
Q-SBL5	58.6	67.1
R-SBL5	62.3	72.7
R-SBL6	55.6	69.8
Q-SBL6	47.8	49.6
Q-SBL7	57.8	59.0
Q-SBL8	61.2	85.2
Q-SBL9	76.2	96.9

5-2. Thermal Stability

[0201] The thermal stability of the bispecific antibody candidates was measured using differential scanning calorimetry (Microcal PEAQ-DSC Automated, Malvern). At this time, the protein concentration used for measurement was up to 1 mg/mL. The sample was heated from 25° C. to 110° C. at a rate of 200° C./hr. Normalized heat capacity (Cp) data were calibrated with respect to the buffer solution baseline. Data were analyzed with Microcal PEAQ-DSC Automated software version 1.60 provided by the manufacturer. The melting point (Tm) was used to determine the temperature stability of the bispecific antibody under 50 mM acetate pH 5.0 conditions. This is data of the thermal stability of the bispecific antibody candidate group. Only sample groups having a purity of 90% or higher based on the result of SEC analysis were tested. FIG. **22** shows the results of DSC analysis of Q-SBL2 and Q-SBL9 as representative examples.

TABLE 29

DSC measurement of bispecific antibodies

Sample ID	Tonset (° C.)	Tm1 (° C.)	Tm2 (° C.)
Q-SBL1	59.53	65.61	73.57
Q-SBL2	55.37	63.99	73.70
Q-SBL3	59.49	65.39	73.37
Q-SBL4	59.17	65.35	73.30
R-SBL1	57.09	62.71	73.80
R-SBL2	56.46	62.37	73.73
R-SBL3	56.87	62.14	73.71
R-SBL4	57.69	62.40	73.86
Q-SBL5	52.70	61.86	73.22
R-SBL5	52.94	61.14	73.31
R-SBL6	51.66	60.29	73.28
Q-SBL6	54.75	61.36	74.16
Q-SBL7	55.19	62.07	73.97
Q-SBL8	56.68	61.74	74.00
Q-SBL9	53.82	63.47	73.03

5-3. Dual Antigen Binding ELISA

[0202] The specific ELISA process is as follows. A 96-well high-adsorption ELISA plate was coated with rhVEGF 165 (R&D Systems) using 1× PBS pH 7.4 and the coating concentration was 0.5 μg/ml (100 μl/well). Coating was performed overnight at 4° C. and washed 5 times with 0.05% PBS-T. The result was blocked with 200 μl/well of 2% BSA, incubated at 37° C. for 2 hours, and washed 5 times with 0.05% PBS-T. Equal amounts of the bispecific antibody serially diluted (10 to 0.0005 μg/ml) in 2% BSA and rhHER2-his (R&D Systems) diluted to 1 μg/ml were mixed and incubated at 37° C. for 1 hour. The microplate was washed 5 times with 0.05% PBS-T, 100 μl of the mixture of the heterodimeric antibody and rhHER2-his sample was added to each well and incubated at 37° C. for 2 hours, followed by washing 5 times with 0.05% PBS-T. Then, HRP-conjugated anti-his antibody (Abcam) diluted 1:10000 with PBS containing 2% BSA was added in an amount of 100 μl/well, incubated at 37° C. for 1 hour, and washed 5 times with 0.05% PBS-T. The colorimetric substrate TMB (Bio-Rad) was added at 100 μl/well and was allowed to develop color at room temperature for 5 minutes. 1M H_2SO_4 was added at 100 μl/well and color development was terminated. Absorbance was measured at a wavelength of 450 nm using a SpectraMax ABS Plus (Molecular Devices) instrument. The results of the comparison in EC_{50} between Q-SBL1, Q-SBL2, Q-SBL3, and Q-SBL4 and of the comparison in EC_{50} between R-SBL1, R-SBL2, R-SBL3, and R-SBL4 are as follows. The EC_{50} of Q-SBL5, R-SBL5, and R-SBL6 with changes in the hinge region, compared to Q-SBL1 and R-SBL1, are as follows. Finally, the result of comparison in EC_{50} between Q-SBL6, Q-SBL7, and Q-SBL8, which are candidates depending on the location of the knob/hole, and Q-SBL1, is shown as follows. Only sample groups having a purity of 90% or higher based on the result of SEC analysis were tested. FIGS. **14**A and **14**B are 4-parameter fitting graphs of the dual antigen binding affinity assay.

TABLE 30

EC_{50} of bispecific antibodies in dual antigen binding ELISA

Sample ID	EC_{50} (μg/ml)	R^2
Q-SBL1	0.018	0.996
Q-SBL2	0.019	1.000
Q-SBL3	0.019	0.996
R-SBL1	0.019	0.992
R-SBL2	0.016	0.999
R-SBL3	0.020	0.996
Q-SBL5	0.017	0.996
R-SBL5	0.016	0.998
R-SBL6	0.017	0.996
Q-SBL6	0.026	0.997
Q-SBL7	0.023	0.999
Q-SBL8	0.017	0.999
Q-SBL9	0.029	0.999

5-4. HUVEC Proliferation Assay

[0203] In order to determine the cell proliferation inhibitory effect of the bispecific antibodies, human umbilical vein endothelial cells (HUVECs) were purchased from Lonza and used in experiments. HUVEC (Lonza) cell culture was performed using EBM-2 (Lonza) containing EGM-2 Single Quot (Lonza) and the HUVEC cells used for test were cells of passage 5 or less. The cells were subcultured in a 37° C., 5% CO_2 incubator and the cell confluence was controlled within 80% in a 25-T flask. To analyze the proliferation inhibitory effect of vascular endothelial cells, the vascular endothelial cells were cultured in EBM-2 medium containing 0.25% FBS (Lonza) at a density of 4,000 cells/well on a 96-well plate for 6 hours. Antibodies of various concentrations were pre-treated with VEGF on a 96-well plate and allowed to react at room temperature for 15 minutes. The culture medium in the 96-well plate containing HUVEC cells was replaced with the EBM-2 culture medium containing 0.25% fetal calf serum. Then, each plate well was treated with various concentrations of antibody and 20 ng/ml of VEGF. After culturing for 70 hours, the result was treated with WST-8 (DOJINDO) for 5 hours and absorbance was measured at a wavelength of 450 nm to compare the degree of cell proliferation under respective conditions (FIG. **24**).

TABLE 31

IC_{50} of bispecific antibodies in HUVEC proliferation assay

Sample ID	IC_{50} (nM)
Q-SBL1	5.347
Q-SBL2	5.014
Q-SBL3	3.472
R-SBL1	6.535
R-SBL2	5.032
R-SBL3	7.772
Q-SBL5	6.437
R-SBL5	7.988
R-SBL6	6.770
Q-SBL9	5.481

[0204] When the cultured human vascular endothelial cell was treated with VEGF alone or in combination with a bispecific antibody, the growth rate of vascular endothelial cells decreased compared to the only VEGF treatment group, and nine bispecific antibodies excluding the bispecific antibody R-SBL3 had an IC_{50} of 10 nM or less. All sample groups were tested only when the purity was 90% or higher based on the result of SEC analysis.

5-5. C1q Binding Assay

[0205] The Fc region of IgG1 interacts with Fcγ receptors (FcγR, Fcγ Receptor) and complement proteins (C1q, Complement component 1q) to induce an immune effector function. This plays an important role in increasing the efficacy by removing target cells through antibody-dependent cell-mediated cytotoxicity (ADCC), antibody-dependent cellular phagocytosis (ADCP), or complement-dependent cytotoxicity (CDC). Therefore, an ELISA experiment was conducted to determine the C1q binding activity of the bispecific antibody candidate group including the IgG1 backbone. The specific ELISA process was as follows. A 96-well high-adsorption ELISA plate was coated with 100 μl/well of the bispecific antibody serially diluted in 1×PBS pH 7.4 (540.5~0.3 nM). Coating was performed overnight at 4° C. and washed 5 times with 0.05% PBS-T. The result was blocked with 200 μl/well of 5% BSA (PBS), incubated at 25° C. for 2 hours, and washed 5 times with 0.05% PBS-T. The C1q protein was diluted to 5 μg/ml with 5% BSA (PBS-T), 100 μl of the result was added to each well and incubated at 25° C. for 2 hours. The microplate was washed 5 times with 0.05% PBS-T, HRP-conjugated anti-his antibody (Abcam) diluted 1:2000 with PBS containing 2% BSA was added in an amount of 100 μl/well, and incubated at 25° C. for 1 hour. The result was washed 5 times with 0.05% PBS-T and the colorimetric substrate TMB (Bio-Rad) was added at 100 μl/well and allowed to develop color for 5 minutes at room temperature. 1M H_2SO_4 was added at 100 μl/well and color development was terminated. Absorbance was measured at a wavelength of 450 nm using a SpectraMax ABS Plus (Molecular Devices) instrument. The result of comparison in EC_{50} between Trastuzumab, Q-SBL2, and Q-SBL9 is as follows. Only sample groups having a purity of 90% or higher based on the result of SEC analysis were tested. FIG. **25** is a 4-parameter fitting graph of the C1q binding ELISA.

TABLE 32

EC_{50} of bispecific antibodies in C1q binding ELISA

Sample ID	EC_{50} (nM)	R^2
Trastuzumab	5.115	0.991
Q-SBL2	7.289	0.990
Q-SBL9	8.270	0.987

[0206] Although specific configurations of the present invention have been described in detail, those skilled in the art will appreciate that this description is provided to set forth preferred embodiments for illustrative purposes and should not be construed as limiting the scope of the present invention. Therefore, the substantial scope of the present invention is defined by the accompanying claims and equivalents thereto.

INDUSTRIAL APPLICABILITY

[0207] The novel bispecific or multispecific antibody format according to the present invention is capable of simultaneously binding to two or more targets and of inhibiting or improving the activity of the desired targets, thus being highly effective in treating or diagnosing diseases compared to a single-target antibody.

[0208] In addition, the novel bispecific or multispecific antibody format according to the present invention forms a heterodimer in a state in which non-specific binding between heavy and light chains is rarely present and forms almost no homodimer. Therefore, the novel bispecific or multispecific antibody format is highly expressed in animal cells and the purification process thereof is not much different from that of monoclonal antibodies. The novel bispecific or multispecific antibody exhibits superior or comparable stability to general monoclonal antibodies. The bispecific antibody format is useful for development of various bispecific antibody drugs and the treatment of patients with complicated diseases.

SEQUENCE LISTING

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<223> OTHER INFORMATION: Bevacizumab VH

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Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
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Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
            20                  25                  30

Gly Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Gly Trp Ile Asn Thr Tyr Thr Gly Glu Pro Thr Tyr Ala Ala Asp Phe
    50                  55                  60

Lys Arg Arg Phe Thr Phe Ser Leu Asp Thr Ser Lys Ser Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ala Lys Tyr Pro His Tyr Tyr Gly Ser Ser His Trp Tyr Phe Asp Val
            100                 105                 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
        115                 120


<210> SEQ ID NO 2
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Bevacizumab VL

<400> SEQUENCE: 2

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Ser Ala Ser Gln Asp Ile Ser Asn Tyr
            20                  25                  30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Val Leu Ile
        35                  40                  45

Tyr Phe Thr Ser Ser Leu His Ser Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Tyr Ser Thr Val Pro Trp
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
            100                 105


<210> SEQ ID NO 3
<211> LENGTH: 120
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Trastuzumab VH

<400> SEQUENCE: 3

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
            20                  25                  30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
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        35                  40                  45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser
        115                 120


<210> SEQ ID NO 4
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Trastuzumab VL

<400> SEQUENCE: 4

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
            20                  25                  30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
        35                  40                  45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
            100                 105


<210> SEQ ID NO 5
<211> LENGTH: 98
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 CH1

<400> SEQUENCE: 5

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys
1               5                   10                  15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
            20                  25                  30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
        35                  40                  45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
    50                  55                  60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
65                  70                  75                  80

Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
                85                  90                  95
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Lys Val


<210> SEQ ID NO 6
<211> LENGTH: 110
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 CH2

<400> SEQUENCE: 6

Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
1               5                   10                  15

Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
            20                  25                  30

Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr
        35                  40                  45

Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
    50                  55                  60

Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His
65                  70                  75                  80

Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
                85                  90                  95

Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys
            100                 105                 110


<210> SEQ ID NO 7
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 CH3

<400> SEQUENCE: 7

Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp
1               5                   10                  15

Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe
            20                  25                  30

Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu
        35                  40                  45

Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe
    50                  55                  60

Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly
65                  70                  75                  80

Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr
                85                  90                  95

Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
            100                 105


<210> SEQ ID NO 8
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 CH3-Knob

<400> SEQUENCE: 8

Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu
1               5                   10                  15
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Glu Met Thr Lys Asn Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe
            20                  25                  30

Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu
        35                  40                  45

Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe
    50                  55                  60

Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly
65                  70                  75                  80

Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr
                85                  90                  95

Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
            100                 105


<210> SEQ ID NO 9
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 CH3-Hole

<400> SEQUENCE: 9

Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu
1               5                   10                  15

Glu Met Thr Lys Asn Gln Val Ser Leu Ser Cys Ala Val Lys Gly Phe
            20                  25                  30

Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu
        35                  40                  45

Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe
    50                  55                  60

Phe Leu Val Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly
65                  70                  75                  80

Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr
                85                  90                  95

Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
            100                 105


<210> SEQ ID NO 10
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 CH3-Knob S354C

<400> SEQUENCE: 10

Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu
1               5                   10                  15

Glu Met Thr Lys Asn Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe
            20                  25                  30

Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu
        35                  40                  45

Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe
    50                  55                  60

Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly
65                  70                  75                  80

Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr
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                85                  90                  95

Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
            100                 105


<210> SEQ ID NO 11
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 CH3-Hole Y349C

<400> SEQUENCE: 11

Gly Gln Pro Arg Glu Pro Gln Val Cys Thr Leu Pro Pro Ser Arg Glu
1               5                   10                  15

Glu Met Thr Lys Asn Gln Val Ser Leu Ser Cys Ala Val Lys Gly Phe
            20                  25                  30

Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu
        35                  40                  45

Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe
    50                  55                  60

Phe Leu Val Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly
65                  70                  75                  80

Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr
                85                  90                  95

Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
            100                 105


<210> SEQ ID NO 12
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 CH3-Knob Y349C

<400> SEQUENCE: 12

Gly Gln Pro Arg Glu Pro Gln Val Cys Thr Leu Pro Pro Ser Arg Glu
1               5                   10                  15

Glu Met Thr Lys Asn Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe
            20                  25                  30

Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu
        35                  40                  45

Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe
    50                  55                  60

Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly
65                  70                  75                  80

Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr
                85                  90                  95

Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
            100                 105


<210> SEQ ID NO 13
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 CH3-Hole S354C

<400> SEQUENCE: 13
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Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu
1               5                   10                  15

Glu Met Thr Lys Asn Gln Val Ser Leu Ser Cys Ala Val Lys Gly Phe
            20                  25                  30

Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu
        35                  40                  45

Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe
    50                  55                  60

Phe Leu Val Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly
65                  70                  75                  80

Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr
                85                  90                  95

Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
            100                 105


<210> SEQ ID NO 14
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CL-Kappa

<400> SEQUENCE: 14

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
1               5                   10                  15

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
            20                  25                  30

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
        35                  40                  45

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
    50                  55                  60

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
65                  70                  75                  80

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
                85                  90                  95

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
            100                 105


<210> SEQ ID NO 15
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CL-Kappa-C214S

<400> SEQUENCE: 15

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
1               5                   10                  15

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
            20                  25                  30

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
        35                  40                  45

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
    50                  55                  60

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
```

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65                  70                  75                  80

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
                85                  90                  95

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Ser
            100                 105


<210> SEQ ID NO 16
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CL-Lambda

<400> SEQUENCE: 16

Gly Gln Pro Lys Ala Asn Pro Thr Val Thr Leu Phe Pro Pro Ser Ser
1               5                   10                  15

Glu Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp
            20                  25                  30

Phe Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Gly Ser Pro
        35                  40                  45

Val Lys Ala Gly Val Glu Thr Thr Lys Pro Ser Lys Gln Ser Asn Asn
    50                  55                  60

Lys Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys
65                  70                  75                  80

Ser His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val
                85                  90                  95

Glu Lys Thr Val Ala Pro Thr Glu Cys Ser
            100                 105


<210> SEQ ID NO 17
<211> LENGTH: 106
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: CL-Lambda-C214S

<400> SEQUENCE: 17

Gly Gln Pro Lys Ala Asn Pro Thr Val Thr Leu Phe Pro Pro Ser Ser
1               5                   10                  15

Glu Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp
            20                  25                  30

Phe Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Gly Ser Pro
        35                  40                  45

Val Lys Ala Gly Val Glu Thr Thr Lys Pro Ser Lys Gln Ser Asn Asn
    50                  55                  60

Lys Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys
65                  70                  75                  80

Ser His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val
                85                  90                  95

Glu Lys Thr Val Ala Pro Thr Glu Ser Ser
            100                 105


<210> SEQ ID NO 18
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<212> TYPE: PRT
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<223> OTHER INFORMATION: IgG4 CH1

<400> SEQUENCE: 18

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Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
1               5                   10                  15

Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
            20                  25                  30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
        35                  40                  45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
    50                  55                  60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Lys Thr
65                  70                  75                  80

Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys
                85                  90                  95

Arg Val
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<210> SEQ ID NO 19
<211> LENGTH: 98
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG4 CH1 C131S

<400> SEQUENCE: 19

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Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Arg
1               5                   10                  15

Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
            20                  25                  30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
        35                  40                  45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
    50                  55                  60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Lys Thr
65                  70                  75                  80

Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys
                85                  90                  95

Arg Val
```

<210> SEQ ID NO 20
<211> LENGTH: 98
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgD CH1

<400> SEQUENCE: 20

```
Ala Pro Thr Lys Ala Pro Asp Val Phe Pro Ile Ile Ser Gly Cys Arg
1               5                   10                  15

His Pro Lys Asp Asn Ser Pro Val Val Leu Ala Cys Leu Ile Thr Gly
            20                  25                  30

Tyr His Pro Thr Ser Val Thr Val Thr Trp Tyr Met Gly Thr Gln Ser
        35                  40                  45

Gln Pro Gln Arg Thr Phe Pro Glu Ile Gln Arg Arg Asp Ser Tyr Tyr
    50                  55                  60
```

-continued

```
Met Thr Ser Ser Gln Leu Ser Thr Pro Leu Gln Gln Trp Arg Gln Gly
65                  70                  75                  80

Glu Tyr Lys Cys Val Val Gln His Thr Ala Ser Lys Ser Lys Lys Glu
                85                  90                  95

Ile Phe
```

<210> SEQ ID NO 21
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 Hinge wild type

<400> SEQUENCE: 21

```
Glu Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro
1               5                   10                  15
```

<210> SEQ ID NO 22
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 Hinge del EPKSC

<400> SEQUENCE: 22

```
Asp Lys Thr His Thr Cys Pro Pro Cys Pro
1               5                   10
```

<210> SEQ ID NO 23
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG1 Hinge EPKSS

<400> SEQUENCE: 23

```
Glu Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro
1               5                   10                  15
```

<210> SEQ ID NO 24
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG4 Hinge

<400> SEQUENCE: 24

```
Glu Ser Lys Tyr Gly Pro Pro Cys Pro Pro Cys Pro
1               5                   10
```

<210> SEQ ID NO 25

<400> SEQUENCE: 25

000

<210> SEQ ID NO 26

<400> SEQUENCE: 26

000

<210> SEQ ID NO 27

-continued

<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker1

<400> SEQUENCE: 27

Gly Gly Gly Gly Ser
1   5

<210> SEQ ID NO 28
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker2

<400> SEQUENCE: 28

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
1   5   10

<210> SEQ ID NO 29
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker3

<400> SEQUENCE: 29

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
1   5   10   15

<210> SEQ ID NO 30
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker4

<400> SEQUENCE: 30

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
1   5   10   15

Gly Gly Gly Ser
            20

<210> SEQ ID NO 31
<211> LENGTH: 562
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL1 HC first arm

<400> SEQUENCE: 31

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1   5   10   15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
            20   25   30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35   40   45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50   55   60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr

-continued

```
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala Ser Thr
225                 230                 235                 240

Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser
                245                 250                 255

Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
            260                 265                 270

Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
        275                 280                 285

Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
    290                 295                 300

Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys
305                 310                 315                 320

Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Asp
                325                 330                 335

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
            340                 345                 350

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
        355                 360                 365

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
    370                 375                 380

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
385                 390                 395                 400

Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
                405                 410                 415

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
            420                 425                 430

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
        435                 440                 445

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
    450                 455                 460

Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
465                 470                 475                 480
```

-continued

```
Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
                485                 490                 495

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
            500                 505                 510

Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp
        515                 520                 525

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
    530                 535                 540

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
545                 550                 555                 560

Gly Lys


<210> SEQ ID NO 32
<211> LENGTH: 331
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL1 LC first arm

<400> SEQUENCE: 32

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
            20                  25                  30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
        35                  40                  45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
            100                 105                 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
        115                 120                 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
    130                 135                 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145                 150                 155                 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
                165                 170                 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
            180                 185                 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
        195                 200                 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
    210                 215                 220

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
225                 230                 235                 240

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
                245                 250                 255
```

-continued

```
Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
            260                 265                 270

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
        275                 280                 285

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
    290                 295                 300

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
305                 310                 315                 320

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Ser
                325                 330


<210> SEQ ID NO 33
<211> LENGTH: 557
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL2 HC first arm

<400> SEQUENCE: 33

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
            20                  25                  30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Ala Ser Thr Lys Gly Pro Ser Val
225                 230                 235                 240

Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala
                245                 250                 255

Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
            260                 265                 270
```

-continued

```
Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
        275                 280                 285

Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
    290                 295                 300

Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys
305                 310                 315                 320

Pro Ser Asn Thr Lys Val Asp Lys Lys Val Asp Lys Thr His Thr Cys
                325                 330                 335

Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu
            340                 345                 350

Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu
        355                 360                 365

Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys
    370                 375                 380

Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys
385                 390                 395                 400

Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu
                405                 410                 415

Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys
            420                 425                 430

Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys
        435                 440                 445

Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser
    450                 455                 460

Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Ser Cys Ala Val Lys
465                 470                 475                 480

Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln
                485                 490                 495

Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly
            500                 505                 510

Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln
        515                 520                 525

Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn
    530                 535                 540

His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
545                 550                 555


<210> SEQ ID NO 34
<211> LENGTH: 326
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL2 LC first arm

<400> SEQUENCE: 34

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
            20                  25                  30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
        35                  40                  45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60
```

-continued

```
Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
            100                 105                 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
        115                 120                 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
    130                 135                 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145                 150                 155                 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
                165                 170                 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
            180                 185                 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
        195                 200                 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Arg Thr Val Ala Ala
    210                 215                 220

Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
225                 230                 235                 240

Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
                245                 250                 255

Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
            260                 265                 270

Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
        275                 280                 285

Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
    290                 295                 300

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
305                 310                 315                 320

Phe Asn Arg Gly Glu Ser
                325
```

<210> SEQ ID NO 35
<211> LENGTH: 567
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL3 HC first arm

<400> SEQUENCE: 35

```
Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
            20                  25                  30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80
```

-continued

```
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
225                 230                 235                 240

Gly Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser
                245                 250                 255

Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys
            260                 265                 270

Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu
        275                 280                 285

Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu
    290                 295                 300

Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr
305                 310                 315                 320

Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val
                325                 330                 335

Asp Lys Lys Val Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro
            340                 345                 350

Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys
        355                 360                 365

Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val
    370                 375                 380

Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp
385                 390                 395                 400

Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr
                405                 410                 415

Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp
            420                 425                 430

Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu
        435                 440                 445

Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg
    450                 455                 460

Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys
465                 470                 475                 480

Asn Gln Val Ser Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp
```

-continued

```
                485                 490                 495

Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys
            500                 505                 510

Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser
        515                 520                 525

Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser
    530                 535                 540

Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser
545                 550                 555                 560

Leu Ser Leu Ser Pro Gly Lys
                565


<210> SEQ ID NO 36
<211> LENGTH: 336
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL3 LC first arm

<400> SEQUENCE: 36

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
            20                  25                  30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
        35                  40                  45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
            100                 105                 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
        115                 120                 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
    130                 135                 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145                 150                 155                 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
                165                 170                 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
            180                 185                 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
        195                 200                 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
    210                 215                 220

Gly Gly Gly Gly Ser Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe
225                 230                 235                 240

Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys
                245                 250                 255

Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val
```

-continued

```
            260                 265                 270

Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln
        275                 280                 285

Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser
    290                 295                 300

Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His
305                 310                 315                 320

Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Ser
                325                 330                 335


<210> SEQ ID NO 37
<211> LENGTH: 572
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL4 HC first arm

<400> SEQUENCE: 37

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
            20                  25                  30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
225                 230                 235                 240

Gly Ser Gly Gly Gly Gly Ser Ala Ser Thr Lys Gly Pro Ser Val Phe
                245                 250                 255

Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu
            260                 265                 270

Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp
```

-continued

```
        275                 280                 285

Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu
    290                 295                 300

Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser
305                 310                 315                 320

Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro
                325                 330                 335

Ser Asn Thr Lys Val Asp Lys Lys Val Asp Lys Thr His Thr Cys Pro
            340                 345                 350

Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe
        355                 360                 365

Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val
    370                 375                 380

Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe
385                 390                 395                 400

Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro
                405                 410                 415

Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr
            420                 425                 430

Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val
        435                 440                 445

Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala
    450                 455                 460

Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg
465                 470                 475                 480

Glu Glu Met Thr Lys Asn Gln Val Ser Leu Ser Cys Ala Val Lys Gly
                485                 490                 495

Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro
            500                 505                 510

Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser
        515                 520                 525

Phe Phe Leu Val Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln
    530                 535                 540

Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His
545                 550                 555                 560

Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
                565                 570


<210> SEQ ID NO 38
<211> LENGTH: 341
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL4 LC first arm

<400> SEQUENCE: 38

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
            20                  25                  30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
        35                  40                  45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
```

-continued

```
    50                  55                  60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
            100                 105                 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
        115                 120                 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
    130                 135                 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145                 150                 155                 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
                165                 170                 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
            180                 185                 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
        195                 200                 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
    210                 215                 220

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Arg Thr Val Ala Ala Pro
225                 230                 235                 240

Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr
                245                 250                 255

Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys
            260                 265                 270

Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu
        275                 280                 285

Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser
    290                 295                 300

Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala
305                 310                 315                 320

Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe
                325                 330                 335

Asn Arg Gly Glu Ser
            340


&lt;210&gt; SEQ ID NO 39
&lt;211&gt; LENGTH: 567
&lt;212&gt; TYPE: PRT
&lt;213&gt; ORGANISM: Artificial Sequence
&lt;220&gt; FEATURE:
&lt;223&gt; OTHER INFORMATION: Q-SBL5 HC first arm

&lt;400&gt; SEQUENCE: 39

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
            20                  25                  30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
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    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala Ser Thr
225                 230                 235                 240

Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser
                245                 250                 255

Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
            260                 265                 270

Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
        275                 280                 285

Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
    290                 295                 300

Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys
305                 310                 315                 320

Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu
                325                 330                 335

Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro
            340                 345                 350

Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys
        355                 360                 365

Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val
    370                 375                 380

Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp
385                 390                 395                 400

Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr
                405                 410                 415

Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp
            420                 425                 430

Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu
        435                 440                 445

Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg
    450                 455                 460
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Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys
465 470 475 480

Asn Gln Val Ser Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp
485 490 495

Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys
500 505 510

Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser
515 520 525

Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser
530 535 540

Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser
545 550 555 560

Leu Ser Leu Ser Pro Gly Lys
565

<210> SEQ ID NO 40
<211> LENGTH: 331
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL5 LC first arm

<400> SEQUENCE: 40

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
20 25 30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
100 105 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
115 120 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
130 135 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145 150 155 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
165 170 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
180 185 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
195 200 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
210 215 220

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
225 230 235 240

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Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
                245                 250                 255

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
            260                 265                 270

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
        275                 280                 285

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
    290                 295                 300

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
305                 310                 315                 320

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Ser
                325                 330


<210> SEQ ID NO 41
<211> LENGTH: 562
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL6 HC first arm

<400> SEQUENCE: 41

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
            20                  25                  30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala Ser Thr
225                 230                 235                 240

Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser
                245                 250                 255
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Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
            260                 265                 270

Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
        275                 280                 285

Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
    290                 295                 300

Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys
305                 310                 315                 320

Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Asp
                325                 330                 335

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
            340                 345                 350

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
        355                 360                 365

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
    370                 375                 380

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
385                 390                 395                 400

Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
                405                 410                 415

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
            420                 425                 430

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
        435                 440                 445

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
    450                 455                 460

Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
465                 470                 475                 480

Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
                485                 490                 495

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
            500                 505                 510

Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp
        515                 520                 525

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
    530                 535                 540

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
545                 550                 555                 560

Gly Lys


<210> SEQ ID NO 42
<211> LENGTH: 331
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL6 LC first arm

<400> SEQUENCE: 42

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
            20                  25                  30
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Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
100 105 110

Pro Gln Val Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn
115 120 125

Gln Val Ser Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile
130 135 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145 150 155 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys
165 170 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
180 185 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
195 200 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
210 215 220

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
225 230 235 240

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
245 250 255

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
260 265 270

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
275 280 285

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
290 295 300

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
305 310 315 320

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Ser
325 330

<210> SEQ ID NO 43
<211> LENGTH: 562
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL7 HC first arm

<400> SEQUENCE: 43

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
20 25 30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

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Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala Ser Thr
225                 230                 235                 240

Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser
                245                 250                 255

Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
            260                 265                 270

Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
        275                 280                 285

Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
    290                 295                 300

Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys
305                 310                 315                 320

Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Asp
                325                 330                 335

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
            340                 345                 350

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
        355                 360                 365

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
    370                 375                 380

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
385                 390                 395                 400

Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
                405                 410                 415

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
            420                 425                 430

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
        435                 440                 445

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
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    450                 455                 460

Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
465                 470                 475                 480

Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
                485                 490                 495

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
            500                 505                 510

Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp
        515                 520                 525

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
    530                 535                 540

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
545                 550                 555                 560

Gly Lys


<210> SEQ ID NO 44
<211> LENGTH: 331
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL7 LC first arm

<400> SEQUENCE: 44

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
            20                  25                  30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
        35                  40                  45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
            100                 105                 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
        115                 120                 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
    130                 135                 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145                 150                 155                 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
                165                 170                 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
            180                 185                 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
        195                 200                 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
    210                 215                 220

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
225                 230                 235                 240
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Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
                245                 250                 255

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
            260                 265                 270

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
        275                 280                 285

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
    290                 295                 300

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
305                 310                 315                 320

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Ser
                325                 330


<210> SEQ ID NO 45
<211> LENGTH: 562
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL8 HC first arm

<400> SEQUENCE: 45

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
            20                  25                  30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala Ser Thr
225                 230                 235                 240

Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser
                245                 250                 255
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Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
260 265 270

Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
275 280 285

Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
290 295 300

Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys
305 310 315 320

Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Asp
325 330 335

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
340 345 350

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
355 360 365

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
370 375 380

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
385 390 395 400

Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
405 410 415

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
420 425 430

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
435 440 445

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
450 455 460

Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
465 470 475 480

Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
485 490 495

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
500 505 510

Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp
515 520 525

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
530 535 540

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
545 550 555 560

Gly Lys

<210> SEQ ID NO 46
<211> LENGTH: 331
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL8 LC first arm

<400> SEQUENCE: 46

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
20 25 30

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```
Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
        35                  40                  45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
            100                 105                 110

Pro Gln Val Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn
        115                 120                 125

Gln Val Ser Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile
    130                 135                 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145                 150                 155                 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys
                165                 170                 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
            180                 185                 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
        195                 200                 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
    210                 215                 220

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
225                 230                 235                 240

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
                245                 250                 255

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
            260                 265                 270

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
        275                 280                 285

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
    290                 295                 300

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
305                 310                 315                 320

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Ser
                325                 330


<210> SEQ ID NO 47
<211> LENGTH: 562
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL9 HC first arm

<400> SEQUENCE: 47

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
            20                  25                  30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45
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```
Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Ala Ser Thr Lys Gly Pro Ser Val
225                 230                 235                 240

Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala
                245                 250                 255

Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
            260                 265                 270

Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
        275                 280                 285

Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
    290                 295                 300

Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys
305                 310                 315                 320

Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Ser Asp
                325                 330                 335

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
            340                 345                 350

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
        355                 360                 365

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
    370                 375                 380

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
385                 390                 395                 400

Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
                405                 410                 415

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
            420                 425                 430

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
        435                 440                 445

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Cys
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450 455 460

Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
465 470 475 480

Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
485 490 495

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
500 505 510

Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp
515 520 525

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
530 535 540

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
545 550 555 560

Gly Lys

<210> SEQ ID NO 48
<211> LENGTH: 326
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Q-SBL9 LC first arm

<400> SEQUENCE: 48

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
20 25 30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
100 105 110

Pro Gln Val Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn
115 120 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
130 135 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145 150 155 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
165 170 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
180 185 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
195 200 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Arg Thr Val Ala Ala
210 215 220

Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
225 230 235 240

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Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
                245                 250                 255

Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
            260                 265                 270

Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
        275                 280                 285

Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
    290                 295                 300

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
305                 310                 315                 320

Phe Asn Arg Gly Glu Ser
                325


<210> SEQ ID NO 49
<211> LENGTH: 562
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: R-SBL1 HC first arm

<400> SEQUENCE: 49

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
            20                  25                  30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala Ser Thr
225                 230                 235                 240

Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Arg Ser Thr Ser
                245                 250                 255
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Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
260 265 270

Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
275 280 285

Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
290 295 300

Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys
305 310 315 320

Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Asp
325 330 335

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
340 345 350

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
355 360 365

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
370 375 380

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
385 390 395 400

Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
405 410 415

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
420 425 430

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
435 440 445

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
450 455 460

Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
465 470 475 480

Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
485 490 495

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
500 505 510

Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp
515 520 525

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
530 535 540

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
545 550 555 560

Gly Lys

<210> SEQ ID NO 50
<211> LENGTH: 331
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: R-SBL1 LC first arm

<400> SEQUENCE: 50

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
20 25 30

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Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
        35                  40                  45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
            100                 105                 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
        115                 120                 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
    130                 135                 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145                 150                 155                 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
                165                 170                 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
            180                 185                 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
        195                 200                 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
    210                 215                 220

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
225                 230                 235                 240

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
                245                 250                 255

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
            260                 265                 270

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
        275                 280                 285

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
    290                 295                 300

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
305                 310                 315                 320

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Ser
                325                 330
```

<210> SEQ ID NO 51
<211> LENGTH: 557
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: R-SBL2 HC first arm

<400> SEQUENCE: 51

```
Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
            20                  25                  30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45
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Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Ala Ser Thr Lys Gly Pro Ser Val
225                 230                 235                 240

Phe Pro Leu Ala Pro Ser Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala
                245                 250                 255

Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
            260                 265                 270

Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
        275                 280                 285

Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
    290                 295                 300

Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys
305                 310                 315                 320

Pro Ser Asn Thr Lys Val Asp Lys Arg Val Asp Lys Thr His Thr Cys
                325                 330                 335

Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu
            340                 345                 350

Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu
        355                 360                 365

Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys
    370                 375                 380

Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys
385                 390                 395                 400

Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu
                405                 410                 415

Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys
            420                 425                 430

Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys
        435                 440                 445

Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser
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    450                 455                 460

Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Ser Cys Ala Val Lys
465                 470                 475                 480

Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln
                485                 490                 495

Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly
            500                 505                 510

Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln
        515                 520                 525

Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn
    530                 535                 540

His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
545                 550                 555


<210> SEQ ID NO 52
<211> LENGTH: 326
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: R-SBL2 LC first arm

<400> SEQUENCE: 52

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
            20                  25                  30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
        35                  40                  45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
            100                 105                 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
        115                 120                 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
    130                 135                 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145                 150                 155                 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
                165                 170                 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
            180                 185                 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
        195                 200                 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Arg Thr Val Ala Ala
    210                 215                 220

Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
225                 230                 235                 240

Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
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245 250 255

Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
260 265 270

Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
275 280 285

Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
290 295 300

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
305 310 315 320

Phe Asn Arg Gly Glu Ser
325

<210> SEQ ID NO 53
<211> LENGTH: 567
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: R-SBL3 HC first arm

<400> SEQUENCE: 53

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
20 25 30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
100 105 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
115 120 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
130 135 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145 150 155 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
165 170 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
180 185 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
195 200 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
210 215 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
225 230 235 240

Gly Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser
245 250 255

Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys

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            260                 265                 270

Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu
        275                 280                 285

Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu
    290                 295                 300

Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr
305                 310                 315                 320

Lys Thr Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val
                325                 330                 335

Asp Lys Arg Val Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro
            340                 345                 350

Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys
        355                 360                 365

Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val
    370                 375                 380

Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp
385                 390                 395                 400

Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr
                405                 410                 415

Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp
            420                 425                 430

Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu
        435                 440                 445

Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg
    450                 455                 460

Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys
465                 470                 475                 480

Asn Gln Val Ser Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp
                485                 490                 495

Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys
            500                 505                 510

Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser
        515                 520                 525

Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser
    530                 535                 540

Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser
545                 550                 555                 560

Leu Ser Leu Ser Pro Gly Lys
                565


&lt;210&gt; SEQ ID NO 54
&lt;211&gt; LENGTH: 336
&lt;212&gt; TYPE: PRT
&lt;213&gt; ORGANISM: Artificial Sequence
&lt;220&gt; FEATURE:
&lt;223&gt; OTHER INFORMATION: R-SBL3 LC first arm

&lt;400&gt; SEQUENCE: 54

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
            20                  25                  30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
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-continued

35 40 45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
100 105 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
115 120 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
130 135 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145 150 155 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
165 170 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
180 185 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
195 200 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
210 215 220

Gly Gly Gly Gly Ser Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe
225 230 235 240

Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys
245 250 255

Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val
260 265 270

Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln
275 280 285

Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser
290 295 300

Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His
305 310 315 320

Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Ser
325 330 335

<210> SEQ ID NO 55
<211> LENGTH: 572
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: R-SBL4 HC first arm

<400> SEQUENCE: 55

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
20 25 30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val

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    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly
225                 230                 235                 240

Gly Ser Gly Gly Gly Gly Ser Ala Ser Thr Lys Gly Pro Ser Val Phe
                245                 250                 255

Pro Leu Ala Pro Ser Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu
            260                 265                 270

Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp
        275                 280                 285

Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu
    290                 295                 300

Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser
305                 310                 315                 320

Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys Pro
                325                 330                 335

Ser Asn Thr Lys Val Asp Lys Arg Val Asp Lys Thr His Thr Cys Pro
            340                 345                 350

Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe
        355                 360                 365

Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val
    370                 375                 380

Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe
385                 390                 395                 400

Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro
                405                 410                 415

Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr
            420                 425                 430

Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val
        435                 440                 445

Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala
    450                 455                 460
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Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg
465 470 475 480

Glu Glu Met Thr Lys Asn Gln Val Ser Leu Ser Cys Ala Val Lys Gly
485 490 495

Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro
500 505 510

Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser
515 520 525

Phe Phe Leu Val Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln
530 535 540

Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His
545 550 555 560

Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
565 570

<210> SEQ ID NO 56
<211> LENGTH: 341
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: R-SBL4 LC first arm

<400> SEQUENCE: 56

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
20 25 30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
100 105 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
115 120 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
130 135 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145 150 155 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
165 170 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
180 185 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
195 200 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
210 215 220

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Arg Thr Val Ala Ala Pro
225 230 235 240

-continued

Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr
245 250 255

Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys
260 265 270

Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu
275 280 285

Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser
290 295 300

Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala
305 310 315 320

Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe
325 330 335

Asn Arg Gly Glu Ser
340

<210> SEQ ID NO 57
<211> LENGTH: 567
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: R-SBL5 HC first arm

<400> SEQUENCE: 57

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
20 25 30

Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
100 105 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
115 120 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
130 135 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145 150 155 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
165 170 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
180 185 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
195 200 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
210 215 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala Ser Thr
225 230 235 240

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```
Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Arg Ser Thr Ser
                245                 250                 255

Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
            260                 265                 270

Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
        275                 280                 285

Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
    290                 295                 300

Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys
305                 310                 315                 320

Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu
                325                 330                 335

Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro
            340                 345                 350

Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys
        355                 360                 365

Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val
    370                 375                 380

Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp
385                 390                 395                 400

Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr
                405                 410                 415

Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp
            420                 425                 430

Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu
        435                 440                 445

Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg
    450                 455                 460

Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys
465                 470                 475                 480

Asn Gln Val Ser Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp
                485                 490                 495

Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys
            500                 505                 510

Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser
        515                 520                 525

Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser
    530                 535                 540

Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser
545                 550                 555                 560

Leu Ser Leu Ser Pro Gly Lys
                565
```

<210> SEQ ID NO 58
<211> LENGTH: 331
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: R-SBL5 LC first arm

<400> SEQUENCE: 58

```
Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15
```

-continued

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
20 25 30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
100 105 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
115 120 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
130 135 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145 150 155 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
165 170 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
180 185 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
195 200 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
210 215 220

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
225 230 235 240

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
245 250 255

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
260 265 270

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
275 280 285

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
290 295 300

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
305 310 315 320

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Ser
325 330

<210> SEQ ID NO 59
<211> LENGTH: 564
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: R-SBL6 HC first arm

<400> SEQUENCE: 59

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Asn Ile Lys Asp Thr
20 25 30

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```
Tyr Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Ala Arg Ile Tyr Pro Thr Asn Gly Tyr Thr Arg Tyr Ala Asp Ser Val
    50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ser Arg Trp Gly Gly Asp Gly Phe Tyr Ala Met Asp Tyr Trp Gly Gln
            100                 105                 110

Gly Thr Leu Val Thr Val Ser Ser Ala Ser Pro Arg Glu Pro Gln Val
        115                 120                 125

Cys Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
    130                 135                 140

Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
145                 150                 155                 160

Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
                165                 170                 175

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val
            180                 185                 190

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
        195                 200                 205

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
    210                 215                 220

Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala Ser Thr
225                 230                 235                 240

Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Arg Ser Thr Ser
                245                 250                 255

Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
            260                 265                 270

Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
        275                 280                 285

Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
    290                 295                 300

Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys
305                 310                 315                 320

Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu
                325                 330                 335

Ser Lys Tyr Gly Pro Pro Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu
            340                 345                 350

Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu
        355                 360                 365

Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser
    370                 375                 380

His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu
385                 390                 395                 400

Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr
                405                 410                 415

Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn
            420                 425                 430
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```
Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro
        435                 440                 445

Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln
    450                 455                 460

Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val
465                 470                 475                 480

Ser Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val
                485                 490                 495

Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro
            500                 505                 510

Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr
        515                 520                 525

Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val
    530                 535                 540

Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu
545                 550                 555                 560

Ser Pro Gly Lys


<210> SEQ ID NO 60
<211> LENGTH: 331
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: R-SBL6 LC first arm

<400> SEQUENCE: 60

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Asp Val Asn Thr Ala
            20                  25                  30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
        35                  40                  45

Tyr Ser Ala Ser Phe Leu Tyr Ser Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60

Ser Arg Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln His Tyr Thr Thr Pro Pro
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Pro Arg Glu
            100                 105                 110

Pro Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn
        115                 120                 125

Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
    130                 135                 140

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
145                 150                 155                 160

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
                165                 170                 175

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
            180                 185                 190

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
        195                 200                 205

Ser Leu Ser Pro Gly Lys Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
```

-continued

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    210                 215                 220

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
225                 230                 235                 240

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
                245                 250                 255

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
            260                 265                 270

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
        275                 280                 285

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
    290                 295                 300

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
305                 310                 315                 320

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Ser
                325                 330


<210> SEQ ID NO 61
<211> LENGTH: 453
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: LC-1 HC second arm

<400> SEQUENCE: 61

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
            20                  25                  30

Gly Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Gly Trp Ile Asn Thr Tyr Thr Gly Glu Pro Thr Tyr Ala Ala Asp Phe
    50                  55                  60

Lys Arg Arg Phe Thr Phe Ser Leu Asp Thr Ser Lys Ser Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ala Lys Tyr Pro His Tyr Tyr Gly Ser Ser His Trp Tyr Phe Asp Val
            100                 105                 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
        115                 120                 125

Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
    130                 135                 140

Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
145                 150                 155                 160

Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
                165                 170                 175

Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
            180                 185                 190

Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
        195                 200                 205

Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
    210                 215                 220

Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu
```

-continued

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225                 230                 235                 240

Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr
                245                 250                 255

Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val
            260                 265                 270

Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val
        275                 280                 285

Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser
    290                 295                 300

Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu
305                 310                 315                 320

Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala
                325                 330                 335

Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro
            340                 345                 350

Gln Val Tyr Thr Leu Pro Pro Cys Arg Glu Glu Met Thr Lys Asn Gln
        355                 360                 365

Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala
    370                 375                 380

Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr
385                 390                 395                 400

Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu
                405                 410                 415

Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser
            420                 425                 430

Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser
        435                 440                 445

Leu Ser Pro Gly Lys
    450


<210> SEQ ID NO 62
<211> LENGTH: 214
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: LC-1 LC second arm

<400> SEQUENCE: 62

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Ser Ala Ser Gln Asp Ile Ser Asn Tyr
            20                  25                  30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Val Leu Ile
        35                  40                  45

Tyr Phe Thr Ser Ser Leu His Ser Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Tyr Ser Thr Val Pro Trp
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala
            100                 105                 110

Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
```

-continued

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        115                 120                 125

Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
    130                 135                 140

Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
145                 150                 155                 160

Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
                165                 170                 175

Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
            180                 185                 190

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
        195                 200                 205

Phe Asn Arg Gly Glu Cys
    210


<210> SEQ ID NO 63
<211> LENGTH: 453
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: LC-3 HC second arm

<400> SEQUENCE: 63

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
            20                  25                  30

Gly Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Gly Trp Ile Asn Thr Tyr Thr Gly Glu Pro Thr Tyr Ala Ala Asp Phe
    50                  55                  60

Lys Arg Arg Phe Thr Phe Ser Leu Asp Thr Ser Lys Ser Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ala Lys Tyr Pro His Tyr Tyr Gly Ser Ser His Trp Tyr Phe Asp Val
            100                 105                 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
        115                 120                 125

Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
    130                 135                 140

Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
145                 150                 155                 160

Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
                165                 170                 175

Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
            180                 185                 190

Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
        195                 200                 205

Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
    210                 215                 220

Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu
225                 230                 235                 240

Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr
```

-continued

```
                245                 250                 255

Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val
            260                 265                 270

Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val
        275                 280                 285

Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser
    290                 295                 300

Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu
305                 310                 315                 320

Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala
                325                 330                 335

Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro
            340                 345                 350

Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln
        355                 360                 365

Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala
    370                 375                 380

Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr
385                 390                 395                 400

Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu
                405                 410                 415

Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser
            420                 425                 430

Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser
        435                 440                 445

Leu Ser Pro Gly Lys
    450
```

<210> SEQ ID NO 64
<211> LENGTH: 453
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: LC-3-H-K HC second arm

<400> SEQUENCE: 64

```
Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1               5                   10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
            20                  25                  30

Gly Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
        35                  40                  45

Gly Trp Ile Asn Thr Tyr Thr Gly Glu Pro Thr Tyr Ala Ala Asp Phe
    50                  55                  60

Lys Arg Arg Phe Thr Phe Ser Leu Asp Thr Ser Lys Ser Thr Ala Tyr
65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
                85                  90                  95

Ala Lys Tyr Pro His Tyr Tyr Gly Ser Ser His Trp Tyr Phe Asp Val
            100                 105                 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
        115                 120                 125

Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly
```

-continued

```
    130                 135                 140

Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
145                 150                 155                 160

Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
                165                 170                 175

Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
            180                 185                 190

Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val
        195                 200                 205

Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys
    210                 215                 220

Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu
225                 230                 235                 240

Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr
                245                 250                 255

Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val
            260                 265                 270

Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val
        275                 280                 285

Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser
    290                 295                 300

Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu
305                 310                 315                 320

Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala
                325                 330                 335

Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro
            340                 345                 350

Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln
        355                 360                 365

Val Ser Leu Ser Cys Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala
    370                 375                 380

Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr
385                 390                 395                 400

Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu
                405                 410                 415

Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser
            420                 425                 430

Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser
        435                 440                 445

Leu Ser Pro Gly Lys
    450


<210> SEQ ID NO 65
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Signal sequence

<400> SEQUENCE: 65

Met Glu Trp Ser Trp Val Phe Leu Phe Phe Leu Ser Val Thr Thr Gly
1               5                   10                  15

Val His Ser
```

-continued

```
<210> SEQ ID NO 66
<211> LENGTH: 98
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgG2 CH1

<400> SEQUENCE: 66

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
1               5                   10                  15

Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
            20                  25                  30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
        35                  40                  45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
    50                  55                  60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr
65                  70                  75                  80

Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys
                85                  90                  95

Thr Val


<210> SEQ ID NO 67
<211> LENGTH: 105
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgM CH1

<400> SEQUENCE: 67

Gly Ser Ala Ser Ala Pro Thr Leu Phe Pro Leu Val Ser Cys Glu Asn
1               5                   10                  15

Ser Pro Ser Asp Thr Ser Ser Val Ala Val Gly Cys Leu Ala Gln Asp
            20                  25                  30

Phe Leu Pro Asp Ser Ile Thr Phe Ser Trp Lys Tyr Lys Asn Asn Ser
        35                  40                  45

Asp Ile Ser Ser Thr Arg Gly Phe Pro Ser Val Leu Arg Gly Gly Lys
    50                  55                  60

Tyr Ala Ala Thr Ser Gln Val Leu Leu Pro Ser Lys Asp Val Met Gln
65                  70                  75                  80

Gly Thr Asp Glu His Val Val Cys Lys Val Gln His Pro Asn Gly Asn
                85                  90                  95

Lys Glu Lys Asn Val Pro Leu Pro Val
            100                 105


<210> SEQ ID NO 68
<211> LENGTH: 104
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgM CH3

<400> SEQUENCE: 68

Thr Ala Ile Arg Val Phe Ala Ile Pro Pro Ser Phe Ala Ser Ile Phe
1               5                   10                  15

Leu Thr Lys Ser Thr Lys Leu Thr Cys Leu Val Thr Asp Leu Thr Thr
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-continued

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            20                  25                  30

Tyr Asp Ser Val Thr Ile Ser Trp Thr Arg Gln Asn Gly Glu Ala Val
        35                  40                  45

Lys Thr His Thr Asn Ile Ser Glu Ser His Pro Asn Ala Thr Phe Ser
    50                  55                  60

Ala Val Gly Glu Ala Ser Ile Cys Glu Asp Asp Trp Asn Ser Gly Glu
65                  70                  75                  80

Arg Phe Thr Cys Thr Val Thr His Thr Asp Leu Pro Ser Pro Leu Lys
                85                  90                  95

Gln Thr Ile Ser Arg Pro Lys Gly
            100
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1. A bispecific antibody comprising:

a first arm binding to a first antigen and comprising VH1-CHa-Fc1 and VL1-CLb; and

a second arm binding to a second antigen and comprising VH2-CH1-Fc2 and VL2-CL,

wherein the VH1 and the VH2 are each heavy-chain variable regions including the same or different antigen-binding regions,

the VL1 and VL2 are each light-chain variable regions including the same or different antigen-binding regions,

the CHa comprises i) an IgG heavy-chain constant region or an IgD heavy-chain constant region CH1, and an IgG heavy-chain constant region CH2 or CH3, or ii) an IgM heavy-chain constant region CH3,

the CLb comprises i) a CL1 including an IgG light-chain constant region λ or κ, and at least one selected from the group consisting of IgG heavy-chain constant regions CH1, CH2, and CH3, or ii) an IgM heavy-chain constant region CH3,

the CH1 is an IgG heavy-chain constant region CH1 and CL is an IgG light-chain constant region CL, and

the Fc1 of the first arm is linked to the Fc2 of the second arm to form a heavy-chain constant region dimer.

2. The bispecific antibody according to claim **1**, wherein the CHa and the CLb are derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM.

3. The bispecific antibody according to claim **1**,

wherein, in the first arm, the CHa comprises a heavy-chain constant region CH1 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM, and a heavy-chain constant region CH3 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM in order from an N-terminus to a C-terminus; or

wherein, in the first arm, the CHa comprises a heavy-chain constant region CH3 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM, and a heavy-chain constant region CH1 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM in order from the N-terminus to the C-terminus.

4. (canceled)

5. The bispecific antibody according to claim **1**, wherein the CHa of the first arm comprises a heavy-chain constant region CH3 derived from IgM.

6. The bispecific antibody according to claim **1**,

wherein, in the first arm, the CLb comprises a light-chain constant region CL1 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM, and a heavy-chain constant region CH3 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM in order from the N-terminus to the C-terminus; or

wherein, in the first arm, the CLb comprises a CH3 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM and a CL1 derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM in order from the N-terminus to the C-terminus.

7. (canceled)

8. The bispecific antibody according to claim **1**, wherein, in the first arm, the CLb comprises a heavy-chain constant region CH3 derived from IgM.

9. The bispecific antibody according to claim **1**, wherein the CH1 of CHa and CL1 of CLb are linked without a disulfide bond.

10. The bispecific antibody according to claim **1**, wherein the CHa and CLb each comprise CH3 and the CH3 forms a dimer through a disulfide bond.

11. The bispecific antibody according to claim **1**,

wherein the CHa and CLb each comprise CH3, and one of the dimers formed by bonding CH3 comprises at least one selected from the group consisting of Y349C, S354C, T366S, T366W, L368A, and Y407V, and comprises at least one selected from the group consisting of S354C, Y349C, T366W, T366S, L368A and Y407V, to form a knob-in-hole structure; or

wherein one of the CH3 dimers of the Fc comprises at least one selected from the group consisting of Y349C, S354C, T366S, T366W, L368A, and Y407V, and at least one selected from the group consisting of S354C, Y349C, T366W, T366S, L368A and Y407V, to form a knob-in-hole structure in the dimer.

12. The bispecific antibody according to claim **1**, wherein the CH1 of CHa and the CL1 of CLb are linked without a disulfide bond, and i) any one of CH3 of CHa and CH3 of CLb comprises at least one selected from the group consisting of Y349C, T366S, L368A and Y407V, and the other comprises S354C and/or T366W, or ii) any one of CH3 of CHa and CH3 of CLb comprises at least one selected from the group consisting of S354C, T366S, L368A and Y407V, and the other comprises Y349C and/or T366W.

13. (canceled)

14. A bispecific antibody comprising:

a first arm binding to a first antigen and comprising VH1-CHa-Fc1 and VL1-CLb; and

a second arm binding to a second antigen and comprising VH2-CH1-Fc2 and VL2-CL,

wherein the VH1 and the VH2 are each heavy-chain variable regions including the same or different antigen-binding regions,

the VL1 and VL2 are each light-chain variable regions including the same or different antigen-binding regions,
the CHa comprises an IgG heavy-chain constant region CH3 and an IgG heavy-chain constant region CH1,
the CLb comprises CL1 including an IgG light-chain constant region or x, and CH3 including an IgG heavy-chain constant region,
the CH1 is an IgG heavy-chain constant region CH1 and CL is an IgG light-chain constant region CL, and
the Fc1 of the first arm is linked to the Fc2 of the second arm to form a heavy-chain constant region dimer.

15. The bispecific antibody according to claim **14**, wherein the IgG heavy-chain constant region CH1 is derived from IgG1, IgG2, IgG3, IgG4, IgD or IgM.

16. The bispecific antibody according to claim **14**,
wherein the CH1 of CHa and CL1 of CLb are linked without a disulfide bond, or
wherein the CH3 of CHa and CL3 of CLb form a dimer through a disulfide bond.

17. (canceled)

18. The bispecific antibody according to claim **14**,
wherein one of the dimers formed by bonding the CH3 of CHa to the CH3 of CLb comprises at least one selected from the group consisting of T366W, S354C, and Y349C, and the other comprises at least one selected from the group consisting of S354C, Y349C, T366S, L368A, and Y407V, to form a knob-in-hole structure, or
wherein one of the dimers formed by bonding the CH3 of CHa to the CH3 of CLb comprises at least one selected from the group consisting of S354C, T366S, L368A, and Y407V, and the other comprises Y349C and T366W, to form a knob-in-hole structure.

19. (canceled)

20. The bispecific antibody according to claim **14**, wherein the CH3 of CHa or CH3 of CLb comprises a sequence represented by SEQ ID NOS. 8 to 13.

21. The bispecific antibody according to claim **14**, wherein the CH1 of CHa and the CL1 of CLb are linked without a disulfide bond, and i) any one of the CH3 of CHa and CH3 of CLb comprises at least one selected from the group consisting of Y349C, T366S, L368A and Y407V, and the other comprises S354C and/or T366W, or ii) any one of the CH3 of CHa and CH3 of CLb comprises at least one selected from the group consisting of S354C, T366S, L368A and Y407V, and the other comprises Y349C and/or T366W.

22. The bispecific antibody according to claim **14**, wherein the CH3 and CH1 of Cha, and CH3 and CL1 of CLb are linked via a linker.

23. (canceled)

24. The bispecific antibody according to claim **22**, wherein monomers of the CH3 dimer formed by the Fc1 of first arm and the Fc2 of second arm are linked through a disulfide bond or without a disulfide bond.

25. The bispecific antibody according to claim **14**, wherein the first arm and the second arm are linked via a hinge.

26. (canceled)

27. A multispecific antibody comprising the bispecific antibody according to claim **1**.

28. (canceled)

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